

Classroom Slides on Plasticity

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Before we start

Notation:

- Throughout this classroom Notes/Slides:
 - ▶ (i) Lightface Italic Latin and Greek letters generally denote **scalars**.
 - ★ Ex. a, b, α, β , Temperature: θ , Internal energy U , Vector position component: x, x_1 , etc.
 - ▶ (ii) Boldface Italic lower and uppercase Latin and Greek letters generally denote **vectors**
 - ★ Ex. a, b, α, β , Heat flux: q , Position vectors: x, X , force: F , etc.
 - ▶ (iii) Boldface lower and uppercase Latin and Greek letters generally denote **second-order tensors**
 - ★ Ex. Tensors: A, B, α, β , Stress tensor: σ , Deformation gradient: F , Strain tensor: ϵ , etc.
 - ▶ (iv) **High order tensors**:
 - ★ Third-order tensors: $A, B, \dots$
 - ★ Fourth-order tensors: $A, B, \dots$
 - ▶ (v) **Sets and spaces** are denoted by the Formal Script or by Regular font
 - ★ Ex.: “General” linear vector space: $\mathcal{V}$, Real space: $\mathcal{R}$, etc.
 - ★ Ex.: Linear vector space of functions: Space of the continuous functions C^m , p -integrable space L^p , etc.

Before we start

Operations & Notations

Table: Some examples of operations used in this text.

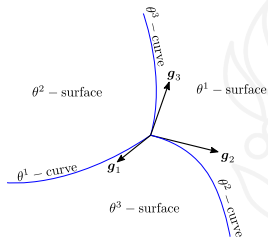
Operation/Notation	Compact	Index	Index Einstein summation convention	Matrix
Internal product between two vectors $\mathbf{u}$ and $\mathbf{v}$	$\alpha = \mathbf{u} \cdot \mathbf{v}$	$\alpha = \sum_{i=1}^3 u_i v_i$	$\alpha = u_i v_i, \quad i = 1, 2, 3$	$\alpha = [\mathbf{u}]^T [\mathbf{v}]$
Product of the tensor $\mathbf{A}$ over vector $\mathbf{u}$	$\mathbf{v} = \mathbf{A}\mathbf{u}$	$v_i = \sum_{j=1}^3 A_{ij} u_j, \quad i = 1, 2, 3$	$v_i = A_{ij} u_j, \quad i, j = 1, 2, 3$	$[\mathbf{v}] = [\mathbf{A}] [\mathbf{u}]$
Product of two tensors $\mathbf{A}$ and $\mathbf{B}$	$\mathbf{C} = \mathbf{A}\mathbf{B}$	$C_{ij} = \sum_{k=1}^3 A_{ik} B_{kj}, \quad i, j = 1, 2, 3$	$C_{ij} = A_{ik} B_{kj}, \quad i, j, k = 1, 2, 3$	$[\mathbf{C}] = [\mathbf{A}] [\mathbf{B}]$
Diadic or tensorial product between two vectors $\mathbf{u}$ and $\mathbf{v}$	$\mathbf{T} = \mathbf{u} \otimes \mathbf{v}$	$T_{ij} = u_i v_j, \quad i, j = 1, 2, 3$	$T_{ij} = u_i v_j, \quad i, j = 1, 2, 3$	$[\mathbf{T}] = [\mathbf{u}] [\mathbf{v}]^T$

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Before we start

Coordinate System

- Curvilinear Coordinate System:



General Curvilinear Coordinate System

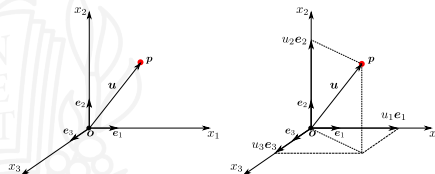
- Examples (cases):

- ▶ Cartesian coordinate system:

- ★ Special coordinate system where the covariant and contravariant bases are exactly the same!

- ▶ Cylindrical coordinate system;
 - ▶ Spherical coordinate system;
 - ▶ etc...

- Cartesian coordinate system:



- Let $u \in \mathcal{R}^3$. Cartesian components:

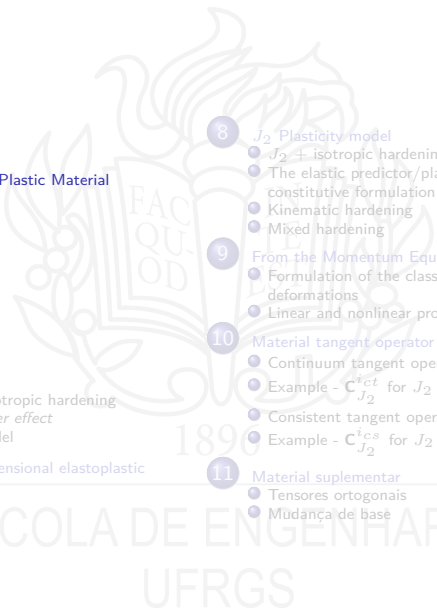
$$u = u_1 e_1 + u_2 e_2 + u_3 e_3 = u_i e_i \quad (1.1)$$

- u_i are the components of the vector u with relation to the orthonormal basis e_i :

$$u_i = u \cdot e_i \rightarrow \begin{cases} u_1 = u \cdot e_1 \\ u_2 = u \cdot e_2 \\ u_3 = u \cdot e_3 \end{cases}$$

$u = u \cdot e$

Outline

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 - Conservation of Mass
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 - Thermodynamic laws
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Introduction

Introduction

- We start our discussion into the plasticity field describing, phenomenologically and mathematically, one of the most simple inelastic material that is the *Elastic, Perfectly Plastic* material.
- Although very simple, the particular mathematical ingredients of this special one dimensional will help us to understand more complex models, even under *multiaxial effects*.
- An elastic perfectly plastic materials presents two distinct mechanical responses that are: a linear elastic reversible phase and plastic phase where permanent irreversible effects take place.
- No viscous effects are considered: It is a rate independent model!
- No relations to the energy potentials will be presented in this moment. Further in this course we revisit this model inside the thermodynamic framework.

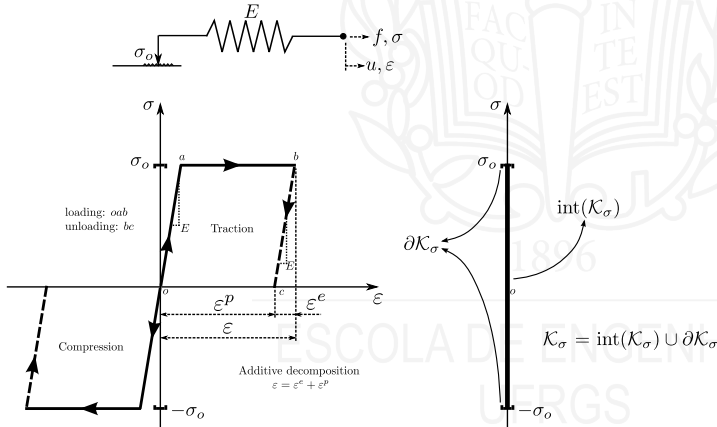
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Mathematical description of the rheological model

Elastic, perfectly plastic rheological model, stress-strain response, and the admissible stress set $\mathcal{K}_\sigma$:

- The model:

Rheological model of an Elastic Perfectly Plastic Material



- Assume that the total deformation ε may be additively decomposed into an elastic deformation ε^e , associated with the linear elastic reversible phase, and a plastic (permanent or inelastic) deformation ε^p one writes

$$\varepsilon = \varepsilon^e + \varepsilon^p. \quad (2.1)$$

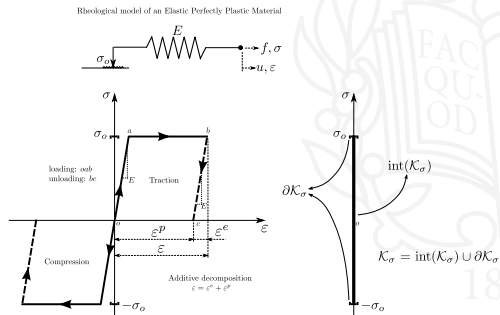
- Assuming a linear elastic isotropic material

$$\sigma = E\varepsilon^e = E(\varepsilon - \varepsilon^p). \quad (2.2)$$

Mathematical description of the rheological model

Elastic, perfectly plastic rheological model, stress-strain response, and the admissible stress set $\mathcal{K}_\sigma$:

- The model:



This condition motivates the definition of the so-called *yield function* such that

$$f(\sigma) = |\sigma| - \sigma_o \leq 0, \quad (2.4)$$

$f: \mathcal{R} \rightarrow \mathcal{R}_o^-$, and the following *admissible stress set* can be defined

$$\mathcal{K}_\sigma = \{\sigma \in \mathcal{R} \mid f(\sigma) = |\sigma| - \sigma_o \leq 0\}. \quad (2.5)$$

$\mathcal{K}_\sigma$ is a *closed convex set*. The set $\mathcal{K}_\sigma$ can be interpreted as

$$\mathcal{K}_\sigma = \text{int}(\mathcal{K}_\sigma) \cup \partial\mathcal{K}_\sigma \quad (2.6)$$

being $\text{int}(\mathcal{K}_\sigma)$ the interior of the set $\mathcal{K}_\sigma$ and $\partial\mathcal{K}_\sigma$ the boundary of $\mathcal{K}_\sigma$, ie

- The stress σ cannot be greater than σ_o in absolute value, so

$$\sigma \in [-\sigma_o, \sigma_o] \quad \text{or} \quad |\sigma| \leq \sigma_o. \quad (2.3)$$

and

$$\text{int}(\mathcal{K}_\sigma) = \{\sigma \in \mathcal{R} \mid f(\sigma) = |\sigma| - \sigma_o < 0\} \quad (2.7)$$

$$\partial\mathcal{K}_\sigma = \{-\sigma_o, \sigma_o\}. \quad (2.8)$$

Mathematical description of the rheological model

Elastic, perfectly plastic rheological model, stress-strain response, and the admissible stress set $\mathcal{K}_\sigma$:

- 2 If $\sigma < \sigma_o$ there is no change in ε^P . That is $\dot{\varepsilon}^P = 0$.

Therefore

$$\dot{\varepsilon}^P = 0 \leftrightarrow f(\sigma) < 0. \quad (2.9)$$

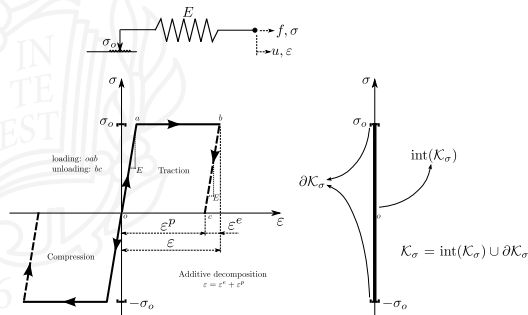
Now from Eq. (2.2) one has the following rate, assuming that E does not change with time,

$$\dot{\sigma} = E \left(\dot{\varepsilon} - \dot{\varepsilon}^P \right) \rightarrow \dot{\sigma} = E \dot{\varepsilon} \quad (2.10)$$

but from Eq. (2.1)

$$\dot{\varepsilon} = \dot{\varepsilon}^e + \dot{\varepsilon}^P \rightarrow \dot{\varepsilon} = \dot{\varepsilon}^e \rightarrow \dot{\sigma} = E \dot{\varepsilon}^e. \quad (2.11)$$

Rheological model of an Elastic Perfectly Plastic Material



- In terms of the set defined in 2.7 this condition can be stated as

$$\sigma \in \text{int}(\mathcal{K}_\sigma) \rightarrow \dot{\varepsilon}^P = 0 \rightarrow \dot{\sigma} = E \dot{\varepsilon}^e. \quad (2.12)$$

Mathematical description of the rheological model

Elastic, perfectly plastic rheological model, stress-strain response, and the admissible stress set $\mathcal{K}_\sigma$:

3 The third consequence comes from the interpretations of the preceding two:

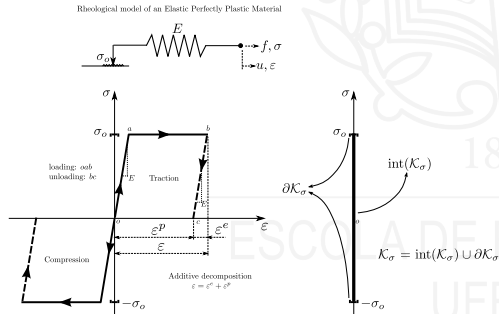
- a From 1: $\nexists \sigma$ such that $f(\sigma) = |\sigma| - \sigma_o > 0$ that is, $\sigma \in \mathcal{K}_\sigma$;
- b From 2: $\sigma \in \text{int}(\mathcal{K}_\sigma) \rightarrow \dot{\varepsilon}^P = 0$;

- So, a change in ε^P takes place when $f(\sigma) = |\sigma| - \sigma_o = 0$ or

$$\dot{\varepsilon}^P \neq 0 \leftrightarrow f(\sigma) = 0 \rightarrow \sigma \in \partial\mathcal{K}_\sigma. \quad (2.13)$$

Meaning that for the rheological model in analysis a change/slip takes place when $\sigma \in \partial\mathcal{K}_\sigma$. The slip will be in the direction (sense) of the applied stress.

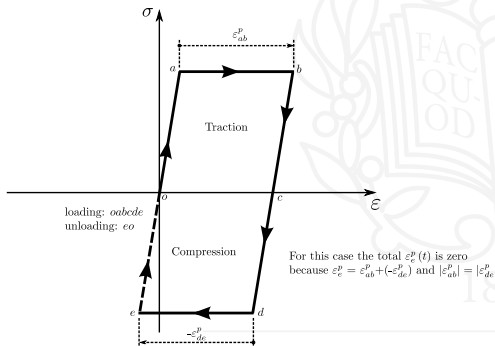
- The condition for the evolution of the plastic deformation was established. However, it is still not possible to determine an explicit evolution for $\dot{\varepsilon}^P$. We need to introduce another variable to do that.



The absolute slip rate - Who is $\dot{\epsilon}^p$?

The absolute slip rate

Notice that:



This is a loading case where the final plastic deformation is zero but the material experience changes in the microstructure.

- Now, let us define the *absolute slip rate* $\dot{\gamma}$ as

$$\dot{\gamma} \geq 0. \quad (2.14)$$

- The absolute slip rate is related with the plastic strain by

$$\dot{\epsilon}^p = \begin{cases} \dot{\gamma} & \text{if } \sigma = \sigma_o \\ -\dot{\gamma} & \text{if } \sigma = -\sigma_o \end{cases} \quad (2.15)$$

or

$$\dot{\epsilon}^p = \dot{\gamma} \text{sign}(\sigma) \quad (2.16)$$

where

$$\text{sign}(x) = \begin{cases} 1 & \text{if } x \geq 0 \\ -1 & \text{if } x < 0 \end{cases} \quad (2.17)$$

is the so-called signal function. The absolute slip rate $\dot{\gamma}$ is still an incognito to be determined.

The absolute slip rate - $\dot{\gamma} \geq 0$

Determination of $\dot{\gamma}$ - loading and unloading conditions

To determine $\dot{\gamma}$ it is necessary to enforce the conditions 1 to 3 and some unilateral conditions. They are:

- From 1:

$$\sigma \in \mathcal{K}_\sigma \rightarrow f(\sigma) = |\sigma| - \sigma_o \leq 0$$

and from 2-3:

$$\dot{\varepsilon}^p = 0 \rightarrow \dot{\gamma} = 0$$

$$\dot{\varepsilon}^p \neq 0 \rightarrow \dot{\gamma} > 0$$

Therefore we have to enforce

$$\dot{\gamma} \geq 0 \text{ and } f(\sigma) \leq 0 \quad (2.18)$$

- Now, from 2

$$\dot{\varepsilon}^p = 0 \rightarrow \dot{\gamma} = 0 \leftrightarrow f(\sigma) < 0 \rightarrow \sigma \in \text{int}(\mathcal{K}_\sigma)$$

and 3:

$$\dot{\varepsilon}^p \neq 0 \rightarrow \dot{\gamma} > 0 \leftrightarrow f(\sigma) = 0 \rightarrow \sigma \in \partial\mathcal{K}_\sigma.$$

- Therefore we have to enforce

$$\dot{\gamma} = 0 \rightarrow f(\sigma) < 0$$

$$\dot{\gamma} > 0 \rightarrow f(\sigma) = 0$$

that can be summarized in only one condition, that is

$$\dot{\gamma} f(\sigma) = 0. \quad (2.19)$$

Loading and unloading conditions

Some important conditions

- Now, suppose that $\varepsilon(t)$ and $\varepsilon^p(t)$ are known for a given $t \in (0, t_f]$. As a consequence, the stress $\sigma(t)$ is also known since $\sigma(t) = E[\varepsilon(t) - \varepsilon^p(t)]$, Eq. (2.2). Also, let us assume that

$$\sigma(t) \in \partial\mathcal{K}_\sigma \leftrightarrow f(\sigma(t)) = 0. \quad (2.20)$$

Keeping this condition in mind, let us investigate what are the possibilities for the existence of $\dot{f}$ when $\sigma(t) \in \partial\mathcal{K}_\sigma$. At a first glance, there are three possibilities, $\dot{f} > 0$, $\dot{f} = 0$, and $\dot{f} < 0$. Let us investigate each one of them.

Loading and unloading conditions

Condition $\dot{f}(\sigma(t)) > 0$

i Suppose that $\dot{f}(\sigma(t)) > 0$: As a first movement towards analysis we write

$$\dot{f}(\sigma(t)) = \lim_{\Delta t \rightarrow 0} \frac{f(\sigma(t + \Delta t)) - f(\sigma(t))}{\Delta t} > 0$$

given $\Delta t > 0$ the condition for a positive rate becomes $f(\sigma(t + \Delta t)) - f(\sigma(t)) > 0$. But notice that, if $\sigma(t) \in \partial\mathcal{K}_\sigma$ then $f(\sigma(t)) = 0$. Therefore to guarantee $\dot{f}(\sigma(t)) > 0$ one has to enforce $f(\sigma(t + \Delta t)) > 0$ but from condition 1, on page 8, there is no σ such that $f(\sigma) = |\sigma| - \sigma_o > 0$. One concludes, therefore, that

$$\nexists \dot{f}(\sigma(t)) > 0.$$

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Loading and unloading conditions

Condition $\dot{f}(\sigma(t)) = 0$

ii Suppose that $\dot{f}(\sigma(t)) = 0$: Using the same arguments as in (a) we write

$$\dot{f}(\sigma(t)) = \lim_{\Delta t \rightarrow 0} \frac{f(\sigma(t + \Delta t)) - f(\sigma(t))}{\Delta t} = 0$$

given $\Delta t > 0$. Therefore, to enforce to $\dot{f}(\sigma(t)) = 0$ one has to enforce that $f(\sigma(t + \Delta t)) = 0$. Despite the signal of $\sigma(t + \Delta t)$ the stress state must lie on the yield surface, $\sigma(t + \Delta t) \in \partial\mathcal{K}_\sigma$. This condition says that $\sigma(t + \Delta t)$ is also a loading state, so one concludes

$$\dot{f}(\sigma(t)) = 0 \rightarrow \text{loading condition.}$$

Loading and unloading conditions

Condition $\dot{f}(\sigma(t)) < 0$

- iii Finally, assume that $\dot{f}(\sigma(t)) < 0$: Also given $\Delta t > 0$ this situation implies that $f(\sigma(t + \Delta t)) < 0$. As the point moved from $f(\sigma(t)) = 0$ to $f(\sigma(t + \Delta t)) < 0$, or from the yielding surface to the elastic region, this condition is interpreted as an unloading stress state, meaning that

$$\dot{f}(\sigma(t)) < 0 \rightarrow \text{unloading condition.}$$

Loading and unloading conditions

Condition $\dot{f}(\sigma(t)) < 0$

- As summary

$$\dot{f}(\sigma(t)) \leq 0 \text{ with } \begin{cases} \dot{f}(\sigma(t)) = 0 \rightarrow \text{loading condition} \\ \dot{f}(\sigma(t)) < 0 \rightarrow \text{unloading condition} \end{cases} . \quad (2.21)$$

- Also, relating the loading and unloading conditions on $\dot{f}(\sigma(t))$ with those derived for the *absolute slip rate*, on page 11, motivate us to write

$$\dot{\gamma} = 0 \rightarrow \dot{f}(\sigma(t)) < 0 \quad (2.22)$$

$$\dot{\gamma} > 0 \rightarrow \dot{f}(\sigma(t)) = 0 \quad (2.23)$$

that can be encapsulated in only one condition, that is

$$\dot{\gamma} \dot{f}(\sigma) = 0. \quad (2.24)$$

- This last condition is known as the *consistency condition* or *persistence condition*. It means that the stress must persist on the yield surface for an increase of the *absolute slip rate*.

Determination of the *absolute slip rate* $\dot{\gamma}$

Absolute slip rate $\dot{\gamma}$ determination

- The determination of $\dot{\gamma}$, of course for loading purposes, comes from the analysis of $\dot{f}(\sigma) = 0$. Using the chain rule the rate of the yield function can be written as

$$\frac{df}{dt} = \frac{df}{d\sigma} \frac{d\sigma}{dt} = 0. \quad (2.25)$$

- Let us analyze the first term $\frac{df}{d\sigma}$. Based on Eq. (2.4) it yields

$$\frac{df}{d\sigma} = \frac{d|\sigma|}{d\sigma} = \begin{cases} 1 & \text{if } \sigma \geq 0 \\ -1 & \text{if } \sigma < 0 \end{cases} \quad (2.26)$$

what is exactly the signal function, Eq. (2.17), ie

$$\frac{df}{d\sigma} = \text{sign}(\sigma). \quad (2.27)$$

- The second term in Eq. (2.25) is determined recalling that $\sigma(t) = E[\varepsilon(t) - \varepsilon^p(t)]$, Eq. (2.2), and Eq. (2.16). Thus

$$\begin{aligned} \dot{\sigma} &= E(\dot{\varepsilon} - \dot{\varepsilon}^p) \\ &= E\dot{\varepsilon} - E\dot{\gamma}\text{sign}(\sigma). \end{aligned} \quad (2.28)$$

- Now substituting Eq. (2.27) and Eq. (2.28) into Eq. (2.25)

$$\begin{aligned} \frac{df}{dt} &= \frac{df}{d\sigma} \frac{d\sigma}{dt} \\ &= \text{sign}(\sigma) [E\dot{\varepsilon} - E\dot{\gamma}\text{sign}(\sigma)] \\ &= E[\dot{\varepsilon}\text{sign}(\sigma) - \dot{\gamma}] = 0 \end{aligned}$$

what leads to

$$\dot{\gamma} = \dot{\varepsilon}\text{sign}(\sigma). \quad (2.29)$$

Some considerations

Some notes...

- Notice that the formulation presented here appears in terms of deformation variables $\{\varepsilon, \varepsilon^p\}$. The deformation is, in fact, the independent variable in this model. This formulation is called *Strain Driven Formulation*.
- Concluding this section, let us present some mathematical consequences of the model presented so far. Substituting Eq. (2.29) in Eq. (2.16) we have

$$\begin{aligned}\dot{\varepsilon}^p &= \dot{\gamma} \text{sign}(\sigma) \\ &= \dot{\varepsilon} [\text{sign}(\sigma)]^2 \\ &= \dot{\varepsilon}\end{aligned}\tag{2.30}$$

showing that for the elastic perfectly plastic material the *plastic flux* $\dot{\varepsilon}^p$ is exactly the same to the total rate of deformation $\dot{\varepsilon}$. Also, as $\varepsilon = \varepsilon^e + \varepsilon^p$ then

$$\dot{\varepsilon}^e = \dot{\varepsilon} - \dot{\varepsilon}^p = 0\tag{2.31}$$

so for a perfectly plastic material there is no evolution of the elastic strain (as expected!).

Accumulated plastic strain

Accumulated plastic strain

- Let us introduce now an auxiliary variable that accounts for the *total* or *accumulated plastic strain* experienced by a point of the body from time $t \in (t_o, t_f]$. The accumulated plastic strain is an integral over time history defined by

$$\bar{\epsilon}^p := \int_{t_o}^{t_f} |\dot{\epsilon}^p| dt \quad (2.32)$$

the module operator is to account for tensile and compression strain states.

- The accumulated plastic strain plays a very important role in the description of hardening in materials. Moreover, it is a very important variable to understand the nature of degradation while modeling the degradation variable (damage) in metals.

- Substituting Eq. (2.16) in Eq. (2.32) it is possible to write

$$\begin{aligned} \bar{\epsilon}^p &= \int_{t_o}^{t_f} |\dot{\gamma} \text{sign}(\sigma)| dt \\ &= \int_{t_o}^{t_f} |\dot{\gamma}| |\text{sign}(\sigma)| dt \\ &= \int_{t_o}^{t_f} \dot{\gamma} dt \end{aligned} \quad (2.33)$$

because $|\text{sign}(\sigma)| = 1$ and $\dot{\gamma} \geq 0$. Moreover, we can show that

$$\begin{aligned} \dot{\bar{\epsilon}}^p &= |\dot{\epsilon}^p| \\ &= \dot{\gamma} \end{aligned} \quad (2.34)$$

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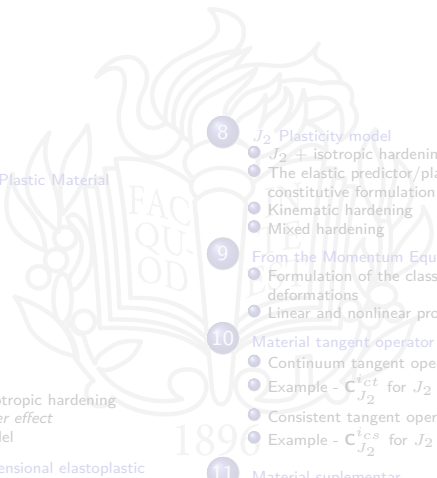
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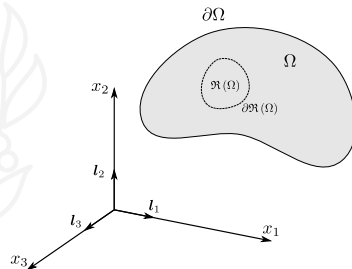
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Body as a continuum

Body an material particles

- A body $\Omega \subset \mathcal{R}^3$ can be viewed as set of *particles*, *material particles* or *material points*, that has *continuous distribution* in space. The body Ω may change its *configuration*, i.e., it may occupies different geometric regions along with time.
- Figure shows a configuration of a body Ω , with *sufficiently regular* (*smooth*) boundary $\partial\Omega$, occupying a region of the *Euclidean space* at a given time t . $\mathfrak{R}(\Omega) \subset \Omega$ is a region of the body with *sufficiently regular* boundary $\partial\mathfrak{R}(\Omega)$.



- For sake of a representation, an orthogonal Cartesian system is presented in this figure. Most part of the figures in this slides make reference to Cartesian systems to define some quantities, however the reader must realize that other coordinate systems may also be used.

Body as a continuum

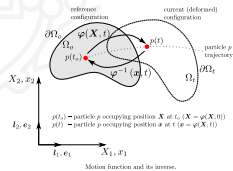
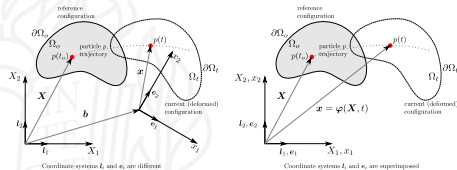
Preliminaries on motion description

- Suppose that a body occupies a certain region in the space at time t_o . In order to fully describe the movement of each particle of this body let us introduce an orthogonal Cartesian system defined by the orthonormal basis l_1 , l_2 , and l_3 such that each material point can be written using

$$\mathbf{X} = X_i l_i = X_1 l_1 + X_2 l_2 + X_3 l_3. \quad (3.1)$$

The body at the configuration t_o will be called Ω_o or

$$\Omega(\mathbf{X}, t_o) = \Omega_o(\mathbf{X}), \quad \Omega_o(\mathbf{X}) \subset \mathcal{R}^3. \quad (3.2)$$



- Assume a movement φ is imposed over Ω_o transforming the body Ω_o into Ω_t , the *current configuration*. In order to describe the movement let us introduce a material point labeled $p(t_o)$ in body Ω_o , so that $p(t_o)$ is referenced by the *material position vector* $\mathbf{X}$, see Figure. Under the imposed movement $p(t_o) \rightarrow p(t)$. Let us reference the point $p(t)$ by the *spatial position vector* $\mathbf{x}$. In further developments we use the notation $p_o \equiv p(t_o)$, $p_t \equiv p(t)$ in order to simplify the reading.

Kinematics

Preliminaries on motion description

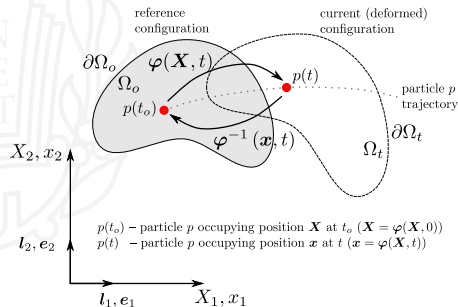
- Let us define the mapping $\varphi : \Omega_o \rightarrow \Omega_t$. $\varphi(\mathbf{X}, t)$ is the *motion function* which maps the *reference configuration* onto the *current configuration*. In other words, $\varphi(\mathbf{X}, t)$ maps each material point $\mathbf{X}$ and time t to the spatial point $\mathbf{x}$, i.e.,

$$\mathbf{x} = \varphi(\mathbf{X}, t). \quad (3.3)$$

So, the spatial point $\mathbf{x}$ is the position of the material point $\mathbf{X}$ at a time t , see figure. The spatial point $\mathbf{x}$ can be decomposed with relation to its orthonormal basis such that $\mathbf{x} = x_i \mathbf{e}_i$.

- A very important required property of the motion function is the *one-to-one onto* mapping (or *bijective* function). In other words, the inverse of the mapping, $\varphi^{-1} : \Omega_t \rightarrow \Omega_o$, exists. Therefore one writes

$$\mathbf{X} = \varphi^{-1}(\mathbf{x}, t). \quad (3.4)$$



Motion function and its inverse.

Kinematics

Material and Spatial descriptions

- When a body, a continuum, is under motion the quantities associated with it may change with time. Such quantities can be scalars fields ϕ , as the temperature for instance, vector fields, as the velocity $\mathbf{v}$ and tensors fields $\mathbf{T}$, as the stress tensor. There are two possible approaches to accompany such changes along time, they are:
- MATERIAL DESCRIPTION (LAGRANGE DESCRIPTION, REFERENCE DESCRIPTION): Following the particles, i.e., expressing ϕ , $\mathbf{v}$, and $\mathbf{T}$ as functions of the particles, i.e., material coordinates, and time:

$$\phi = \phi(X_1, X_2, X_3, t) = \phi(\mathbf{X}, t) \quad (3.5a)$$

$$\mathbf{v} = \mathbf{v}(X_1, X_2, X_3, t) = \mathbf{v}(\mathbf{X}, t) \quad (3.5b)$$

$$\mathbf{T} = \mathbf{T}(X_1, X_2, X_3, t) = \mathbf{T}(\mathbf{X}, t) \quad (3.5c)$$

Material coordinates $\{X\} = (X_1, X_2, X_3)$ are *independent variables* in a material description.

- SPATIAL DESCRIPTION (EULERIAN DESCRIPTION): Fixing at a location, i.e., expressing ϕ , $\mathbf{v}$, and $\mathbf{T}$ as functions of a fixed position in space. That is

$$\phi = \phi(x_1, x_2, x_3, t) = \phi(\mathbf{x}, t) \quad (3.6a)$$

$$\mathbf{v} = \mathbf{v}(x_1, x_2, x_3, t) = \mathbf{v}(\mathbf{x}, t) \quad (3.6b)$$

$$\mathbf{T} = \mathbf{T}(x_1, x_2, x_3, t) = \mathbf{T}(\mathbf{x}, t) \quad (3.6c)$$

Kinematics

Material description of a spatial field

- Let $\bullet(x, t)$ represent a spatial quantity. The *material description* of the *spatial quantity* is recoverable by changing the position arguments of such function by evoking Eq. (3.3). So, the *material description* of the *spatial quantity* is defined by

$$\phi_m(\mathbf{X}, t) = \phi(\varphi(\mathbf{X}, t), t) \quad (3.7a)$$

$$\mathbf{v}_m(\mathbf{X}, t) = \mathbf{v}(\varphi(\mathbf{X}, t), t) \quad (3.7b)$$

$$\mathbf{T}_m(\mathbf{X}, t) = \mathbf{T}(\varphi(\mathbf{X}, t), t) \quad (3.7c)$$

in which the subscript m stands for *material quantities* whereas s for *spatial quantities*. By writing in this form we desire to call the attention of the reader that we are interested in accompany the material particle p along its movement.

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Kinematics

Spatial description of the material quantity

- On the other hand, let $\bullet(\mathbf{X}, t)$ represent a material quantity. Employing Eq. (3.4) one may write the *spatial description* of the *material quantity*, that is:

$$\phi_s(\mathbf{x}, t) = \phi(\varphi^{-1}(\mathbf{x}, t), t) \quad (3.8a)$$

$$\mathbf{v}_s(\mathbf{x}, t) = \mathbf{v}(\varphi^{-1}(\mathbf{x}, t), t). \quad (3.8b)$$

$$\mathbf{T}_s(\mathbf{x}, t) = \mathbf{T}(\varphi^{-1}(\mathbf{x}, t), t) \quad (3.8c)$$

- The use of subscript m or s will be employed in this text only when strictly necessary to not overstress the notation. The reader can go to the example on page 32 given in the following section to account for the transformations needed to evaluate the fields in the reference and spatial configurations.

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Kinematics

Material and spatial derivatives

- Let ϕ be a scalar field associated with a certain quantity (density, temperature etc.) of a body. Assume that such field be expressed in terms of the referenced $\mathbf{X}$, that is the material description of $\phi = \phi(\mathbf{X}, t)$. The material derivative of ϕ is

$$\frac{d\phi(\mathbf{X}, t)}{dt} = \dot{\phi}(\mathbf{X}, t) = \left. \frac{\partial \phi(\mathbf{X}, t)}{\partial t} \right|_{\mathbf{X} \text{ fixed}} \quad (3.9)$$

since $\dot{\mathbf{X}} = 0$.

- Assume now that the same quantity is written in terms of the current, spatial, description, that is $\phi = \phi(\mathbf{x}, t)$. Notice that the derivative with relation to time now yields

$$\begin{aligned} \frac{d\phi(\mathbf{x}, t)}{dt} = \dot{\phi}(\mathbf{x}, t) &= \left. \frac{\partial \phi(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x} \text{ fixed}} + \frac{\partial \phi(\mathbf{x}, t)}{\partial \mathbf{x}} \cdot \frac{d\mathbf{x}}{dt} \\ &= \left. \frac{\partial \phi(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x} \text{ fixed}} + \nabla_{\mathbf{x}} \phi(\mathbf{x}, t) \cdot \dot{\mathbf{x}} \end{aligned} \quad (3.10)$$

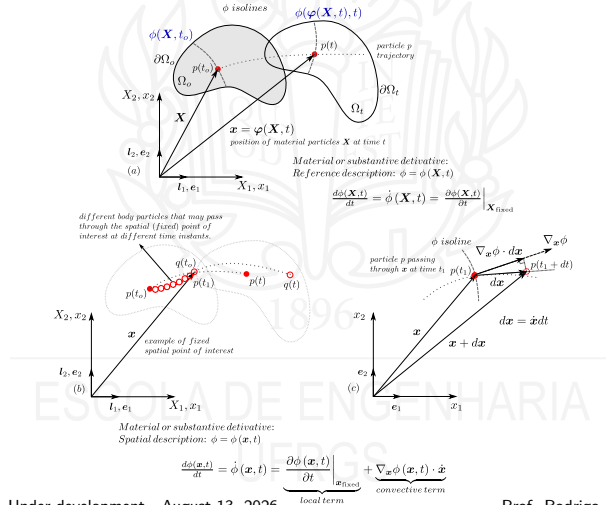
notice that in Eq. (3.10) $\mathbf{x} = \mathbf{x}(\mathbf{X}, t)$, see also Eq. (3.16). The term $\dot{\mathbf{x}} = \frac{d\mathbf{x}}{dt}$ is the velocity.

- Notice that, in general we have

$$\left. \frac{\partial \phi(\mathbf{X}, t)}{\partial t} \right|_{\mathbf{X} \text{ fixed}} \neq \left. \frac{\partial \phi(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x} \text{ fixed}} \quad \text{but} \quad \left. \frac{d\phi(\mathbf{x}, t)}{dt} \right|_{\mathbf{x}=\varphi(\mathbf{X}, t)} = \frac{d\phi(\mathbf{X}, t)}{dt}. \quad (3.11)$$

Kinematics

Physical/Graphical interpretation of material derivatives of fields ϕ written in terms of referential and spatial descriptions. (a) Referential description; (b) Spatial description; and (c) Convective term interpretation of the material derivative of $\phi(\mathbf{x}, t)$.



Kinematics

Material derivative of a vector field

- The material derivative of vector field follows Eq. (3.9) and (3.10). Let $\mathbf{u}$ be a vector field written in terms of the reference configuration $\mathbf{u} = \mathbf{u}(\mathbf{X}, t)$. The material derivative is

$$\frac{d\mathbf{u}(\mathbf{X}, t)}{dt} = \dot{\mathbf{u}}(\mathbf{X}, t) = \left. \frac{\partial \mathbf{u}(\mathbf{X}, t)}{\partial t} \right|_{\mathbf{X}_{\text{fixed}}} . \quad (3.12)$$

Now if $\mathbf{u} = \mathbf{u}(\mathbf{x}, t)$, then

$$\frac{d\mathbf{u}(\mathbf{x}, t)}{dt} = \dot{\mathbf{u}}(\mathbf{x}, t) = \left. \frac{\partial \mathbf{u}(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x}_{\text{fixed}}} + \nabla_{\mathbf{x}} \mathbf{u}(\mathbf{x}, t) \frac{d\mathbf{x}}{dt} \quad (3.13)$$

what in components is $\dot{u}_i(\mathbf{x}, t) = \left. \frac{\partial u_i(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x}_{\text{fixed}}} + \frac{\partial u_i(\mathbf{x}, t)}{\partial x_j} \dot{x}_j$. The term $\frac{d\mathbf{x}}{dt}$ is the velocity and will be discussed latter.

- Some authors, as [Gurtin 1981, Gurtin et al. 2010], refer to $\left. \frac{\partial \phi(\mathbf{X}, t)}{\partial t} \right|_{\mathbf{X}_{\text{fixed}}}$ as the *material derivative* and $\left. \frac{\partial \phi(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x}_{\text{fixed}}}$ as the *spatial derivative* denoting then by $\dot{\phi}$ and ϕ' , respectively. Other authors employ the notation $\frac{D}{Dt}$ to account for the material derivative. The term substantive derivative for the material derivative is employed by [Coimbra 1981].

Example

First ideas...

Example 1

Assume that the motion of a continuum body can be described by the following mapping (see [Reddy 2008] for a similar example)

$$\boldsymbol{x} = \boldsymbol{\varphi}(\boldsymbol{X}, t) = (X_1 + AtX_2)\boldsymbol{e}_1 + (X_2 - AtX_1)\boldsymbol{e}_2 + X_3\boldsymbol{e}_3 \quad (3.14)$$

and the temperature in the continuum is given in spatial description, $\theta = \theta(\boldsymbol{x}, t)$, by

$$\theta(\boldsymbol{x}, t) = x_1 + tx_2. \quad (3.15)$$

Determine:

- ① the inverse of the mapping: $\boldsymbol{\varphi}^{-1}$;
- ② the velocity components;
- ③ the material material time derivatives in both descriptions (material and spatial).

Kinematics

Displacement vector function and displacement gradient

- Let us define the *displacement vector function* $\mathbf{u}(\mathbf{X}, t)$ establishing the correspondence between $p(t_o)$ and $p(t)$. So, the current position $\mathbf{x}$ can be related to the reference position $\mathbf{X}$ using

$$\mathbf{x}(\mathbf{X}, t) = \mathbf{X} + \mathbf{u}(\mathbf{X}, t). \quad (3.16)$$

or by means of Eq. (3.3)

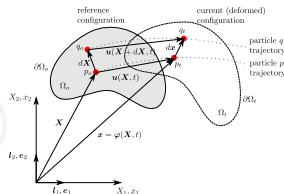
$$\boldsymbol{\varphi}(\mathbf{X}, t) = \mathbf{X} + \mathbf{u}(\mathbf{X}, t). \quad (3.17)$$

- Let $q(t_o)$ be a point in the *vicinity* of $p(t_o)$. This point moves to the *vicinity* of $p(t)$ resulting in $q(t)$. $q(t_o)$ is referenced by $\mathbf{X} + d\mathbf{X}$ and $q(t)$ by $\mathbf{x} + d\mathbf{x}$. By Eq. (3.16):

$$\mathbf{x}(\mathbf{X}, t) + d\mathbf{x}(\mathbf{X}, t) = \mathbf{X} + d\mathbf{X} + \mathbf{u}(\mathbf{X} + d\mathbf{X}, t). \quad (3.18)$$

Now subtracting Eq. (3.18) of Eq. (3.16) leads to

$$\begin{aligned} d\mathbf{x}(\mathbf{X}, t) &= d\mathbf{X} + \mathbf{u}(\mathbf{X} + d\mathbf{X}, t) - \mathbf{u}(\mathbf{X}, t) \\ &= d\mathbf{X} + \boldsymbol{\nabla}_{\mathbf{X}} \mathbf{u} d\mathbf{X} \end{aligned} \quad (3.19)$$



- $\boldsymbol{\nabla}_{\mathbf{X}} \mathbf{u}$ is the gradient of a vector field called *displacement gradient*. The *Cartesian components* are:

$$[\boldsymbol{\nabla}_{\mathbf{X}} \mathbf{u}] = \begin{bmatrix} \frac{\partial u_1}{\partial X_1} & \frac{\partial u_1}{\partial X_2} & \frac{\partial u_1}{\partial X_3} \\ \frac{\partial u_2}{\partial X_1} & \frac{\partial u_2}{\partial X_2} & \frac{\partial u_2}{\partial X_3} \\ \frac{\partial u_3}{\partial X_1} & \frac{\partial u_3}{\partial X_2} & \frac{\partial u_3}{\partial X_3} \end{bmatrix}, \quad (3.20)$$

or in index notation

$$[\boldsymbol{\nabla}_{\mathbf{X}} \mathbf{u}]_{ij} = \frac{\partial u_i}{\partial X_j}. \quad (3.21)$$

Kinematics

Deformation gradient

- Now, from Eq. (3.19) it is possible to write the relation between $d\mathbf{X}$ and $d\mathbf{x}$

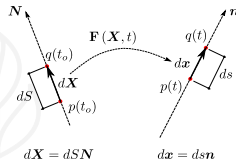
$$\begin{aligned} d\mathbf{x} &= d\mathbf{X} + \nabla_{\mathbf{X}} \mathbf{u} d\mathbf{X} \\ &= (\mathbf{I} + \nabla_{\mathbf{X}} \mathbf{u}) d\mathbf{X} \\ &= \mathbf{F} d\mathbf{X} \end{aligned} \quad (3.22)$$

in which

$$\mathbf{F}(\mathbf{X}, t) = \mathbf{I} + \nabla_{\mathbf{X}} \mathbf{u}(\mathbf{X}, t) \quad (3.23)$$

is called *deformation gradient*.

- The *deformation gradient* $\mathbf{F}$ maps $d\mathbf{X}$ onto $d\mathbf{x}$. Such movement may imply changes at *positions*, *orientations* and *sizes* between $d\mathbf{X}$ and $d\mathbf{x}$.
- The deformation gradient $\mathbf{F}$ is a second order tensor quantity that holds the whole deformation information of the vicinity of each point that belongs to the body!



- It is important to call the attention of the reader that elements belonging to reference configuration are written in terms of basis $\mathbf{l}_i$ and elements of the current configuration in terms of $\mathbf{e}_i$. As a result, the deformation gradient has a *mixed* written, i.e.,

$$\frac{dx_i}{dX_j} \mathbf{e}_i \otimes \mathbf{l}_j = F_{ij} \mathbf{e}_i \otimes \mathbf{l}_j. \quad (3.24)$$

Alternatively one can write

$$\mathbf{F} = \nabla_{\mathbf{X}} \mathbf{x}(\mathbf{X}, t). \quad (3.25)$$

Kinematics

Reference and current configuration gradient relations.

- Before proceed, let us consider now the relation between the gradients taken at the reference configuration to gradients at current configuration for scalar and vector fields. Recalling Eq. (3.7a) one can write for scalar fields

$$\nabla_X \phi(X, t) = \mathbf{F}^T \nabla_x \phi(x, t) \quad (3.26)$$

for scalar fields and Eq. (3.7b)

$$\nabla_X u(X, t) = \nabla_x u(x, t) \mathbf{F} \quad (3.27)$$

for vector fields.

- Exercise: Show Eq. (3.26) and Eq. (3.27).

Kinematics

Inverse of the Motion Function - Jacobian of the mapping

- We demand that the motion function in Eq. (3.3) be a one to one function in order to compute its inverse. The mathematical condition that guarantees the existence of such an inverse function is the non-vanishing of the Jacobian determinant J . The determinant of the Jacobian of the motion mapping function is

$$\begin{aligned} J &= \det \left(\frac{d\mathbf{x}}{d\mathbf{X}} \right) = \det \left(\frac{d\boldsymbol{\varphi}}{d\mathbf{X}} \right) \\ &= \det(\mathbf{F}) \\ &= \frac{dv}{dV} > 0 \end{aligned} \tag{3.28}$$

in which dV is the volume of an element in initial configuration and dv is the volume of the deformed volume element. The geometric interpretation of the Jacobian will be given latter in this chapter when we go trough the changes of volumes and areas due to deformation.

Kinematics

Relation between the lengths of $d\mathbf{X}$ and $d\mathbf{x}$

- Let us define the lengths of $d\mathbf{X}$ and $d\mathbf{x}$ by

$$dS = \|d\mathbf{X}\| = (d\mathbf{X} \cdot d\mathbf{X})^{\frac{1}{2}} \quad (3.29)$$

$$ds = \|d\mathbf{x}\| = (d\mathbf{x} \cdot d\mathbf{x})^{\frac{1}{2}}. \quad (3.30)$$

Therefore using Eq. (3.22) one can write

$$\begin{aligned} d\mathbf{x} \cdot d\mathbf{x} &= \mathbf{F}d\mathbf{X} \cdot \mathbf{F}d\mathbf{X} \\ &= d\mathbf{X} \cdot \mathbf{F}^T \mathbf{F} d\mathbf{X}. \end{aligned} \quad (3.31)$$

Let us define

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} \quad (3.32)$$

and

$$d\mathbf{x} \cdot d\mathbf{x} = d\mathbf{X} \cdot \mathbf{C} d\mathbf{X}. \quad (3.33)$$

- The second order tensor $\mathbf{C}$ is called *right Cauchy-Green deformation tensor*. Based on Eq. (3.23) $\mathbf{C}$ can be expressed as

$$\mathbf{C} = \mathbf{F}^T \mathbf{F} \quad (3.34)$$

$$= \mathbf{I} + \nabla_{\mathbf{X}} \mathbf{u} + (\nabla_{\mathbf{X}} \mathbf{u})^T + (\nabla_{\mathbf{X}} \mathbf{u})^T \nabla_{\mathbf{X}} \mathbf{u}. \quad (3.35)$$

Remark

Notice that if $\mathbf{F}$ is orthogonal then

$$\mathbf{F}^T \mathbf{F} = \mathbf{I}$$

and we have a rigid movement of the body and the lengths remains the same, that is

$$d\mathbf{x} \cdot d\mathbf{x} = d\mathbf{X} \cdot \mathbf{F}^T \mathbf{F} d\mathbf{X} = d\mathbf{X} \cdot d\mathbf{X}.$$

Kinematics

Relation between the lengths of $d\mathbf{X}$ and $d\mathbf{x}$

- Also, Eq. (3.31) can be written as

$$\begin{aligned} d\mathbf{X} \cdot d\mathbf{X} &= \mathbf{F}^{-1} d\mathbf{x} \cdot \mathbf{F}^{-1} d\mathbf{x} \\ &= d\mathbf{x} \cdot \mathbf{F}^{-T} \mathbf{F}^{-1} d\mathbf{x}. \end{aligned} \quad (3.36)$$

and

$$\mathbf{B}^{-1} = \mathbf{F}^{-T} \mathbf{F}^{-1} \longrightarrow \mathbf{B} = \mathbf{F} \mathbf{F}^T$$

where $\mathbf{B}$ is called *left Cauchy-Green deformation tensor*, or *Finger tensor*. Thus

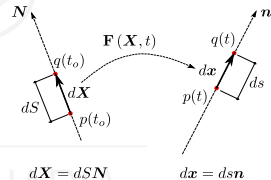
$$\mathbf{B}^{-1} = \mathbf{F}^{-T} \mathbf{F}^{-1} \longrightarrow \mathbf{B} = \mathbf{F} \mathbf{F}^T \quad (3.37)$$

- Let us write the vectors $d\mathbf{X}$ and $d\mathbf{x}$ with help of the lengths dS and ds and the *unitary* vectors $\mathbf{N}$ and $\mathbf{n}$ by

$$d\mathbf{X} = dS \mathbf{N} \quad (3.38)$$

$$d\mathbf{x} = ds \mathbf{n}. \quad (3.39)$$

The graphical interpretation is given in Figure below:



Infinitesimal strain tensor

Infinitesimal strain tensor

- Assuming that the displacements and the displacements gradients are small, in fact infinitesimals, it is possible to set

$$(\nabla_X u)^T \nabla_X u \approx 0 \quad (3.40)$$

and Eq.(3.34) reduces to

$$\begin{aligned} \mathbf{F}^T \mathbf{F} &= \mathbf{I} + \nabla_X u + (\nabla_X u)^T \\ &= \mathbf{I} + 2\varepsilon \end{aligned} \quad (3.41)$$

where

$$\varepsilon = \frac{1}{2} \left[\nabla_X u + (\nabla_X u)^T \right] \quad (3.42)$$

is known as *Infinitesimal strain tensor*. The Cartesian components of ε are

$$\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} \right) \quad (3.43)$$

or

$$[\varepsilon] = \begin{bmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{bmatrix} = \begin{bmatrix} \frac{\partial u_1}{\partial X_1} & \frac{1}{2} \left(\frac{\partial u_1}{\partial X_2} + \frac{\partial u_2}{\partial X_1} \right) & \frac{1}{2} \left(\frac{\partial u_1}{\partial X_3} + \frac{\partial u_3}{\partial X_1} \right) \\ \frac{1}{2} \left(\frac{\partial u_2}{\partial X_1} + \frac{\partial u_1}{\partial X_2} \right) & \frac{\partial u_2}{\partial X_2} & \frac{1}{2} \left(\frac{\partial u_2}{\partial X_3} + \frac{\partial u_3}{\partial X_2} \right) \\ \frac{1}{2} \left(\frac{\partial u_3}{\partial X_1} + \frac{\partial u_1}{\partial X_3} \right) & \frac{1}{2} \left(\frac{\partial u_3}{\partial X_2} + \frac{\partial u_2}{\partial X_3} \right) & \frac{\partial u_3}{\partial X_3} \end{bmatrix} \quad (3.44)$$

that is, the *Infinitesimal strain tensor* is symmetric.

Kinematics

Geometric interpretation of the rectangular/Cartesian components of the Infinitesimal strain tensor

- Consider two material elements named $d\mathbf{X}^a$ and $d\mathbf{X}^b$ that emanate from the same point (at the reference configuration), but in different directions. After a movement they become

$$\begin{aligned}d\mathbf{x}^a &= \mathbf{F}d\mathbf{X}^a \\d\mathbf{x}^b &= \mathbf{F}d\mathbf{X}^b.\end{aligned}$$

The dot product between $d\mathbf{x}^a$ and $d\mathbf{x}^b$ is meaningful. Keeping in mind Eq.(3.41) it reads

$$\begin{aligned}d\mathbf{x}^a \cdot d\mathbf{x}^b &= d\mathbf{X}^a \cdot \mathbf{F}^T \mathbf{F} d\mathbf{X}^b \\&= d\mathbf{X}^a \cdot d\mathbf{X}^b + 2d\mathbf{X}^a \cdot \boldsymbol{\varepsilon} d\mathbf{X}^b.\end{aligned}\tag{3.45}$$

Based on Eq.(3.45) we are going to derive the physical and mathematical interpretation of the Cartesian components of the infinitesimal strain tensor.

Kinematics

Diagonal elements of the infinitesimal strain tensor in Cartesian basis

- Consider a material element where $d\mathbf{X}^a = d\mathbf{X}^b = d\mathbf{X}$. Imposing a deformation gradient $\mathbf{F}$ we have $d\mathbf{x}^a = d\mathbf{x}^b = d\mathbf{x}$. Now, one can write Eq. (3.45), with aid of Eq. (3.38)-(3.39), as

$$\begin{aligned}d\mathbf{x} \cdot d\mathbf{x} &= d\mathbf{X} \cdot d\mathbf{X} + 2d\mathbf{X} \cdot \boldsymbol{\varepsilon} d\mathbf{X} \\ds\mathbf{n} \cdot ds\mathbf{n} &= dS\mathbf{N} \cdot dS\mathbf{N} + 2dS\mathbf{N} \cdot \boldsymbol{\varepsilon} dS\mathbf{N} \\ds^2 &= dS^2 + 2dS^2\mathbf{N} \cdot \boldsymbol{\varepsilon} \mathbf{N}.\end{aligned}\quad (3.46)$$

Under the “Infinitesimal strain” assumption one writes

$$\begin{aligned}ds^2 - dS^2 &= (ds + dS)(ds - dS) \\&\approx 2dS(ds - dS).\end{aligned}\quad (3.47)$$

Then Eq.(3.46)

$$\mathbf{N} \cdot \boldsymbol{\varepsilon} \mathbf{N} = \varepsilon_{NN} = \frac{ds - dS}{dS} \quad (3.48)$$

with no sum in N .

- Notice that Eq. (3.48) says that the diagonal components of the Infinitesimal Strain Tensor represent the variation of length per unity of original length. In particular when $\mathbf{N} = \mathbf{e}_i$ then $\mathbf{e}_i \cdot \boldsymbol{\varepsilon} \mathbf{e}_i = \varepsilon_{ii}$ what leads to the particular interpretation:

- ▶ ε_{11} is the variation of length per unity of original length in the $\mathbf{e}_1$;
- ▶ ε_{22} is the variation of length per unity of original length in the $\mathbf{e}_2$;
- ▶ ε_{33} is the variation of length per unity of original length in the $\mathbf{e}_3$;

Kinematics

Off-diagonal elements of the Cartesian tensor

- Let $d\mathbf{X}^a = dS^a \mathbf{M}$ and $d\mathbf{X}^b = dS^b \mathbf{N}$ be two material orthogonal elements defined in Ω_o , so $\mathbf{M} \perp \mathbf{N}$ with $\|\mathbf{M}\| = \|\mathbf{N}\| = 1$, emanating from the same material point.
- After a movement, deformation gradient $\mathbf{F}$, one has $d\mathbf{x}^a = ds^a \mathbf{m}$ and $d\mathbf{x}^b = ds^b \mathbf{n}$, being $\mathbf{m}$ and $\mathbf{n}$ also unitary vectors. From Eq. (3.45) one reads

$$\begin{aligned}d\mathbf{x}^a \cdot d\mathbf{x}^b &= d\mathbf{X}^a \cdot d\mathbf{X}^b + 2d\mathbf{X}^a \cdot \boldsymbol{\varepsilon} d\mathbf{X}^b \\ds^a ds^b \mathbf{m} \cdot \mathbf{n} &= dS^a dS^b \mathbf{M} \cdot \mathbf{N} + 2dS^a dS^b \mathbf{M} \cdot \boldsymbol{\varepsilon} \mathbf{N} \\ds^a ds^b \cos \theta &= 2dS^a dS^b \mathbf{M} \cdot \boldsymbol{\varepsilon} \mathbf{N}\end{aligned}\quad (3.49)$$

here, θ is the angle between $\mathbf{m}$ and $\mathbf{n}$. Now, writing

$$\theta = \frac{\pi}{2} - \gamma$$

in which γ the complement of θ to the original orthogonal angle

- Using the trigonometric identity:

$$\begin{aligned}\cos\left(\frac{\pi}{2} - \gamma\right) &= \cos \frac{\pi}{2} \cos \gamma + \sin \frac{\pi}{2} \sin \gamma \\&= \sin \gamma.\end{aligned}\quad (3.50)$$

- Invoking the infinitesimal strains assumption

$$\sin \gamma \approx \gamma \quad \frac{ds^a}{dS^a} \approx 1 \quad \frac{ds^b}{dS^b} \approx 1. \quad (3.51)$$

Returning to Eq. (3.49) we simply write

$$2\mathbf{M} \cdot \boldsymbol{\varepsilon} \mathbf{N} = \gamma \quad \rightarrow \quad \varepsilon_{MN} = \frac{1}{2} \gamma_{MN} \quad (3.52)$$

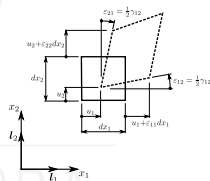
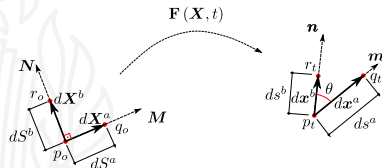
and inside the infinitesimal strain framework γ is known as *shear strain*.

Kinematics

First ideas...

- Therefore, the double value of the off-diagonal components of the Infinitesimal Strain Tensor represent the angle variation of two elements originally perpendiculars in Ω_o . More specifically, when $M = e_i$ and $N = e_j$ ($i \neq j$), then $e_i \cdot \epsilon e_j = \epsilon_{ij}$, one concludes that:

- ▶ ϵ_{12} is one-half of the angle change of two material elements originally collinear to e_1 and e_2 ;
- ▶ ϵ_{13} is one-half of the angle change of two material elements originally collinear to e_1 and e_3 ;
- ▶ ϵ_{23} is one-half of the angle change of two material elements originally collinear to e_2 and e_3 ;

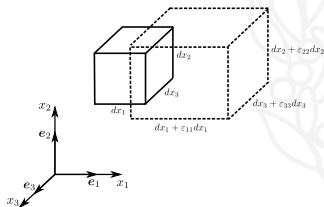


- This last Figure depicts the geometric interpretations of the infinitesimal deformations under a plane state assumption. All deformations presented in this figure are considered positive.

Kinematics

Volumetric deformation.

- Consider the element of volume with sides given by dx_1 , dx_2 , and dx_3 .



- The initial volume is

$$dV = dx_1 dx_2 dx_3. \quad (3.53)$$

- After the deformation, the sides of this volume element changed to $dx_1 + dx_1 \epsilon_{11}$, $dx_2 + dx_2 \epsilon_{22}$ and $dx_3 + dx_3 \epsilon_{33}$.

- Clearly, the new elementary volume is

$$\begin{aligned} dv &= (1 + \epsilon_{11}) dx_1 (1 + \epsilon_{22}) dx_2 (1 + \epsilon_{33}) dx_3 \\ &= (1 + \epsilon_{11} + \epsilon_{22} + \epsilon_{33} + \epsilon_{11}\epsilon_{22} + \epsilon_{11}\epsilon_{33} + \\ &\quad + \epsilon_{22}\epsilon_{33} + \epsilon_{11}\epsilon_{22}\epsilon_{33}) dx_1 dx_2 dx_3 \\ &\approx (1 + \epsilon_{11} + \epsilon_{22} + \epsilon_{33}) dx_1 dx_2 dx_3 \\ &= dV + \delta V \end{aligned}$$

in which δV is the variation of volume $\delta V = dv - dV$. Dividing the variation by the initial volume one arises at the *volumetric deformation* definition

$$\epsilon_v = \frac{\delta V}{dV} = \epsilon_{11} + \epsilon_{22} + \epsilon_{33}. \quad (3.54)$$

- The volumetric deformation ϵ_v is the sum of the diagonal components of the infinitesimal strain tensor. Therefore, using the trace operator it can be written as

$$\epsilon_v = \text{tr}(\epsilon). \quad (3.55)$$

Kinematics

Finite Deformation Tensors

- Why the infinitesimal strain tensor fails to represent Finite (Large) Deformations?
 - ▶ Why the term $(\nabla_{\mathbf{x}} \mathbf{u})^T \nabla_{\mathbf{x}} \mathbf{u}$ in Eq. 3.34 (or $\nabla_{\mathbf{x}} \mathbf{u} (\nabla_{\mathbf{x}} \mathbf{u})^T$ in Eq. 3.37) can not be disregarded!
- Before go further into the definitions of finite deformation tensors let us introduce some important theorems regarding second order tensors. They are the Spectral Decomposition and the Polar Decomposition Theorems.

1896

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SPECTRAL DECOMPOSITION THEOREM

Very important theorem in solid mechanics

Theorem 2

SPECTRAL DECOMPOSITION THEOREM: [Gurtin 1981] Let $\mathcal{V}$ denote the space of all symmetric second order tensors. Let $\mathbf{S} \in \mathcal{V}$. Then there is an orthonormal basis for consisting entirely of eigenvectors of $\mathbf{S}$. Moreover, for any such basis $\mathbf{m}_1$, $\mathbf{m}_2$, and $\mathbf{m}_3$ the corresponding eigenvalues λ_1 , λ_2 , and λ_3 , when ordered, form the entire spectrum of $\mathbf{S}$ and

$$\mathbf{S} = \lambda_i \mathbf{m}_i \otimes \mathbf{m}_i. \quad (3.56)$$

Conversely, if $\mathbf{S}$ has the form of Eq. (3.56) with $\{\mathbf{m}_i\}$ orthonormal, then λ_1 , λ_2 , and λ_3 are the eigenvalues of $\mathbf{S}$ and $\mathbf{m}_1$, $\mathbf{m}_2$, and $\mathbf{m}_3$ are the corresponding eigenvectors. Further,

- 1 $\mathbf{S}$ has exactly three distinct eigenvalues if and only if the characteristic spaces of $\mathbf{S}$ are three mutually perpendicular lines through $\mathbf{0}$;
- 2 $\mathbf{S}$ has exactly two distinct eigenvalues if and only if $\mathbf{S}$ admits the representation

$$\mathbf{S} = \lambda_1 \mathbf{m} \otimes \mathbf{m} + \lambda_2 (\mathbf{I} - \mathbf{m} \otimes \mathbf{m}). \quad (3.57)$$

In this case λ_1 and λ_2 are the two distinct eigenvalues and the corresponding characteristic spaces are $\mathbf{m}$ and $\mathbf{m}^\perp$, respectively. Alternatively, one can write, for $\lambda_1 = \lambda_2$,

$$\mathbf{S} = \lambda_1 (\mathbf{m}_1 \otimes \mathbf{m}_1 + \mathbf{m}_2 \otimes \mathbf{m}_2) + \lambda_3 (\mathbf{m}_3 \otimes \mathbf{m}_3).$$

- 3 $\mathbf{S}$ has exactly one eigenvalue ($\lambda = \lambda_1 = \lambda_2 = \lambda_3$) if and only if

$$\mathbf{S} = \lambda \mathbf{m}_i \otimes \mathbf{m}_i = \lambda \mathbf{I}. \quad (3.58)$$

POLAR DECOMPOSITION THEOREM

Another important theorem.

Theorem 3

POLAR DECOMPOSITION THEOREM:[Gurtin 1981] Let $\mathbf{T}$ be an invertible tensor with $\det(\mathbf{T}) > 0$. Then there are symmetric, positive-definite tensors $\mathbf{U}$ and $\mathbf{V}$ and a rotation $\mathbf{R}$ (orthogonal tensor with $\det(\mathbf{R}) = +1$) such that

$$\mathbf{T} = \mathbf{R}\mathbf{U} = \mathbf{V}\mathbf{R}. \quad (3.59)$$

Moreover, $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{R}$ are unique.

Proof.

Left as an exercise. □

- It follows immediately from this theorem that

$$\mathbf{U} = \mathbf{R}^T \mathbf{V} \mathbf{R} \text{ or } \mathbf{V} = \mathbf{R} \mathbf{U} \mathbf{R}^T. \quad (3.60)$$

- ▶ $\mathbf{R}\mathbf{U}$ is the right polar decomposition of $\mathbf{T}$;
- ▶ $\mathbf{V}\mathbf{R}$ is the left polar decomposition of $\mathbf{T}$;
- ▶ $\mathbf{U}$ is known as the *right stretch tensor* whereas;
- ▶ $\mathbf{V}$ is *left stretch tensor*.

Geometric Interpretation

A geometric interpretation using as example the deformation gradient tensor

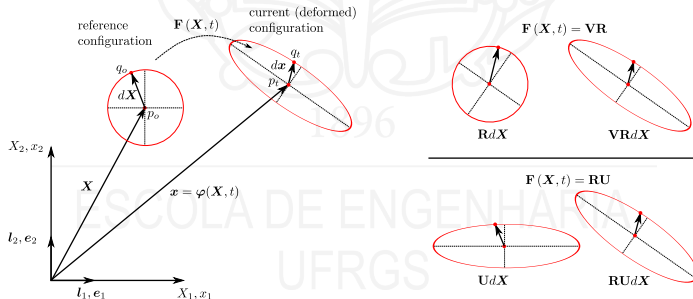
- Let us investigate the use of the polar decomposition theorem applied to the deformation gradient, that is $\mathbf{T} = \mathbf{F}$. The mapping from a reference configuration to a deformed configuration is

$$d\mathbf{x} = \mathbf{F}d\mathbf{X}$$

$$= \mathbf{R}\mathbf{U}d\mathbf{X} \quad (3.61)$$

$$= \mathbf{V}\mathbf{R}d\mathbf{X} \quad (3.62)$$

- Note that $\mathbf{R}$ does not change the length (magnitude) of the vector, but only its orientation. The figure below shows the geometric interpretation of the polar decomposition for $\mathbf{F}$.



Cauchy-Green deformation tensors

$\mathbf{C}$ is the right Cauchy-Green deformation tensor while $\mathbf{B}$ is the left Cauchy-Green deformation tensor.

Now, let us now establish an relation between $\mathbf{U}$ and $\mathbf{V}$ to $\mathbf{C}$ and $\mathbf{B}$, respectively.

① First $\mathbf{U}$ and $\mathbf{C}$:

$$\begin{aligned}\mathbf{C} &= \mathbf{F}^T \mathbf{F} = (\mathbf{R}\mathbf{U})^T \mathbf{R}\mathbf{U} \\ &= \mathbf{F}^T \mathbf{F} = \mathbf{U}\mathbf{R}^T \mathbf{R}\mathbf{U} \\ &= \mathbf{U}\mathbf{U} = \mathbf{U}^2.\end{aligned}\tag{3.63}$$

② Now $\mathbf{V}$ and $\mathbf{B}$:

$$\begin{aligned}\mathbf{B} &= \mathbf{F}\mathbf{F}^T = \mathbf{V}\mathbf{R}(\mathbf{V}\mathbf{R})^T \\ &= \mathbf{V}\mathbf{R}\mathbf{R}^T \mathbf{V} \\ &= \mathbf{V}\mathbf{V} = \mathbf{V}^2.\end{aligned}\tag{3.64}$$

Kinematics

Unified definitions of finite strain tensors

- The mathematical representation of deformation is not unique. As pointed out before it can be presented in a material basis (Lagrangian version) or in a spatial basis (Eulerian version). Moreover, it is possible to write families of material or spatial strain tensors by:

- ▶ Material Strain Tensor Family

$$\mathbf{E}^{(m)} = \begin{cases} \frac{1}{m} (\mathbf{U}^m - \mathbf{I}) & m \neq 0 \\ \ln(\mathbf{U}) & m = 0 \end{cases} = \sum_{i=1}^3 \begin{cases} \frac{1}{m} [(\lambda_i)^m - 1] & m \neq 0 \\ \ln(\lambda_i) & m = 0 \end{cases} \mathbf{N}_i \otimes \mathbf{N}_i; \quad (3.65)$$

- ▶ Spatial Strain Tensor Family:

$$\boldsymbol{\varepsilon}^{(m)} = \begin{cases} \frac{1}{m} (\mathbf{V}^m - \mathbf{I}) & m \neq 0 \\ \ln(\mathbf{V}) & m = 0 \end{cases} = \sum_{i=1}^3 \begin{cases} \frac{1}{m} [(\lambda_i)^m - 1] & m \neq 0 \\ \ln(\lambda_i) & m = 0 \end{cases} \mathbf{n}_i \otimes \mathbf{n}_i. \quad (3.66)$$

where λ_i ($i = 1, 2, 3$) are the eigenvalues of $\mathbf{U}$ (or $\mathbf{V}$ (they are the same)) and $\mathbf{N}_i$ and $\mathbf{n}_i$ are the principal directions (eigenvectors) of $\mathbf{U}$ and $\mathbf{V}$ respectively.

Kinematics

Some finite strain tensors can be now classified according a given exponent and name. Some important, most employed, finite strain tensors are:

Material strain tensors:

- *Green-Lagrange* strain tensor - $m = 2$

$$\mathbf{E}^{(2)} = \mathbf{E} = \frac{1}{2} (\mathbf{U}^2 - \mathbf{I}) = \frac{1}{2} (\mathbf{C} - \mathbf{I}) \quad (3.67)$$

- *Biot* strain tensor - $m = 1$:

$$\mathbf{E}^{(1)} = \mathbf{H} = \mathbf{U} - \mathbf{I} \quad (3.68)$$

- *Hencky* strain tensor - $m = 0$:

$$\mathbf{E}^{(0)} = \ln(\mathbf{U}) \quad (3.69)$$

Spatial strain tensors:

- *Almansi* strain tensor - $m = -2$:

$$\boldsymbol{\epsilon}^{(-2)} = \frac{1}{2} (\mathbf{I} - \mathbf{V}^{-2}) \quad (3.70)$$

- spatial *Biot* strain tensor - $m = -1$:

$$\boldsymbol{\epsilon}^{(-1)} = \mathbf{h} = \mathbf{I} - \mathbf{V}^{-1} \quad (3.71)$$

- spatial *Hencky* strain tensor - $m = 0$:

$$\boldsymbol{\epsilon}^{(0)} = \ln(\mathbf{V}) \quad (3.72)$$

Kinematics

Geometric interpretation of the rectangular components of Finite Deformation Tensors

- *Green-Lagrange* strain tensor
- *Almansi* strain tensor

Find the interpretations for each finite strain tensor (including C and B!)



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Kinematics

A note on compatibility

- The discussion that follows assumes a infinitesimal strain state, but it can be conveniently extended to finite deformations tensors. When the displacement functions for u_1 , u_2 , and u_3 are known (given) the six deformation components can be computed. For infinitesimal strains one can directly employ Eq. (3.42), which requires only that the field u_i be smooth in order to guaranty the existence of the partial derivatives $\frac{\partial u_i}{\partial X_j}$, i.e $u_i \in C^1$.
- However a problem arises when one tries to determine the displacement field from a given strain field. This problem is due the existence of six deformation components for only three displacements. As a result we need to introduce more equations, restrictions over the displacement and deformation field will be necessary. Such restriction are the known as the *equations of compatibility*.

Theorem 4

COMPATIBILITY THEOREM: *If $\varepsilon_{ij} \in C^2$, ε_{ij} has continuous second partial derivatives, in a simply connected region, then the necessary and sufficient conditions for the existence of single-valued continuous functions u_1 , u_2 , and u_3 satisfying the six equations in the $\varepsilon_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial X_j} + \frac{\partial u_j}{\partial X_i} \right)$ are:*

$$\begin{aligned} \frac{\partial^2 \varepsilon_{11}}{\partial X_2^2} + \frac{\partial^2 \varepsilon_{22}}{\partial X_1^2} &= 2 \frac{\partial^2 \varepsilon_{12}}{\partial X_1 \partial X_2} & \frac{\partial^2 \varepsilon_{11}}{\partial X_2 \partial X_3} &= \frac{\partial}{\partial X_1} \left(-\frac{\partial \varepsilon_{23}}{\partial X_1} + \frac{\partial \varepsilon_{31}}{\partial X_2} + \frac{\partial \varepsilon_{12}}{\partial X_3} \right) \\ \frac{\partial^2 \varepsilon_{22}}{\partial X_3^2} + \frac{\partial^2 \varepsilon_{33}}{\partial X_2^2} &= 2 \frac{\partial^2 \varepsilon_{23}}{\partial X_2 \partial X_3} & \frac{\partial^2 \varepsilon_{22}}{\partial X_1 \partial X_3} &= \frac{\partial}{\partial X_2} \left(\frac{\partial \varepsilon_{23}}{\partial X_1} - \frac{\partial \varepsilon_{31}}{\partial X_2} + \frac{\partial \varepsilon_{12}}{\partial X_3} \right) \\ \frac{\partial^2 \varepsilon_{33}}{\partial X_1^2} + \frac{\partial^2 \varepsilon_{11}}{\partial X_3^2} &= 2 \frac{\partial^2 \varepsilon_{31}}{\partial X_3 \partial X_1} & \frac{\partial^2 \varepsilon_{33}}{\partial X_1 \partial X_2} &= \frac{\partial}{\partial X_3} \left(\frac{\partial \varepsilon_{23}}{\partial X_1} + \frac{\partial \varepsilon_{31}}{\partial X_2} - \frac{\partial \varepsilon_{12}}{\partial X_3} \right) \end{aligned} \quad (3.73)$$

- For more details see [Lai et al., 2010] page 102 and Appendix 3.1. The compatibility equations can be written in compact form as

$$\nabla_{\mathbf{X}} \times \nabla_{\mathbf{X}} \times \boldsymbol{\varepsilon} = \mathbf{0}. \quad (3.74)$$

Kinematics

Velocity vector, velocity gradient tensor, spin, and the rate of deformation tensor

- Let us define the vector field $\mathbf{v}$ as the spatial description of the velocity. It means the velocity of a material particle that, at time t , occupies the spatial point $\mathbf{x}$. It reads

$$\mathbf{v} = \mathbf{v}(\mathbf{x}, t). \quad (3.75)$$

being

$$\mathbf{v} \equiv \frac{d\mathbf{x}}{dt} \quad \text{or} \quad \mathbf{v} \equiv \dot{\mathbf{x}}.$$

- Note that, by using Eq. 3.16 an alternative definition of the velocity may be stated by using the displacement vector, that is

$$\mathbf{v} \equiv \frac{d\mathbf{x}}{dt} = \frac{d(\mathbf{X} + \mathbf{u})}{dt} = \frac{d\mathbf{u}}{dt} \equiv \dot{\mathbf{u}}.$$

- Following the description presented in [Malvern 1969], the relative velocity components $d\mathbf{v}$ of a particle q relative to the particle p are given by

$$d\mathbf{v} = \frac{\partial \mathbf{v}}{\partial \mathbf{x}} d\mathbf{x} \quad \text{or} \quad dv_i = \frac{\partial v_i}{\partial x_j} dx_j. \quad (3.76)$$

- The tensor $\frac{\partial \mathbf{v}}{\partial \mathbf{x}}$ is called as the *velocity gradient tensor* and denoted by $\mathbf{L}$, i.e.

$$\mathbf{L} = \frac{\partial \mathbf{v}}{\partial \mathbf{x}} = \nabla_{\mathbf{x}} \mathbf{v} \rightarrow d\mathbf{v} = \mathbf{L} d\mathbf{x}. \quad (3.77)$$

Kinematics

Velocity vector, velocity gradient tensor, spin, and the rate of deformation tensor

- The velocity gradient tensor can be additively decomposed into its symmetric and anti-symmetric (or skew) tensors by

$$\mathbf{L} = \underbrace{\frac{1}{2} [\nabla_{\mathbf{x}} \mathbf{v} + (\nabla_{\mathbf{x}} \mathbf{v})^T]}_{\mathbf{D}} + \underbrace{\frac{1}{2} [\nabla_{\mathbf{x}} \mathbf{v} - (\nabla_{\mathbf{x}} \mathbf{v})^T]}_{\mathbf{W}} \quad (3.78)$$

in which the symmetric part is denoted by $\mathbf{D}$ and known as the *rate of deformation tensor* and the anti-symmetric part is denoted by $\mathbf{W}$ and named as the *spin tensor* or *vorticity tensor* (or still *rate of rotation tensor*).

Kinematics

An example

Example 5

INTERPRETATION OF THE RATE OF DEFORMATION TENSOR: Let us investigate further the rate of deformation and rate of rotation tensors. Rate of deformation meaning: Let us take the rate of the length of a current ds , that is, the length of $d\mathbf{X}$ at time t . We know from Eq. (3.30) that

$$(ds)^2 = d\mathbf{x} \cdot d\mathbf{x}.$$

The rate of this equation yields

$$\frac{d}{dt} [(ds)^2] = 2d\mathbf{x} \cdot \frac{d}{dt} (d\mathbf{x}). \quad (3.79)$$

Let us investigate further the term $\frac{d}{dt} (d\mathbf{x})$. Invoking Eqs. (3.22) and (3.25) one writes

$$\frac{d}{dt} (d\mathbf{x}) = \frac{d}{dt} (\nabla_{\mathbf{X}} \mathbf{x} (\mathbf{X}, t) d\mathbf{X}) = \frac{d}{dt} (\nabla_{\mathbf{X}} \mathbf{x} (\mathbf{X}, t)) d\mathbf{X} + \nabla_{\mathbf{X}} \mathbf{x} (\mathbf{X}, t) \frac{d}{dt} (d\mathbf{X})$$

and as $d\mathbf{X}$ does not vary with time and the second vanishes. Therefore

$$\frac{d}{dt} (d\mathbf{x}) = \frac{d}{dt} (\nabla_{\mathbf{X}} \mathbf{x} (\mathbf{X}, t)) d\mathbf{X} = \nabla_{\mathbf{X}} \left(\frac{d\mathbf{x} (\mathbf{X}, t)}{dt} \right) d\mathbf{X} \rightarrow d\mathbf{v} = \nabla_{\mathbf{X}} \mathbf{v} (\mathbf{X}, t) d\mathbf{X} \quad (3.80)$$

Note that

$$\nabla_{\mathbf{X}} \mathbf{v} \neq \nabla_{\mathbf{x}} \mathbf{v} = \mathbf{L}. \quad (3.81)$$

Example 6

But from Eq. (3.77) we have $d\mathbf{v} = \nabla_{\mathbf{x}} \mathbf{v} d\mathbf{x}$ and from Eq. (3.80) $d\mathbf{v} = \nabla_{\mathbf{X}} \mathbf{v} (\mathbf{X}, t) d\mathbf{X}$, therefore one identifies

$$\nabla_{\mathbf{x}} \mathbf{v} d\mathbf{x} = \nabla_{\mathbf{X}} \mathbf{v} d\mathbf{X} \rightarrow \frac{d}{dt} (d\mathbf{x}) = \mathbf{L} d\mathbf{x}. \quad (3.82)$$

Now returning to Eq. (3.79) one writes

$$\frac{d}{dt} [(ds)^2] = 2d\mathbf{x} \cdot \mathbf{L} d\mathbf{x}. \quad (3.83)$$

Using the decomposition in Eq. (3.78)

$$\frac{d}{dt} [(ds)^2] = 2d\mathbf{x} \cdot \mathbf{D} d\mathbf{x} + 2d\mathbf{x} \cdot \mathbf{W} d\mathbf{x} \quad (3.84)$$

but $d\mathbf{x} \cdot \mathbf{W} d\mathbf{x} = 0$ since $d\mathbf{x} \perp \mathbf{W} d\mathbf{x}$ resulting in

$$\frac{d}{dt} [(ds)^2] = 2d\mathbf{x} \cdot \mathbf{D} d\mathbf{x}. \quad (3.85)$$

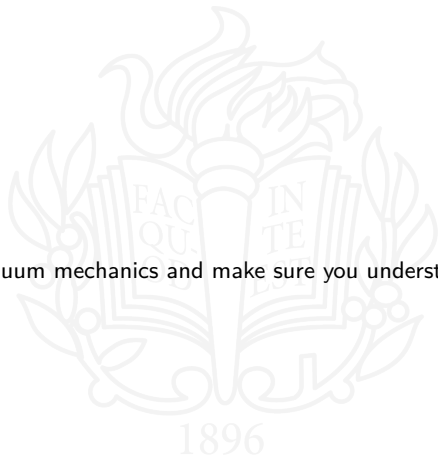
Again by Eq. (3.79)

$$2d\mathbf{x} \cdot \frac{d}{dt} (d\mathbf{x}) = 2d\mathbf{x} \cdot \mathbf{D} d\mathbf{x} \rightarrow \frac{d}{dt} (d\mathbf{x}) = \mathbf{D} d\mathbf{x}. \quad (3.86)$$

Based on last result one concludes that the rate of a deformed element, increment of velocity, emanating from a particle p at the current configuration is given by the application of the rate of deformation tensor on current increment $d\mathbf{x}$.

Summary

- Review your course of continuum mechanics and make sure you understand each and every slide presented here...



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Classroom Slides on Plasticity

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Draft version - - - Under development

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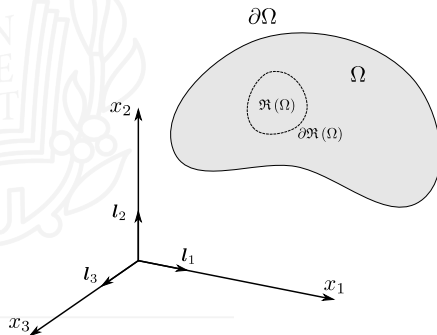
Outline

- 
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 - Kinematics of deformation
 - **Conservation of Mass**
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Body as a continuum

The *Conservation of Mass* law or principle (originally stated by Antoine Lavoisier) implies that *mass can neither be created nor destroyed, although it may be rearranged in space, or the entities associated with it may be changed in form.*

- A body $\Omega \subset \mathcal{R}^3$ can be viewed as set of *particles*, *material particles* or *material points*, that has *continuous distribution* in space. The body Ω may change its *configuration*, i.e., it may occupies different geometric regions along with time.
- Figure shows a configuration of a body Ω , with *sufficiently regular (smooth)* boundary $\partial\Omega$, occupying a region of the *Euclidean space* at a given time t . $\mathfrak{R}(\Omega) \subset \Omega$ is a region of the body with *sufficiently regular* boundary $\partial\mathfrak{R}(\Omega)$.



- For sake of a representation, an orthogonal Cartesian system is presented in this figure. Most part of the figures in this slides make reference to Cartesian systems to define some quantities, however the reader must realize that other coordinate systems may also be used.

Mass balance equation

Mass and mass density

- From the continuum point of view the existence of the quantity called *mass* implies in the existence of a scalar field called *mass density*, denoted by ρ . The mass of a certain region $\mathcal{R}$ of the body Ω , denoted by $m_{\mathcal{R}(\Omega)}$, is

$$m_{\mathcal{R}(\Omega)} = \int_{\mathcal{R}(\Omega)} \rho(\mathbf{x}, t) \, dv. \quad (3.87)$$

This result is equivalent to express

$$\rho(\mathbf{x}, t) = \lim_{\mathcal{R}(\Omega) \rightarrow \mathbf{x}} \frac{m_{\mathcal{R}(\Omega)}}{V_{\mathcal{R}(\Omega)}} \quad (3.88)$$

in which $\mathbf{x}$ is a arbitrary point such that $\mathbf{x} \in \mathcal{R}(\Omega)$. Moreover, $\rho(\mathbf{x}, t)$ is assumed to be smooth ($\rho \in C^1$). The *principle of mass conservation* states that the rate of change of mass does not vary. It reads

$$\frac{d}{dt} m_{\mathcal{R}(\Omega)} = \frac{d}{dt} \int_{\mathcal{R}(\Omega)} \rho(\mathbf{x}, t) \, dv = 0. \quad (3.89)$$

- It is possible define a mass element dm . As a result of Eq. 3.88 one may write

$$dm = \rho(\mathbf{x}, t) \, dv. \quad (3.90)$$

Continuity equation at current configuration

Continuity equation at current configuration

- The conservation of mass, Eq. 3.89, can be written using the transport formula (see [Gurtin 1981, page78]), as

$$\frac{d}{dt} \int_{\mathcal{R}(\Omega)} \rho(\mathbf{x}, t) dv = 0 \rightarrow \int_{\mathcal{R}(\Omega)} (\dot{\rho} + \rho \nabla_{\mathbf{x}} \cdot \dot{\mathbf{x}}) dv = 0. \quad (3.91)$$

and having an arbitrary region (arbitrary integration limits) one writes

$$\dot{\rho} + \rho \nabla_{\mathbf{x}} \cdot \dot{\mathbf{x}} = 0. \quad (3.92)$$

Equation 3.92 is one of the forms of the *continuity equation* and expresses the local conservation of mass at any point of a continuum medium.

- Let $\phi = \rho$ in Eq. 3.10 that combining with Eq. 3.92 yields

$$\left. \frac{\partial \rho(\mathbf{x}, t)}{\partial t} \right|_{\mathbf{x} \text{ const}} + \nabla_{\mathbf{x}} \cdot (\rho \dot{\mathbf{x}}) = 0. \quad (3.93)$$

Equation 3.93 is another the form of the *continuity equation*.

Continuity equation at current configuration

Incompressibility

- Incompressibility condition:

- ▶ If the density of the vicinity of each material point remains constant during a movement, the continuity equation Eq. (3.92), assumes the simply form

$$\nabla_x \cdot \dot{x} = 0. \quad (3.94)$$

- ▶ The continuity equations just stated must be imposed at the current configuration. It is also possible state a mass conservation law at reference configuration.

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Material form of the continuity equation

Material form of the continuity equation

The material form of the continuity equation can be derived easily if one defines ρ_o as the mass density at reference configuration Ω_o , i.e. $\rho_o = \rho_o(\mathbf{X})$. By the conservation of mass one states that

$$\int_{\mathcal{R}(\Omega)} \rho(\mathbf{x}, t) dv = \int_{\mathcal{R}(\Omega_o)} \rho_o(\mathbf{X}) dV \quad (3.95)$$

Here the region $\mathcal{R}(\Omega)$ is the region occupied at time t which occupied $\mathcal{R}(\Omega_o)$ at time t_o . By using Eq. (3.28) and recalling the discussion about material and spatial quantities, slide on page 29, it is possible to change variables and write

$$\int_{\mathcal{R}(\Omega)} \rho(\mathbf{x}, t) dv = \int_{\mathcal{R}(\Omega_o)} \rho(\mathbf{x}, t)|_{\mathbf{x}=\varphi(\mathbf{X}, t)} J dV.$$

Therefore,

$$\int_{\mathcal{R}(\Omega_o)} [\rho(\varphi(\mathbf{X}, t), t) J - \rho_o(\mathbf{X})] dV = 0 \quad (3.96)$$

and using Eq. (3.28)

$$\rho J = \rho_o \rightarrow J = \frac{\rho_o}{\rho} \rightarrow \det(\mathbf{F}) = \frac{\rho_o}{\rho} \quad (3.97)$$

Another local form of conservation of mass can be derived combining Eq. (3.89) and Eq. (3.96) yielding

$$\frac{d}{dt}(\rho J) = 0. \quad (3.98)$$

Departing from Eq. (3.95) or Eq. (3.96) and keeping in mind Eq. (3.28) one writes

$$\frac{d}{dt}(\rho dv) = 0. \quad (3.99)$$

Reynolds transport theorem

Reynolds transport theorem

- Now we want to investigate further the *material time derivative* of a volume integral with a rather more general integrand of that presented in Eq. 3.89. Let

$$\frac{d}{dt} \int_{\mathcal{R}(\Omega)} \rho(\mathbf{x}, t) \mathcal{A}(\mathbf{x}, t) dv$$

where $\mathcal{A}$ is a specific property of the material, as for instance the specific energy, and may be a scalar, vector or tensor field. Notice that the integrand is expressed in spatial coordinates so it is the region $\mathcal{R}(\Omega)$ and its boundary $\partial\mathcal{R}(\Omega)$. This last *material time derivative* of a volume integral can be interpreted as the *rate of change of the amount of $\mathcal{A}$ possessed by the material instantaneously inside the region $\mathcal{R}(\Omega)$* .

Theorem 7

REYNOLDS TRANSPORT THEOREM: *It states that the rate of change of the amount of $\mathcal{A}$ possessed by the material instantaneously inside the region $\mathcal{R}(\Omega)$ is given by the rate of change of the total amount of $\mathcal{A}$ in region $\mathcal{R}(\Omega)$ plus the rate of outward flux of $\mathcal{A}$ carried by mass transport through the control surface $\partial\mathcal{R}(\Omega)$. Mathematically it reads*

$$\frac{d}{dt} \int_{\mathcal{R}(\Omega)} \rho \mathcal{A} dv = \int_{\mathcal{R}(\Omega)} \frac{\partial}{\partial t} (\rho \mathcal{A}) dv + \int_{\partial\mathcal{R}(\Omega)} \rho \mathcal{A} \dot{\mathbf{x}} \cdot \mathbf{n} da. \quad (3.100)$$

Reynolds transport theorem

- The divergence theorem yields a useful result:

$$\int_{\partial \mathcal{R}(\Omega)} \rho \mathcal{A} \dot{\mathbf{x}} \cdot \mathbf{n} \, da = \int_{\mathcal{R}(\Omega)} \nabla_{\mathbf{x}} \cdot (\rho \mathcal{A} \dot{\mathbf{x}}) \, dv.$$

Substituting back in Eq. 3.100 yields

$$\begin{aligned} \frac{d}{dt} \int_{\mathcal{R}(\Omega)} \rho \mathcal{A} \, dv &= \int_{\mathcal{R}(\Omega)} \left[\frac{\partial}{\partial t} (\rho \mathcal{A}) + \nabla_{\mathbf{x}} \cdot (\rho \mathcal{A} \dot{\mathbf{x}}) \right] \, dv \\ &= \int_{\mathcal{R}(\Omega)} \mathcal{A} \left[\frac{\partial}{\partial t} (\rho) + \nabla_{\mathbf{x}} \cdot (\rho \dot{\mathbf{x}}) \right] \, dv + \int_{\mathcal{R}(\Omega)} \rho \left[\frac{\partial}{\partial t} \mathcal{A} + \nabla_{\mathbf{x}} \mathcal{A} \cdot \dot{\mathbf{x}} \right] \, dv. \end{aligned}$$

The first term of the LHS vanishes according to Eq. 3.93 while the second term is

$$\int_{\mathcal{R}(\Omega)} \rho \left[\frac{\partial}{\partial t} \mathcal{A} + \nabla_{\mathbf{x}} \mathcal{A} \cdot \dot{\mathbf{x}} \right] \, dv = \int_{\mathcal{R}(\Omega)} \rho \frac{d}{dt} \mathcal{A} \, dv$$

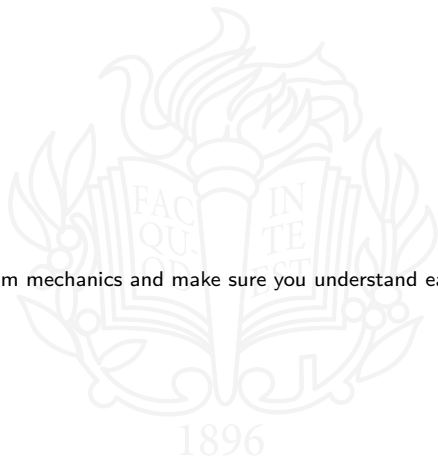
since $\frac{\partial}{\partial t} \mathcal{A} + \nabla_{\mathbf{x}} \mathcal{A} \cdot \dot{\mathbf{x}}$ is the total derivative of quantity $\mathcal{A}$, see Eq. 3.10. Thus, the Reynolds transport theorem is

$$\frac{d}{dt} \int_{\mathcal{R}(\Omega)} \rho \mathcal{A} \, dv = \int_{\mathcal{R}(\Omega)} \rho \frac{d}{dt} \mathcal{A} \, dv = \int_{\mathcal{R}(\Omega)} \rho \dot{\mathcal{A}} \, dv. \quad (3.101)$$

See [Malvern 1969], pages 210-212, for more details.

Summary

- Review your course of continuum mechanics and make sure you understand each and every slide presented here...



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Classroom Slides on Plasticity

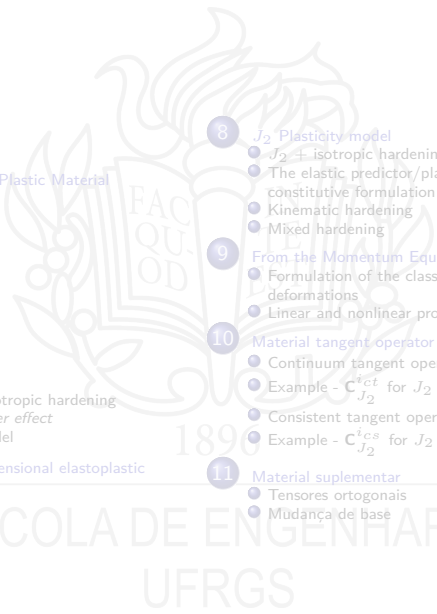
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Draft version - - - Under development

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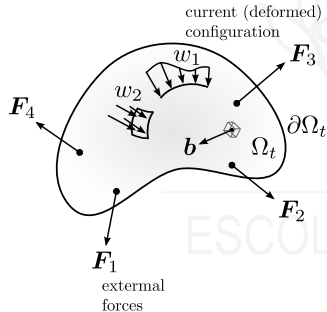
Outline

- 
- 1 Before we start:
 - 2 Motivation: One-dimensional Elastic, Perfectly Plastic Material
 - A first model
 - 3 Review of Continuum Mechanics
 - Kinematics of deformation
 - Conservation of Mass
 - **Equilibrium and Stresses**
 - Thermodynamic laws
 - 4 Constitutive theory
 - 5 Revisiting one-dimensional models
 - Elastic, perfectly plastic model
 - One-dimensional elasto-plastic model with isotropic hardening
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 - 9 From the Momentum Equation to the Infinitesimal Tangent Modulus
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 - Consistent tangent operator - $\mathbf{C}^{ics}$
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 - 11 Material suplementar
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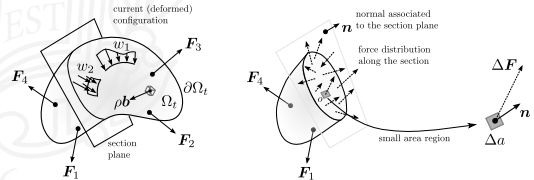
Stress state - equilibrium of a point in terms of stresses

Equilibrium

- Our discussion begins assuming a body under equilibrium where forces are prescribed on its surface and submitted to a field forces, as gravitational field, called body forces.
- Ω is the current region occupied by this body and let us state that the body is under *equilibrium conditions*.



- Let us separate the body through an imaginary and arbitrary cutting section, separating it (initially) into two parts. By passing this section, a set of forces per unit area, a distribution of density of forces, must arise in both sections to keep the body in equilibrium.



- This distribution of forces per unit area acting over each cross section may vary from point to point, but it is not arbitrary.

Cauchy traction vector

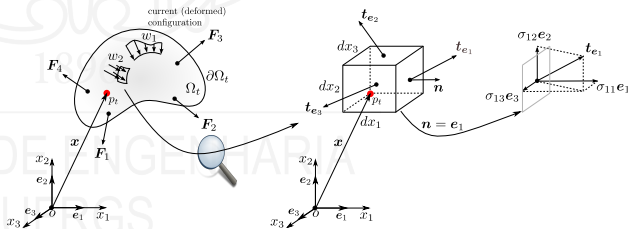
Traction vector

- To the force field per unit area we define the so-called traction vector $\mathbf{t}$ which will be associated with a unitary normal vector $\mathbf{n}$ to the section plane.
- To define the so-called Cauchy traction vector (or true traction vector), defined at the current (deformed) configuration, one must allow that the cutting section plane be divided into small current area elements (or regions) Δa . Let us analyze one of this elements, let us say in the vicinity of point o ;
- The traction vector $\mathbf{t}_n$, also known as *Cauchy traction vector* is defined as

$$\mathbf{t}_n = \lim_{\Delta a \rightarrow 0} \frac{\Delta \mathbf{F}}{\Delta a}. \quad (3.102)$$

- Considering now a Cartesian system x_1, x_2 , and x_3 generated by orthogonal vectors $\mathbf{e}_1, \mathbf{e}_2$ and $\mathbf{e}_3$. It is possible to decompose the traction vector $\mathbf{t}_n$ on such directions as

$$\mathbf{t}_n = t_{nx_1} \mathbf{e}_1 + t_{nx_2} \mathbf{e}_2 + t_{nx_3} \mathbf{e}_3. \quad (3.103)$$



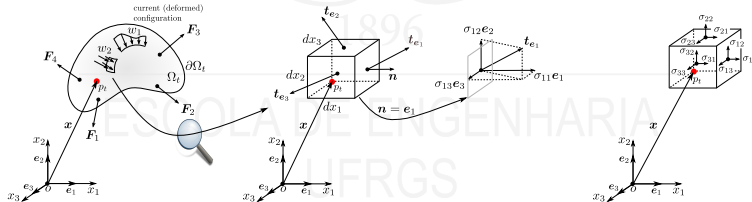
Traction vector and stress

Traction vector and stress

- Let us particularize a little bit more the traction vector components. Let the normal vector $\mathbf{n}$ be given by each of orthogonal vectors. Let us define three traction vectors named $\mathbf{t}_{e_1}$, $\mathbf{t}_{e_2}$ and $\mathbf{t}_{e_3}$ which act on section planes x_2x_3 , x_1x_3 , and x_1x_2 , respectively. Each one of this traction vectors has three components on the directions of $\mathbf{e}_1$, $\mathbf{e}_2$ and $\mathbf{e}_3$. They are:

$$\begin{aligned} \mathbf{t}_{e_1} &= t_{11}\mathbf{e}_1 + t_{12}\mathbf{e}_2 + t_{13}\mathbf{e}_3 = \sigma_{11}\mathbf{e}_1 + \sigma_{12}\mathbf{e}_2 + \sigma_{13}\mathbf{e}_3 \\ \mathbf{t}_{e_2} &= t_{21}\mathbf{e}_1 + t_{22}\mathbf{e}_2 + t_{23}\mathbf{e}_3 = \sigma_{21}\mathbf{e}_1 + \sigma_{22}\mathbf{e}_2 + \sigma_{23}\mathbf{e}_3 \\ \mathbf{t}_{e_3} &= t_{31}\mathbf{e}_1 + t_{32}\mathbf{e}_2 + t_{33}\mathbf{e}_3 = \sigma_{31}\mathbf{e}_1 + \sigma_{32}\mathbf{e}_2 + \sigma_{33}\mathbf{e}_3 \end{aligned} \quad (3.104)$$

That is, the rectangular Cartesian stress components are the components of the vector traction $\mathbf{t}_{e_1}$, $\mathbf{t}_{e_2}$, and $\mathbf{t}_{e_3}$ at the point p_t in terms of x_1 , x_2 , and x_3 . Projecting each one of the stress components on their respective section planes one have the representation:



Stress tensor

Stress tensor

- The *Cauchy* stress tensor is

$$\boldsymbol{\sigma} = \sigma_{ij} \mathbf{e}_i \otimes \mathbf{e}_j. \quad (3.105)$$

The Cartesian components of the stress tensor $\boldsymbol{\sigma}$ are

$$[\boldsymbol{\sigma}] = [\sigma_{ij}] = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}. \quad (3.106)$$

The stress tensor in Eq. 3.106 is known as the *Cauchy stress tensor* or *Cauchy true stress tensor*. It acts on the actual (deformed) configuration of the body.

- A strong connection exists between the traction vector and the Cauchy stress tensor. For a particular normal vector $\mathbf{n}$, the traction vector related to a Cauchy stress at a point is

$$\mathbf{t} = \boldsymbol{\sigma}^T \mathbf{n} \quad (3.107)$$

$$t_i = \sigma_{ji} n_j.$$

- When the coordinate system is named as x , y , and z the orthonormal vectors become $\mathbf{e}_x$, $\mathbf{e}_y$, and $\mathbf{e}_z$, and the components of the stress tensor representation are given in the following way:

$$[\boldsymbol{\sigma}] = \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix}. \quad (3.108)$$

- The components' representation of the stress tensor may change according to the literature. Some authors would rather use the so-called *von Kármán* notation to describe the Cartesian components of the stress tensor, that is,

$$[\boldsymbol{\sigma}] = \begin{bmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{yx} & \sigma_y & \tau_{yz} \\ \tau_{zx} & \tau_{zy} & \sigma_z \end{bmatrix}. \quad (3.109)$$

Linear and angular momentum conservation

Linear momentum and the *Cauchy* law of motion

- Linear momentum:
- *Cauchy* law of motion is the continual analog of the *Newton's* 2nd law applied to particles. The linear momentum conservation law for a particle of mass m states that the rate of change of its linear momentum is equal to the resulting force applied to that mass. Mathematically it is

$$\frac{d}{dt}(m\dot{\mathbf{x}}) = \mathbf{f}. \quad (3.110)$$

Now let us state the continuum form. Let Ω be a body with boundary $\partial\Omega$ where acts surface forces (density of forces) $\mathbf{t} = \mathbf{t}(\mathbf{x}, t)$ and body forces $\mathbf{b} = \mathbf{b}(\mathbf{x}, t)$ (per unity of mass). Let $\ddot{\mathbf{x}}(\mathbf{x}, t)$ be the acceleration, so that $\rho\ddot{\mathbf{x}}dv$ is the inertial force. So it is stated:

- *The rate of change of linear momentum of the particles which instantaneously lie within a fixed region Ω is proportional to the resultant force applied to the material occupying Ω .* This resultant force consists of the resultant of the body forces in Ω together with the resultant of the surface traction $\mathbf{t} = \mathbf{t}(\mathbf{x}, t)$ acting on $\partial\Omega$.

$$\int_{\partial\Omega} \mathbf{t} da + \int_{\Omega} \rho \mathbf{b} dv = \frac{d}{dt} \int_{\Omega} \rho \dot{\mathbf{x}} dv \quad (3.111)$$

Linear momentum and the *Cauchy* law of motion

Linear momentum and the *Cauchy* law of motion

- Now using Eq. 3.107 one writes

$$\int_{\partial\Omega} \mathbf{t} \, da = \int_{\partial\Omega} \boldsymbol{\sigma}^T \mathbf{n} \, da \quad (3.112)$$

and using the *Gauss divergence theorem*

$$\int_{\partial\Omega} \boldsymbol{\sigma}^T \mathbf{n} \, da = \int_{\Omega} \operatorname{div}(\boldsymbol{\sigma}^T) \, dv. \quad (3.113)$$

Moreover, the integral in RHS may be written as

$$\frac{d}{dt} \int_{\Omega} \rho \dot{\mathbf{x}} \, dv \stackrel{\text{Eq. (3.101)}}{=} \int_{\Omega} \rho \frac{d}{dt} \dot{\mathbf{x}} \, dv = \int_{\Omega} \rho \ddot{\mathbf{x}} \, dv$$

- Substituting this last result into Eq. 3.111 yields

$$\int_{\Omega} \left(\operatorname{div}(\boldsymbol{\sigma}^T) + \rho \mathbf{b} - \rho \ddot{\mathbf{x}} \right) \, dv = \mathbf{0} \quad (3.114)$$

and as Ω is an arbitrary region, the integrand must be zero. Hence, the *Cauchy law of motion* is expressed as

$$\operatorname{div}(\boldsymbol{\sigma}^T) + \rho \mathbf{b} = \rho \ddot{\mathbf{x}}. \quad (3.115)$$

Cauchy law of motion

Cauchy law of motion

- The *Cauchy law of motion*

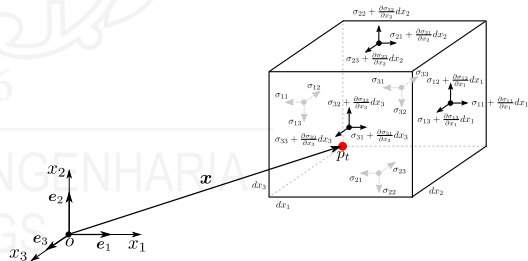
$$\operatorname{div}(\boldsymbol{\sigma}^T) + \rho \mathbf{b} = \rho \ddot{\mathbf{x}}. \quad (3.116)$$

is expressed in index as

$$\frac{\partial \sigma_{ji}}{\partial x_j} + \rho b_i = \rho \ddot{x}_i \quad \text{or} \quad \sigma_{ji,j} + \rho b_i = \rho \ddot{x}_i. \quad (3.117)$$

When the inertial forces are neglected, we have $\rho \ddot{\mathbf{x}} = \mathbf{0}$ and Eq. 3.115 is reduced to $\operatorname{div}(\boldsymbol{\sigma}^T) + \rho \mathbf{b} = \mathbf{0}$, which is known as the *Cauchy equilibrium equation* in terms of stresses. Expanding the equilibrium equation in Cartesian components, one reads

$$\begin{aligned} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + \frac{\partial \sigma_{zx}}{\partial z} + \rho b_x &= 0 \\ \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{zy}}{\partial z} + \rho b_y &= 0. \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + \rho b_z &= 0 \end{aligned} \quad (3.118)$$



Conservation of the Angular momentum

Angular momentum

- The angular momentum conservation law for a particle is

$$\frac{d}{dt} (m\mathbf{x} \times \dot{\mathbf{x}}) = \mathbf{x} \times \mathbf{f} \quad (3.119)$$

where $\mathbf{x}$ is the position vector from an arbitrary origin. The continual generalization comes directly from Eq. 3.111

$$\int_{\partial\Omega} \mathbf{x} \times \mathbf{t} \, da + \int_{\Omega} \mathbf{x} \times \rho \mathbf{b} \, dv = \frac{d}{dt} \int_{\Omega} \mathbf{x} \times \rho \dot{\mathbf{x}} \, dv. \quad (3.120)$$

Again, making use of Eq. 3.107 together with Eq. 3.111 yields

$$\begin{aligned} \int_{\partial\Omega} \mathbf{x} \times \boldsymbol{\sigma}^T \mathbf{n} \, da + \int_{\Omega} \mathbf{x} \times \rho \mathbf{b} \, dv &= \int_{\Omega} \mathbf{x} \times \rho \ddot{\mathbf{x}} \, dv \\ \int_{\partial\Omega} \epsilon_{ijk} x_j \sigma_{lk} n_l \, da + \int_{\Omega} \epsilon_{ijk} x_j \rho b_k \, dv &= \int_{\Omega} \epsilon_{ijk} x_j \rho \ddot{x}_k \, dv \end{aligned} \quad (3.121)$$

where ϵ_{ijk} is the permutation symbol (*Levi-Civita* symbol). It is

$$\epsilon_{ijk} = \begin{cases} 1 & (ijk) = (123), (231), (312) \\ -1 & (ijk) = (132), (321), (213) \\ 0 & \text{else} \end{cases}$$

Conservation of the Angular momentum

Symmetry of the *Cauchy* stress tensor

- To show such a very important results of continuum mechanics consider Eq. (3.121). Using the divergence theorem it is possible transform the surface integral to a volume integral and write

$$\int_{\Omega} \epsilon_{ijk} \left[x_j \left(\frac{\partial \sigma_{lk}}{\partial x_l} + \rho b_k - \rho \dot{v}_k \right) + \delta_{jl} \sigma_{lk} \right] dv = 0 \quad (3.122)$$

what reduces to

$$\int_{\Omega} \epsilon_{ijk} \sigma_{jk} dv = 0 \quad (3.123)$$

and as dv is arbitrary

$$\epsilon_{ijk} \sigma_{jk} = 0 \quad (3.124)$$

which expanded will leads to

$$\sigma_{ij} = \sigma_{ji}. \quad (3.125)$$

Eigenvalues and eigenvectors of a tensor

Eigenvalues and eigenvectors of a second-order tensor

- Given a second-order tensor $\mathbf{T}$, an vector $\mathbf{n}$ is said an eigenvector with associated eigenvalue λ if, and only if, the following equation is satisfied

$$\mathbf{T}\mathbf{n} = \lambda\mathbf{n}. \quad (3.126)$$

As $\lambda\mathbf{n} = \lambda\mathbf{I}\mathbf{n}$ it is possible to write

$$(\mathbf{T} - \lambda\mathbf{I})\mathbf{n} = \mathbf{0} \quad \text{or in components} \quad (T_{ij} - \lambda\delta_{ij})n_j = 0. \quad (3.127)$$

- To fulfill Eq. (3.127) the eigenvalues must satisfy the following scalar equation

$$\det(\mathbf{T} - \lambda\mathbf{I}) = 0 \quad (3.128)$$

that is:

$$\det(\mathbf{T} - \lambda\mathbf{I}) = 0 \rightarrow \begin{vmatrix} T_{11} - \lambda & T_{12} & T_{13} \\ T_{21} & T_{22} - \lambda & T_{23} \\ T_{31} & T_{32} & T_{33} - \lambda \end{vmatrix} = 0. \quad (3.129)$$

Eigenvalues and eigenvectors of a tensor

- The determinant generates a polynomial of third degree which is known as *characteristic polynomial* or *characteristic equation* of the tensor $\mathbf{T}$. After some algebraic mathematical manipulation the following cubic equation

$$\det(\mathbf{T} - \lambda \mathbf{I}) = \lambda^3 - I_{\mathbf{T}}\lambda^2 + II_{\mathbf{T}}\lambda - III_{\mathbf{T}} = 0 \quad (3.130)$$

whose coefficients of this equation are those shown in Eq. (3.131), Eq. (3.135), and Eq. (3.137) unveiling why they are named invariants of a tensor in detriment of other operations.

- The cubic polynomial in Eq. (3.130) has three roots, named λ_1 , λ_2 , and λ_3 (that are the eigenvalues). The eigenvalues of this *characteristic polynomial* may have the following values:
 - ▶ all real and distinct;
 - ▶ all real, being some repeated (or even all repeated);
 - ▶ some real (distinct and/or repeated) and some complex (imaginary);
 - ▶ all complex.

Invariants of a second-order tensors

The use of invariants is very important in continuum mechanics and in particular in solid mechanics, especially in the *constitutive equations* for writing various types of materials.

- First invariant: Trace of a tensor

- ▶ The first invariant of a tensor $\mathbf{T}$, $I_{\mathbf{T}}$ or I_1 , is its trace of the tensor, $\text{tr}(\mathbf{T})$, i.e.

$$I_{\mathbf{T}} = \text{tr}(\mathbf{T}) = \sum_{i=1}^3 T_{ii} = T_{ii} \quad (3.131)$$

To check the invariance of the tensor trace one has to show that

$$\text{tr}([\mathbf{T}]_{\mathbf{e}_i}) = \text{tr}([\mathbf{T}]_{\mathbf{e}'_i}). \quad (3.132)$$

By the use of Eq. (11.5) and the orthogonality of $\mathbf{Q}$ given by Eq. (11.1) one can show that

$$\begin{aligned} \text{tr}([\mathbf{T}]_{\mathbf{e}'_i}) &= T'_{ii} = Q_{ki} T_{kl} Q_{li} \\ &= T_{kl} Q_{ki} Q_{li} \\ &= T_{kl} \delta_{kl} \\ &= T_{kk} = \text{tr}(\mathbf{T}) \end{aligned} \quad (3.133)$$

Invariants of a second-order tensors

Invariants of second-order tensors

- Second invariant:

- ▶ Although the inner product of a tensor by itself, $\mathbf{T} \cdot \mathbf{T}$, is invariant it is not known as the second invariant by large part of the literature. Instead, the second invariant of a second-order tensor, denoted by $II_{\mathbf{T}}$ or I_2 , is given by

$$II_{\mathbf{T}} = \frac{1}{2} \left((\text{tr}(\mathbf{T}))^2 - \mathbf{T} \cdot \mathbf{T} \right) \quad (3.134)$$

or

$$II_{\mathbf{T}} = \frac{1}{2} \left(I_{\mathbf{T}}^2 - \text{tr}(\mathbf{T}^2) \right). \quad (3.135)$$

- ▶ It can be written also as the sum of the diagonal co-factors of the tensor $\mathbf{T}$, that is,

$$\begin{aligned} II_{\mathbf{T}} &= \frac{1}{2} \left(I_{\mathbf{T}}^2 - \text{tr}(\mathbf{T}^2) \right) \\ &= \begin{vmatrix} T_{11} & T_{12} \\ T_{21} & T_{22} \end{vmatrix} + \begin{vmatrix} T_{11} & T_{13} \\ T_{31} & T_{33} \end{vmatrix} + \begin{vmatrix} T_{22} & T_{23} \\ T_{32} & T_{33} \end{vmatrix} \\ &= T_{11}T_{22} + T_{22}T_{33} + T_{11}T_{33} - T_{12}T_{21} - T_{13}T_{31} - T_{23}T_{32}. \end{aligned} \quad (3.136)$$

Invariants of a second-order tensors

Invariants of second-order tensors

- Third invariant - the determinant

- ▶ The third invariant of a second-order tensor is the determinant which is defined as the determinant of the matrix of the Cartesian components of that tensor. Therefore

$$III_{\mathbf{T}} = I_3 = \det \mathbf{T} = \det [\mathbf{T}] \quad (3.137)$$

or

$$III_{\mathbf{T}} = \begin{vmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{vmatrix} = T_{11}T_{22}T_{33} + T_{12}T_{23}T_{31} + T_{13}T_{21}T_{32} - T_{13}T_{22}T_{31} - T_{12}T_{21}T_{33} - T_{11}T_{23}T_{32}$$

- Some examples of stress states in principal directions

uniaxial stress



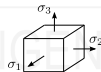
$$[\sigma] = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

biaxial or plane stress



$$[\sigma] = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

triaxial stress



$$[\sigma] = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix}$$

hydrostatic stress



$$[\sigma] = \begin{bmatrix} p & 0 & 0 \\ 0 & p & 0 \\ 0 & 0 & p \end{bmatrix}$$

Eigenvalues and eigenvectors of a Symmetric tensor

- The eigenvalues (λ_1 , λ_2 , and λ_3) of a Symmetric tensors, $\mathbf{S} = \mathbf{S}^T$, are *reals* and the corresponding eigenvector $\mathbf{n}_1$, $\mathbf{n}_2$ and $\mathbf{n}_3$ are *mutually orthogonal*, that is

$$\mathbf{S}\mathbf{n}_i = \lambda_i \mathbf{n}_i \quad i = 1, 2, 3 \text{ (no sum)} \quad (3.138a)$$

$$\mathbf{n}_i \cdot \mathbf{n}_j = \delta_{ij} \quad i, j = 1, 2, 3. \quad (3.138b)$$

This result allows us to say that the unitary eigenvectors $\mathbf{n}_1$, $\mathbf{n}_2$, and $\mathbf{n}_3$ can be used as an alternative base.

- By Eq. (3.138b) the off-diagonal terms are zero for this representation, and the terms that appear in the diagonal are the eigenvalues λ_i , i.e.,

$$[\mathbf{S}]_{\mathbf{n}_i} = \begin{bmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}. \quad (3.139)$$

- Thus, the representation of an symmetric tensor $\mathbf{S}$ is given by

$$\mathbf{S} = \sum_{i=1}^3 \lambda_i \mathbf{n}_i \otimes \mathbf{n}_i \quad (3.140)$$

which is known as *spectral decomposition*.

Examples

Examples

Example 8

Find the eigenvalues and eigenvectors of the following tensor

$$[\mathbf{S}] = \begin{bmatrix} 7 & -2 & 0 \\ -2 & 6 & -2 \\ 0 & -2 & 5 \end{bmatrix}$$

then sketch the basis n_i in terms of the original basis and represent the eigenvalues.

Example 9

Find the eigenvalues and eigenvectors of the following tensor

$$[\mathbf{S}] = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Cubic polynomial - root solution

This solution is provided by *Murray R. Spiegel, Seymour Lipschutz and John Liu, Manual de Fórmulas e Tabelas Matemáticas, Coleção Schaum, terceira edição, Bookman, 2012. (page 23)*

Equação cúbica: $x^3 + a_1x^2 + a_2x + a_3 = 0$

Sejam

$$Q = \frac{3a_2 - a_1^2}{9}, \quad R = \frac{9a_1a_2 - 27a_3 - 2a_1^3}{54},$$
$$S = \sqrt[3]{R + \sqrt{Q^3 + R^2}}, \quad T = \sqrt[3]{R - \sqrt{Q^3 + R^2}}$$

onde $ST = -Q$.

5.3 Soluções

$$\begin{cases} x_1 = S + T - \frac{1}{3}a_1 \\ x_2 = -\frac{1}{2}(S + T) - \frac{1}{3}a_1 + \frac{1}{2}i\sqrt{3}(S - T) \\ x_3 = -\frac{1}{2}(S + T) - \frac{1}{3}a_1 - \frac{1}{2}i\sqrt{3}(S - T) \end{cases}$$

Se a_1, a_2 e a_3 são reais e se $D = Q^3 + R^2$ é o discriminante, então

- (i) uma raiz é real e duas são complexas conjugadas se $D > 0$;
- (ii) todas as raízes são reais e, no mínimo, duas são iguais se $D = 0$ e
- (iii) todas as raízes são reais e desiguais se $D < 0$.

Se $D < 0$, o cálculo é simplificado usando-se trigonometria.

5.4 Soluções

se $D < 0$:

$$\begin{cases} x_1 = 2\sqrt{-Q} \cos(\frac{1}{3}\theta) - \frac{1}{3}a_1 \\ x_2 = 2\sqrt{-Q} \cos(\frac{1}{3}\theta + 120^\circ) - \frac{1}{3}a_1 \\ x_3 = 2\sqrt{-Q} \cos(\frac{1}{3}\theta + 240^\circ) - \frac{1}{3}a_1 \end{cases}$$

onde $\cos \theta = R/\sqrt{-Q^3}$

Spherical and Deviatoric Tensors

Additive decomposition of a tensor into Spherical and Deviatoric Tensors

- Given any second order tensor $\mathbf{T}$, it can be additively decomposed into a spherical, $\mathbf{T}^{\text{sph}}$, and a deviatoric part, $\mathbf{T}^{\text{dev}}$, such that:

$$\mathbf{T} = \mathbf{T}^{\text{sph}} + \mathbf{T}^{\text{dev}} \quad (3.141)$$

- The spherical part is defined as

$$\mathbf{T}^{\text{sph}} = \frac{1}{3} \text{tr}(\mathbf{T}) \mathbf{I} = \alpha \mathbf{I} \quad (3.142)$$

whereas the so-called deviatoric part of $\mathbf{T}$, or deviator tensor, is given by

$$\mathbf{T}^{\text{dev}} = \mathbf{T} - \frac{1}{3} \text{tr}(\mathbf{T}) \mathbf{I}. \quad (3.143)$$

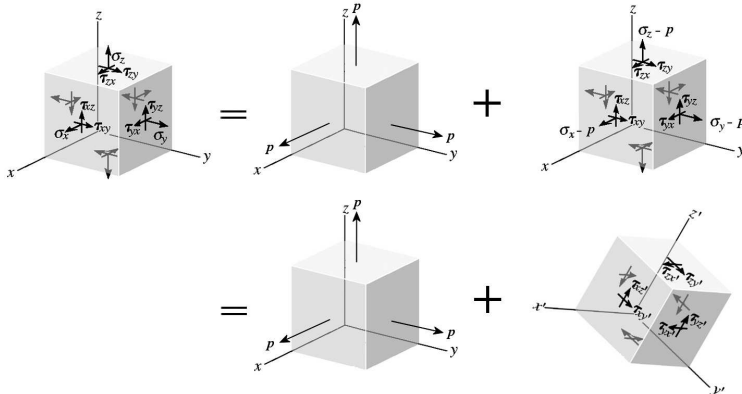
It easy to note that

$$\text{tr}(\mathbf{T}^{\text{dev}}) = 0. \quad (3.144)$$

Spherical and Deviatoric Tensors

Additive decomposition of a tensor into Spherical and Deviatoric Tensors

- Case $\mathbf{T}$ is symmetric, say $\mathbf{T} = \mathbf{S}$, the deviator tensor can be transformed, by orthogonal transformations, into a tensor where the diagonal components are all null.
- An example of the application of such decomposition is given for the stress in continuous mechanics, see the figure below.



External, Internal Power and Kinetic Energy

Energy and power

- When considering a particle in motion with velocity $\dot{x}$, or v , in classical mechanics the *power expended* by a force F acting on such a particle is

$$\mathcal{P}_{expended} = F \cdot \dot{x} \quad (3.145)$$

and F is power conjugate with $\dot{x}$, [Gurtin et al. 2010]. As previously done for linear and angular momentum in this slides (see on page 76 and 79), let us extend this concept to continuum mechanics.

External power

External power

- The power expended by external forces (agents) on a body Ω is the sum of the power due forces (density of forces) $\mathbf{t}$ acting on the surface of $\partial\Omega$ and the power due forces exerted at a distance by systems (sources) external to Ω , supposed to be defined as density of forces acting in the volume of Ω , such as the body forces $\rho\mathbf{b}$ ($\mathbf{b}$ is per unity of mass), see figure on page 72. Both forces are power conjugate with the velocity of the material particle $\dot{\mathbf{x}}$, so that we define

$$\mathcal{P}_{ext} := \int_{\partial\Omega} \mathbf{t} \cdot \dot{\mathbf{x}} \, da + \int_{\Omega} \rho\mathbf{b} \cdot \dot{\mathbf{x}} \, dv \quad (3.146)$$

or in view of Eq. (3.107)

$$\mathcal{P}_{ext} = \int_{\partial\Omega} \boldsymbol{\sigma}\mathbf{n} \cdot \dot{\mathbf{x}} \, da + \int_{\Omega} \rho\mathbf{b} \cdot \dot{\mathbf{x}} \, dv \quad (3.147)$$

and $\mathcal{P}_{ext}$ is the *external power*. More details in [Gurtin et al. 2010].

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Internal power and Kinetic energy

Internal power and Kinetic energy

- The surface integral in Eq. (3.147) may be transformed to the volume integral using an divergence theorem, yielding

$$\begin{aligned}\int_{\partial\Omega} \boldsymbol{\sigma} \mathbf{n} \cdot \dot{\mathbf{x}} \, da &= \int_{\Omega} (\operatorname{div}(\boldsymbol{\sigma}) \cdot \dot{\mathbf{x}} + \boldsymbol{\sigma} \cdot \nabla_{\mathbf{x}} \dot{\mathbf{x}}) \, dv \\ &= \int_{\Omega} (\operatorname{div}(\boldsymbol{\sigma}) \cdot \dot{\mathbf{x}} + \boldsymbol{\sigma} \cdot \mathbf{L}) \, dv \\ &= \int_{\Omega} (\operatorname{div}(\boldsymbol{\sigma}) \cdot \dot{\mathbf{x}} + \boldsymbol{\sigma} \cdot \mathbf{D}) \, dv.\end{aligned}\tag{3.148}$$

- Returning to the analysis of Eq. (3.147),

$$\mathcal{P}_{ext} = \int_{\Omega} [(\operatorname{div}(\boldsymbol{\sigma}) + \rho \mathbf{b}) \cdot \dot{\mathbf{x}} + \boldsymbol{\sigma} \cdot \mathbf{D}] \, dv\tag{3.149}$$

and the first term on RHS may be identified as the LHS of Eq. (3.115), thus

$$\mathcal{P}_{ext} = \int_{\Omega} \rho \ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} \, dv + \int_{\Omega} \boldsymbol{\sigma} \cdot \mathbf{D} \, dv.\tag{3.150}$$

Internal power and Kinetic energy

Internal power and Kinetic energy

- Let us analyze further the terms on RHS of Eq. (3.150). The first term on RHS can be identified as the material derivative of the kinetic energy. That is, being the *kinetic energy* defined as

$$K := \frac{1}{2} \int_{\Omega} \rho \dot{\mathbf{x}} \cdot \dot{\mathbf{x}} \, dv \quad (3.151)$$

its material rate yields

$$\begin{aligned} \frac{d}{dt} K &:= \frac{1}{2} \frac{d}{dt} \int_{\Omega} \rho \dot{\mathbf{x}} \cdot \dot{\mathbf{x}} \, dv \\ &= \int_{\Omega} \rho \dot{\mathbf{x}} \cdot \frac{d}{dt} \dot{\mathbf{x}} \, dv = \int_{\Omega} \rho \ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} \, dv \end{aligned} \quad (3.152)$$

that is the first term on the RHS.

- The second term on the RHS is known as the *internal power* and will be denoted by $\mathcal{P}_{int}$. Thus,

$$\mathcal{P}_{int} := \int_{\Omega} \boldsymbol{\sigma} \cdot \mathbf{D} \, dv. \quad (3.153)$$

Internal power and Kinetic energy

Internal power and Kinetic energy

- The integrand of Eq. (3.153) brings us some particular important interpretations. They are:
 - ① $\boldsymbol{\sigma} \cdot \mathbf{D}$ represents the power expended within Ω , measured per unit volume, and it is referred to as the *stress power*;
 - ② it establishes the rate of deformation tensor $\mathbf{D}$ as the appropriate power conjugate variable for the Cauchy stress $\boldsymbol{\sigma}$, and we say that $\boldsymbol{\sigma}$ is power conjugate to $\mathbf{D}$.

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Power balance

Internal power and Kinetic energy

- After the identifications in Eq. (3.152) and Eq. (3.153) one may write Eq. (3.150) as

$$\mathcal{P}_{ext} = \mathcal{P}_{int} + \frac{d}{dt}K \quad (3.154)$$

or

$$\underbrace{\int_{\partial\Omega} \boldsymbol{\sigma} \mathbf{n} \cdot \dot{\mathbf{x}} \, da + \int_{\Omega} \rho \mathbf{b} \cdot \dot{\mathbf{x}} \, dv}_{\text{external power}} = \underbrace{\int_{\Omega} \boldsymbol{\sigma} \cdot \mathbf{D} \, dv}_{\text{internal power}} + \underbrace{\frac{1}{2} \frac{d}{dt} \int_{\Omega} \rho \dot{\mathbf{x}} \cdot \dot{\mathbf{x}} \, dv}_{\text{kinetic-energy rate}} \quad (3.155)$$

having the following interpretation, see [Gurtin et al. 2010]:

- the external power expended on Ω is balanced by the sum of the internal power expended within Ω and the temporal change in kinetic energy of Ω .*
- Neglecting the inertial terms, so that $\ddot{\mathbf{x}} = 0$, one writes

$$\mathcal{P}_{ext} = \mathcal{P}_{int}. \quad (3.156)$$

Other stress measures

Relation among Cauchy, First Piola Kirchoff, Second Piola Kirchoff, Kirchoff, etc.

- Stress-Strain conjugate measures
- Under construction
- Under construction
- Under construction
- Under construction
- Under construction
- Review your course of continuum mechanics and make sure you understand each and every slide presented here...

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Classroom Slides on Plasticity

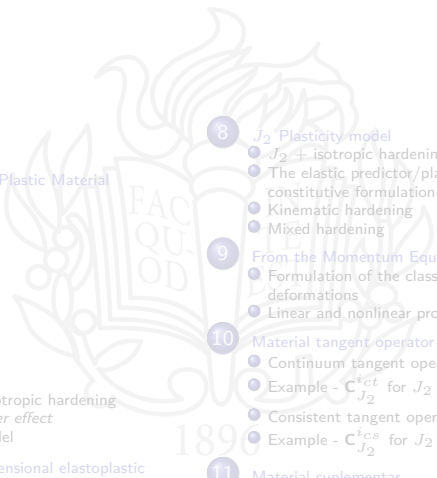
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Draft version - - - Under development

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Outline

- 
- 1 Before we start:
 - 2 Motivation: One-dimensional Elastic, Perfectly Plastic Material
 - A first model
 - 3 Review of Continuum Mechanics
 - Kinematics of deformation
 - Conservation of Mass
 - Equilibrium and Stresses
 - **Thermodynamic laws**
 - 4 Constitutive theory
 - 5 Revisiting one-dimensional models
 - Elastic, perfectly plastic model
 - One-dimensional elasto-plastic model with isotropic hardening
 - Kinematic hardening model - The *Bauschinger effect*
 - Mixed isotropic and kinematic hardening model
 - 6 Numerical integration algorithm of the one-dimensional elastoplastic constitutive formulation
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 - 9 From the Momentum Equation to the Infinitesimal Tangent Modulus
 - Formulation of the classical BVP in terms of equilibrium and under small deformations
 - Linear and nonlinear problem due to the material constitutive relation
 - 10 Material tangent operator derivation
 - Continuum tangent operator - $\mathbf{C}^{ict}$
 - Example - $\mathbf{C}_{J_2}^{ict}$ for J_2 plasticity model
 - Consistent tangent operator - $\mathbf{C}^{ics}$
 - Example - $\mathbf{C}_{J_2}^{ics}$ for J_2 plasticity model
 - 11 Material suplementar
 - Tensores ortogonais
 - Mudança de base

Conservation of energy - First law of thermodynamics

One of the most important laws in physics is the conservation of energy. It states that energy can be neither created nor destroyed, but can change its form.

- This law states that *the rate of external work W_{ext} (external power $\mathcal{P}_{ext}$) done on the system plus the rate of heat Q received by it is equal to the rate of kinetic energy K plus the rate of internal energy E , i.e.,*

$$\frac{d}{dt} (E + K) = \mathcal{P}_{ext} + Q. \quad (3.157)$$

- First, let us define the LHS terms in Eq. (3.157). The rate of internal energy is

$$\frac{d}{dt} E = \frac{d}{dt} \int_{\Omega} \rho e \, dv = \int_{\Omega} \rho \dot{e} \, dv \quad (3.158)$$

in which e is the *specific internal energy* and the rate of kinetic energy assumes the form of Eq. (3.152), i.e.

$$\frac{d}{dt} K = \frac{1}{2} \frac{d}{dt} \int_{\Omega} \rho \dot{\mathbf{x}} \cdot \dot{\mathbf{x}} \, dv = \int_{\Omega} \rho \dot{\mathbf{x}} \cdot \frac{d}{dt} \dot{\mathbf{x}} \, dv = \int_{\Omega} \rho \ddot{\mathbf{x}} \cdot \dot{\mathbf{x}} \, dv \quad (3.159)$$

First law of thermodynamics

First law of thermodynamics

- Now, the RHS of Eq. (3.157). The heat received by the body is the sum of two terms, one due the heat generated in its interior and other due the heat supplied by its surroundings. Therefore this term is written as

$$Q = \int_{\Omega} r \, dv - \int_{\partial\Omega} \mathbf{q} \cdot \mathbf{n} \, da \quad (3.160a)$$

$$= \int_{\Omega} (r - \nabla_{\mathbf{x}} \cdot \mathbf{q}) \, dv \quad (3.160b)$$

in which r is the *volumetric density of the internal heat production*, and $\mathbf{q}$ is the heat flux vector.

- The *volumetric density of the internal heat production* can be a process due a chemical reaction, or a nuclear reaction, acting on Ω or even the passage of electrical current.

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First law of thermodynamics

Local form

- Using of Eq. (3.158), (3.159), (3.160) and (3.155) it is possible to re-write the first law of thermodynamics, Eq. (3.157), in the following convenient local form, that is

$$\int_{\Omega} \rho \dot{e} \, dv - \int_{\Omega} \boldsymbol{\sigma} \cdot \mathbf{D} \, dv - \int_{\Omega} (r - \nabla_{\mathbf{x}} \cdot \mathbf{q}) \, dv = 0$$

or

$$\rho \dot{e} = \boldsymbol{\sigma} \cdot \mathbf{D} + r - \nabla_{\mathbf{x}} \cdot \mathbf{q}. \quad (3.161)$$

- Notice that in the local form of the first law of thermodynamics, Eq. (3.161), one has the writing of the *stress power* $\boldsymbol{\sigma} \cdot \mathbf{D}$ what will be important to establish a link among the conservation of energy and deformation of solid bodies and thus help one to propose well founded constitutive formulations. The second law of thermodynamics complement this idea.
- Under the the small deformation hypothesis $\mathbf{D} \approx \dot{\boldsymbol{\epsilon}}$ yielding

$$\rho \dot{e} = \boldsymbol{\sigma} \cdot \dot{\boldsymbol{\epsilon}} + r - \nabla_{\mathbf{x}} \cdot \mathbf{q}. \quad (3.162)$$

Entropy production - Second law of thermodynamics

The second law of thermodynamics tell us about the production of entropy. Let us introduce a new variable S as the entropy. The entropy is assumed to be the summation of an irreversible S_i part plus a reversible one S_r .

- The second law of thermodynamics states that *the entropy production rate $\frac{dS_i}{dt}$ (irreversible part), must be non-negative*, that is

$$\frac{d}{dt}S_i = \frac{d}{dt}S - \frac{d}{dt}S_r \geq 0. \quad (3.163)$$

- But, what is the role of the inequality (3.163)?

- ▶ The answer is that it plays a very important role in physics because through the imposition of this inequality it is possible to eliminate *non feasible* physical process, that is, such process in which the rate of entropy is negative.

- Introducing the specific entropy s and s_i , as the total and irreversible parts, respectively as

$$\frac{d}{dt}S = \frac{d}{dt} \int_{\Omega} \rho s \, dv = \int_{\Omega} \rho \dot{s} \, dv \quad (3.164)$$

$$\frac{d}{dt}S_i = \frac{d}{dt} \int_{\Omega} \rho s_i \, dv = \int_{\Omega} \rho \dot{s}_i \, dv \quad (3.165)$$

and assuming that the rate of the reversible part is heat received by the body Ω divided by the absolute temperature, see Eq. (3.160a), i.e.

$$\frac{d}{dt}S_r = \int_{\Omega} \frac{r}{\theta} \, dv - \int_{\partial\Omega} \frac{1}{\theta} \mathbf{q} \cdot \mathbf{n} \, da. \quad (3.166)$$

Second law of thermodynamics

Entropy production

- One may write inequality (3.163), with aid of the divergence theorem in Eq. (3.166), as

$$\int_{\Omega} \left[\rho \dot{s} + \nabla_{\mathbf{x}} \cdot \left(\frac{\mathbf{q}}{\theta} \right) - \frac{r}{\theta} \right] dv \geq 0 \quad (3.167)$$

or in its local form

$$\rho \dot{s} + \nabla_{\mathbf{x}} \cdot \left(\frac{\mathbf{q}}{\theta} \right) - \frac{r}{\theta} \geq 0. \quad (3.168)$$

- Now we can return to the First Law, Eq. (3.161), and isolate the volumetric density of the internal heat production, r , and then substitute it in inequality (3.168) and after a straightforward algebraic manipulation re-write the Second Law of thermodynamics as

$$\boldsymbol{\sigma} \cdot \mathbf{D} - \rho (\dot{e} - \theta \dot{s}) - \frac{1}{\theta} \mathbf{q} \cdot \mathbf{g} \geq 0. \quad (3.169)$$

where $\mathbf{g} \equiv \nabla_{\mathbf{x}} \theta$.

Specific free energy - *Helmholtz free energy*

Helmholtz free energy

- Now, defining the specific free energy (*Helmholtz free energy*) as

$$\psi = e - \theta s \quad (3.170)$$

and inserting its rate into the last expression, yields the following dissipation inequality

$$\Phi = \boldsymbol{\sigma} \cdot \mathbf{D} - \rho \left(\dot{\psi} + s\dot{\theta} \right) - \frac{1}{\theta} \mathbf{q} \cdot \mathbf{g} \geq 0. \quad (3.171)$$

The inequality (3.171) is known as the *Clausius-Duhem* inequality.

- The specific free energy (*Helmholtz free energy*) has a particular interest in formulations derived in this course. The *Helmholtz free energy* is a thermodynamic potential that measures the “*useful*” work obtainable from a closed thermodynamic system at a constant temperature.
- A closed thermodynamic system can exchange energy (as heat or work) but not matter with its surroundings.
- Assuming the small strain hypothesis the inequality (3.171) may be written as

$$\Phi = \boldsymbol{\sigma} \cdot \dot{\boldsymbol{\epsilon}} - \rho \left(\dot{\psi} + s\dot{\theta} \right) - \frac{1}{\theta} \mathbf{q} \cdot \mathbf{g} \geq 0. \quad (3.172)$$

Constitutive theory

An introduction to the ideas of the constitutive theory

- So far, we have presented the main laws that governs a body subjected to displacements and forces, and the consequences in terms of deformations and stresses... The link between stresses and strains is through the Constitutive Theory!
- Constitutive Theory - Two main approaches:
 - ▶ The functional approach
 - ★ In a general way, a constitutive law must allow us to calculate the actual values of stress, free energy, entropy, and heat flux as a function of the **WHOLE HISTORY** of the deformation and thermal process (known as *thermodynamic determinism*). It states that the such response must be *material frame indifference* (or material objectivity) and *respect of material symmetries*.
 - ▶ Thermodynamics with internal variables (TIV)
 - ★ Also in a general form, it says that the thermodynamic state depends only on the instantaneous value of the *state variables* and not on their past history.
 - ★ That is, at any instant of a thermodynamic process the thermodynamic state, given by $\sigma(t)$, $\psi(t)$, $s(t)$, and $q(t)$, at a point of the material can be completely determined by the knowledge of a finite number of state variables.

The functional approach

The 3 axioms commented...

- 1 *Thermodynamically compatible determinism*: Implies the existence of constitutive functionals $\mathcal{F}, \mathcal{G}, \mathcal{H}$, and $\mathcal{J}$ of the whole histories of $\mathbf{F}$, θ , and $\mathbf{g}$ of a point of Ω such that the quantities

$$\boldsymbol{\sigma}(t) = \mathcal{F}(\mathbf{F}, \theta, \mathbf{g}), \quad \psi(t) = \mathcal{G}(\mathbf{F}, \theta, \mathbf{g})$$

$$s(t) = \mathcal{H}(\mathbf{F}, \theta, \mathbf{g}), \quad \mathbf{q}(t) = \mathcal{J}(\mathbf{F}, \theta, \mathbf{g})$$

are determined.

- 2 *Principle of material objectivity (Frame indifference)*: It states that *the material response is independent of the observer*. In short, given a change in the observer, $\mathbf{F}^* = \mathbf{Q}\mathbf{F}$, then the vector and tensor variables change as

$$\boldsymbol{\sigma}^* = \mathbf{Q}\boldsymbol{\sigma}\mathbf{Q}^T, \quad g^* = \mathbf{Q}\mathbf{g}, \quad q^* = \mathbf{Q}q$$

and the scalar quantities $\psi(t)$ and $s(t)$ do not change.

Such change implies in some restrictions on the functionals, requiring the satisfaction of:

$$\boldsymbol{\sigma}^*(t) = \mathcal{F}(\mathbf{F}^*, \theta, \mathbf{g}^*)$$

$$\psi(t) = \mathcal{G}(\mathbf{F}^*, \theta, \mathbf{g}^*)$$

$$s(t) = \mathcal{H}(\mathbf{F}^*, \theta, \mathbf{g}^*)$$

$$\mathbf{q}^*(t) = \mathcal{J}(\mathbf{F}^*, \theta, \mathbf{g}^*)$$

- 3 *Respect of material symmetries*: If a continuum presents a symmetry group, represented by $\mathbf{Q}_S$, then

$$\boldsymbol{\sigma}(t) = \mathcal{F}(\mathbf{F}, \theta, \mathbf{g}) = \mathcal{F}(\mathbf{Q}_S \mathbf{F}, \theta, \mathbf{g})$$

$$\psi(t) = \mathcal{G}(\mathbf{F}, \theta, \mathbf{g}) = \mathcal{G}(\mathbf{Q}_S \mathbf{F}, \theta, \mathbf{g})$$

$$s(t) = \mathcal{H}(\mathbf{F}, \theta, \mathbf{g}) = \mathcal{H}(\mathbf{Q}_S \mathbf{F}, \theta, \mathbf{g})$$

$$\mathbf{q}(t) = \mathcal{J}(\mathbf{F}, \theta, \mathbf{g}) = \mathcal{J}(\mathbf{Q}_S \mathbf{F}, \theta, \mathbf{g})$$

hold for any $\mathbf{Q}_S$. A solid is said isotropic if $\mathbf{Q}_S$ is any proper orthogonal group.

Thermodynamics with Internal Variables (TIV)

Sometimes called: The method of local state, the local state method, ...

- It assumes that the thermodynamical state of the continuum at a given point and a given time is completely given by the definition of a set of state variables. This *relax* the first axiom of the functional approach...
- State variables are quantities that characterize the state of the system
 - ▶ For instance, a gas in a confined region is characterized by the pressure p , the volume V and the temperature θ . These are the state variables for this simple system.
- A state function is a function that only depends on the state of the system and not on the manner in which the system is achieved. A state function is a function of state variables.
- While the Internal Energy and the Entropy are state functions the Mechanical Work and the Heat are not! See the discussion in [Ottosen & Ristinmaa, 2005, chapter 20 and 21].
- As the specific free energy (*Helmholtz free energy*), Eq. 3.170, is a function of the specific internal energy and the specific entropy then it is an state function!

Thermodynamics with Internal Variables (TIV)

TIV's framework

- Considering the problems and applications involving elasticity, plasticity (inelasticity) that will be addressed in this course, the chosen state variables, which define the thermodynamic state at any time t , are given by the following set:

$$\{\mathbf{F}, \theta, \mathbf{g}, \alpha_k\} \quad (4.1)$$

where we introduce the dependence on a set of internal variables, named now of α_k , $k = 1, 2, \dots$ where k is k^{th} internal variable introduced into the model. α_k is a general name for somewhat quite general process that takes place under the assumptions of *continuous mechanics*.

- From the mathematical point of view, the components of α_k may be scalars, vectors or tensors.
- In this course the state equations are derived from a *free energy* potential, the *Helmholtz free energy*. Now, let us assume that the *Helmholtz free energy* as a function of a set of variables such that

$$\psi = \psi(\mathbf{F}, \theta, \alpha_k). \quad (4.2)$$

note that the dependence of ψ on the temperature gradient $\mathbf{g}$ is disregarded because it contradicts the second principle of thermodynamics, see the discussion in [Neto et al, 2008, chapter 3] and the references therein..

Thermodynamics with Internal Variables (TIV)

Let us focus in the infinitesimal strain assumption

- Now, let us assume that the *Helmholtz free energy* as a function of a set of state variables such that

$$\psi = \psi(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p, \theta, \boldsymbol{\alpha}_k) = \psi(\boldsymbol{\varepsilon}^e, \theta, \boldsymbol{\alpha}_k) \quad (4.3)$$

where we assume also the infinitesimal strain hypothesis.

- Based on such state variables one can derive:

$$\frac{\partial \psi}{\partial \boldsymbol{\varepsilon}} = \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}^e} = - \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}^p} \quad (4.4)$$

- The imposition of the small strains hypothesis is more related to the scope of this course than an limitation of the thermodynamic potential ψ . One can find the free energy written in terms of large deformation in several books and articles.

Thermo-elastic materials

Thermo-elastic materials: Let us consider the most simple material model for a first analysis...

- Assume a reversible process, then $\Phi = 0$, and for initial argumentation that there is no variation of the gradient of temperature, that is $\mathbf{g} = 0$. Then Eq. 3.172 reduces to:

$$\dot{\psi} = \frac{1}{\rho} \boldsymbol{\sigma} \cdot \dot{\boldsymbol{\varepsilon}} - s \dot{\theta} \quad (4.5)$$

- As the free energy is a state function this last equation suggests

$$\psi = \psi(\boldsymbol{\varepsilon}, \theta).$$

Notice that we can write

$$\dot{\psi} = \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}} \cdot \dot{\boldsymbol{\varepsilon}} + \frac{\partial \psi}{\partial \theta} \dot{\theta}.$$

- By substituting this last result in Eq. 4.5 yields

$$\left(\frac{\partial \psi}{\partial \boldsymbol{\varepsilon}} - \frac{1}{\rho} \boldsymbol{\sigma} \right) \cdot \dot{\boldsymbol{\varepsilon}} + \left(\frac{\partial \psi}{\partial \theta} + s \right) \dot{\theta} = 0. \quad (4.6)$$

Thermo-elastic materials

Thermo-elastic materials: Let us consider the most simple material model for a first analysis...

- As it is possible to control the spatially uniform temperature and the strains in an independently way this last expression implies:

$$\sigma = \rho \frac{\partial \psi}{\partial \varepsilon} \quad \text{and} \quad s = \frac{\partial \psi}{\partial \theta}.$$

This are the state equations for stress and entropy!

- Note that:
 - ▶ In Eq. 4.6 $\dot{\varepsilon}$ and $\dot{\theta}$ are arbitrary factible process;
 - ▶ As s and σ are obtained from ψ by differentiation the specific free energy works as a potential function for both entropy and stress;
 - ▶ The variables θ and ε are observable state variables, in the sense we can measure... We will find later in this course state variables that are not observable (hidden state variables...)]
- More details in [Ottosen & Ristinmaa, 2005, chapter 21].

Potentials doctrine

The power of choice and its consequences...

- An important note before we go any further:
 - ▶ In this course the constitutive theory will be presented within the thermodynamics of irreversible processes approach, using state variables as a tool to describe the thermodynamic state of a point on the body.
 - ▶ Within this approach we will postulate the existence of "potentials" from which equations of state and also the evolution of the variables will be determined.
 - ▶ The reader should be aware that this form of presentation is one of the possible ways to approach the subject. There are several bibliographies that present the subject in a different way, see [Chen & Han, 2007]. However the main results (and their consequences) are the same.

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State variables and state functions

Remarks on State variables and state functions

- Some important remarks before we go further into the analysis:
 - ▶ State variables: are quantities (assumed) that fully describe the state of a thermodynamic system (in our case the solid the body - deformable or not!).
 - ▶ State functions: are functions of state variables and depends (only) on the state of the system and not on the manner in which this system is achieved.
 - ▶ Another point we need explore further (in the future) is the idea of thermodynamic equilibrium and the determination of the values of the state variables and thermodynamic forces at equilibrium.
 - ★ The reader should seek for the idea of thermodynamic equilibrium and non-equilibrium thermodynamics...

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State Variables and Thermodynamic Forces

More about state variables...

- Let us depart from the free energy as in Eq. 4.3, so

$$\psi = \psi(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p, \theta, \boldsymbol{\alpha}_k) = \psi(\boldsymbol{\varepsilon}^e, \theta, \boldsymbol{\alpha}_k)$$

where θ is the absolute temperature and $\boldsymbol{\alpha}_k$, $k = 1, 2, \dots$, is a set of internal variables, where k is k^{th} internal variable introduced into the model.

- Here, $\boldsymbol{\alpha}_k$ is a general name for somewhat quite general process that takes place under the assumptions of *continuous mechanics*. From the mathematical point of view, the components of $\boldsymbol{\alpha}_k$ may be scalars, vectors or tensors.

- Inside the context of this course, examples of internal variables are the isotropic hardening α , and the kinematic hardening $\boldsymbol{\beta}$ (a second order tensor). Also, one will find in the literature the damage D (scalar quantity) as an internal variable...
- Inside the framework of TIV each *State Variable* is associate to its *dual variable*, called *Associate Variables* or *Thermodynamic Forces*. The table bellow, see [Lemaitre, 1992], presents an idea of such variables.

State Variables		Associate Variables or Thermodynamic Forces $\boldsymbol{\beta}_k$
Observable	Internal	
$\boldsymbol{\varepsilon}$		$\boldsymbol{\sigma}$
θ		s
	$\boldsymbol{\varepsilon}^e$	$\boldsymbol{\sigma}$
	$\boldsymbol{\varepsilon}^p$	$-\boldsymbol{\sigma}$
	$\boldsymbol{\alpha}_k$	$\boldsymbol{\beta}_k$

Free energy - general case

Thermo-elastic materials: Let us consider the most simple material model for a first analysis...

- Departing from Eq. 4.3 (repeated here)

$$\psi = \psi(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p, \theta, \boldsymbol{\alpha}_k) = \psi(\boldsymbol{\varepsilon}^e, \theta, \boldsymbol{\alpha}_k) \quad (4.7)$$

- Based on this definition one writes

$$\dot{\psi} = \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}^e} \cdot \dot{\boldsymbol{\varepsilon}}^e + \frac{\partial \psi}{\partial \boldsymbol{\alpha}_k} \star \dot{\boldsymbol{\alpha}}_k + \frac{\partial \psi}{\partial \theta} \dot{\theta}. \quad (4.8)$$

Now, considering $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p$ then

$$\dot{\boldsymbol{\varepsilon}} = \dot{\boldsymbol{\varepsilon}}^e + \dot{\boldsymbol{\varepsilon}}^p \quad (4.9)$$

and substituting Eq.(4.9) into Eq.(4.8) yields

$$\left(\boldsymbol{\sigma} - \rho \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}^e} \right) \cdot \dot{\boldsymbol{\varepsilon}}^e + \boldsymbol{\sigma} \cdot \dot{\boldsymbol{\varepsilon}}^p - \rho \left(\frac{\partial \psi}{\partial \theta} + s \right) \dot{\theta} - \rho \frac{\partial \psi}{\partial \boldsymbol{\alpha}_k} \star \dot{\boldsymbol{\alpha}}_k - \frac{1}{\theta} \mathbf{q} \cdot \nabla \theta \geq 0. \quad (4.10)$$

Feasible processes and arbitrariness

Feasible processes

- The inequality in Eq.(3.172) must hold for any feasible process. Therefore, suppose the following processes:
 - ▶ Suppose that elastic deformations take place with no evolution of internal variables ($\dot{\epsilon}^p = 0, \dot{\alpha}_k = 0$), and under isothermal ($\dot{\theta} = 0$) and homogeneous temperature distribution ($\nabla\theta = 0$), this is a reversible process. Eq.(4.10) gives the following state equation for stresses

$$\sigma - \rho \frac{\partial \psi}{\partial \epsilon^e} = 0 \rightarrow \sigma = \rho \frac{\partial \psi}{\partial \epsilon^e} \quad (4.11)$$

- ▶ Also, suppose a process where thermal deformation takes place in which $\dot{\epsilon}^p = 0, \dot{\alpha}_k = 0, g = 0$. Thus, Eq.(3.172) produces (it is also a reversible process)

$$s - \frac{\partial \psi}{\partial \theta} = 0 \rightarrow s = \frac{\partial \psi}{\partial \theta}. \quad (4.12)$$

- ▶ We can now introduce the concept of *thermodynamic forces* β_k . The thermodynamics forces are defined as

$$\beta_k = \rho \frac{\partial \psi}{\partial \alpha_k}. \quad (4.13)$$

They are the pairs (or *dual variables*) associated to the rate of $\dot{\alpha}_k$ in Eq.(4.10) that produces *power densities* $\left[\frac{Nm}{m^3s} = \frac{W}{m^3} \right]$.

Internal variables evolution

The pseudo-potential of dissipation φ and φ^*

- In this course we are interested in modeling the *mechanical intrinsic dissipation* Φ^{mech} , therefore assuming that $g = 0$ the dissipation is

$$\Phi^{\text{mech}} = \boldsymbol{\sigma} \cdot \dot{\boldsymbol{\varepsilon}}^p - \beta_k \star \dot{\alpha}_k \quad (4.14)$$

- The question that arises is: How can we always satisfy the intrinsic mechanical dissipation Φ^{mech} ? (Could this inequality be violated?)
- To answer this question we do need to employ some important results from convex analysis.
 - ▶ First we postulate the existence of a pseudo-dissipation potential φ which has as arguments the evolution terms $(\dot{\boldsymbol{\varepsilon}}^p, \dot{\alpha}_k)$ and may have as parameters $(\boldsymbol{\varepsilon}^e, \boldsymbol{\alpha}_k)$ terms, i.e.,

$$\varphi = \varphi(\dot{\boldsymbol{\varepsilon}}^p, \dot{\alpha}_k; \boldsymbol{\varepsilon}^e, \boldsymbol{\alpha}_k) \quad (4.15)$$

- ▶ We postulate as well that φ is constructed as a convex function with relation to $(\dot{\boldsymbol{\varepsilon}}^p, \dot{\alpha}_k)$, moreover it is non-negative, and zero valued function at origin, i.e., $\varphi(\mathbf{0}, \mathbf{0}_k) = 0$. The complementary laws are derived using the *normal dissipation hypothesis*

$$\boldsymbol{\sigma} = \frac{\partial \varphi}{\partial \dot{\boldsymbol{\varepsilon}}^p} \quad (4.16)$$

$$\beta_k = -\frac{\partial \varphi}{\partial \dot{\alpha}_k}. \quad (4.17)$$

Legendre-Fenchel transform

Legendre-Fenchel transform

- The normality condition can be represented by the gradient condition, in case of differentiable φ functions, or sub-gradients for non-differentiable φ function, that is:

$$\nabla\varphi = \left[\begin{array}{c} \frac{\partial\varphi}{\partial\dot{\varepsilon}^p} \\ -\frac{\partial\varphi}{\partial\dot{\alpha}_k} \end{array} \right] \xrightarrow[\text{function}]{\text{Non-differentiable}} \partial\varphi = \left[\begin{array}{c} \frac{\partial\dot{\varepsilon}^p\varphi}{\partial\dot{\alpha}_k\varphi} \end{array} \right]. \quad (4.18)$$

- In this format the pseudo-potential seems to not help much to assure the satisfaction of inequality (4.14). Therefore we invoke the *Legendre-Fenchel* transformation. The *Legendre-Fenchel* transform of $\varphi(\dot{\varepsilon}^p, \dot{\alpha}_k; \varepsilon^e, \alpha_k)$ is

$$\varphi^*(\sigma, \beta_k; \varepsilon^e, \alpha_k) = \sup_{(\dot{\varepsilon}^p, \dot{\alpha}_k; \varepsilon^e, \alpha_k)} \{(\sigma \cdot \dot{\varepsilon}^p - \beta_k \star \dot{\alpha}_k) - \varphi(\dot{\varepsilon}^p, \dot{\alpha}_k; \varepsilon^e, \alpha_k)\}. \quad (4.19)$$

for constant values of σ and β_k .

- Assuming $\varphi \in C^1$, then via the transformation in Eq.(4.19) we assure the normality correspondence

$$\nabla\varphi = \left[\begin{array}{c} \frac{\partial\varphi}{\partial\dot{\varepsilon}^p} \\ \frac{\partial\varphi}{\partial\dot{\alpha}_k} \end{array} \right] \xleftrightarrow{\text{Eq.(4.19)}} \nabla\varphi^* = \left[\begin{array}{c} \frac{\partial\varphi^*}{\partial\sigma} \\ \frac{\partial\varphi^*}{\partial\beta_k} \end{array} \right]. \quad (4.20)$$

Legendre-Fenchel transform

Legendre-Fenchel transform

Remark

Important property of the *Legendre-Fenchel* transformation: As by construction φ is a convex function with respect to $(\dot{\epsilon}^p, \dot{\alpha}_k)$, and more, it is non-negative, and zero valued function at origin, i.e., $\varphi(\mathbf{0}, \mathbf{0}_k) = 0$ then so does φ^* in terms of (σ, β_k) via Eq.(4.19).

- Obviously, this result holds outside the context of the thermodynamic framework presented here.

Internal variables evolution

Internal variables evolution

- Via the *Legendre-Fenchel* transform in Eq.(4.19) we have the one to one correspondence

$$\begin{aligned}\sigma &= \frac{\partial \varphi}{\partial \dot{\varepsilon}^p} & \xleftrightarrow{\text{Eq.(4.19)}} & \dot{\varepsilon}^p = \frac{\partial \varphi^*}{\partial \sigma} \\ \beta_k &= -\frac{\partial \varphi}{\partial \dot{\alpha}_k} & \xleftrightarrow{\text{Eq.(4.19)}} & \dot{\alpha}_k = -\frac{\partial \varphi^*}{\partial \beta_k}.\end{aligned}$$

- Notice that, assuming the existence of $\varphi^*(\sigma, \beta_k)$ the internal variable evolution is now given by

$$\dot{\varepsilon}^p = \frac{\partial \varphi^*}{\partial \sigma} \tag{4.21}$$

$$\dot{\alpha}_k = -\frac{\partial \varphi^*}{\partial \beta_k}. \tag{4.22}$$

- Substituting Eq.(4.21) and Eq.(4.22) into Eq.(4.14) yields

$$\Phi^{\text{mech}} = \sigma \cdot \frac{\partial \varphi^*}{\partial \sigma} + \beta_k \star \frac{\partial \varphi^*}{\partial \beta_k} \geq 0. \tag{4.23}$$

and due the properties of $\varphi^*(\sigma, \beta_k)$ aforementioned we have this last inequality always verified.

Another important consideration

Too many assumptions?

- This section ends citing literally a paragraph from reference [Lemaitre & Chaboche, 1990, page 63]:

The whole problem of modeling a phenomenon lies in the determination of the analytical expressions for the thermodynamic potential ψ and for the dissipation potential φ or φ^ , and their identification in characteristic experiments. In fact the values of φ or φ^* are almost impossible to measure as they represent an energy usually dissipated as heat.*

Classroom Slides on Plasticity

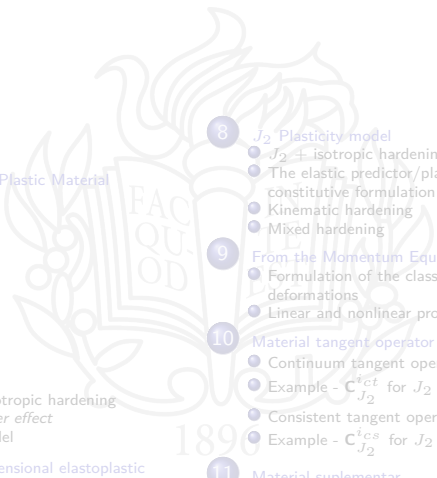
Prof. Rodrigo Rossi¹

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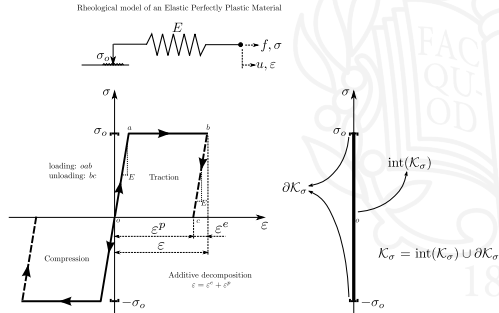
Outline

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 - Conservation of Mass
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Revisiting the Elastic, perfectly plastic model

Let us revisit the Elastic, perfectly plastic rheological model

- The model:



- Regarding the thermodynamic aspects, ϵ^p is the internal variable associated with irreversibility, therefore one may write the Helmholtz free energy as

$$\rho\psi = \rho\psi(\epsilon - \epsilon^p) = \frac{1}{2}E(\epsilon - \epsilon^p)^2. \quad (5.2)$$

Assuming an isothermal process, one writes

$$\sigma = \rho \frac{\partial \psi}{\partial \epsilon} = E(\epsilon - \epsilon^p). \quad (5.3)$$

But notice that, based on Eq. (5.2)

$$\begin{aligned} \rho \frac{\partial \psi}{\partial \epsilon^p} &= -E(\epsilon - \epsilon^p) \\ &= -\sigma \end{aligned} \quad (5.4)$$

- The stress σ cannot be greater than σ_o in absolute value, so

$$\sigma \in [-\sigma_o, \sigma_o] \quad \text{or} \quad |\sigma| \leq \sigma_o. \quad (5.1)$$

yielding

$$\Phi^{\text{mech}} = \sigma \dot{\epsilon}^p \geq 0 \quad (5.5)$$

Summary of the model

The Elastic, perfectly plastic rheological model:

- Elastic-plastic strain decomposition

$$\varepsilon = \varepsilon^e + \varepsilon^p$$

- Elastic law

$$\sigma = \rho \frac{\partial \psi}{\partial \varepsilon} = \rho \frac{\partial \psi}{\partial \varepsilon^e} = E (\varepsilon - \varepsilon^p) = E \varepsilon^e$$

- Yield criterion

$$f(\sigma) = |\sigma| - \sigma_o$$

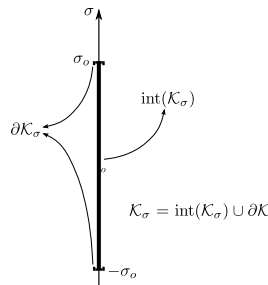
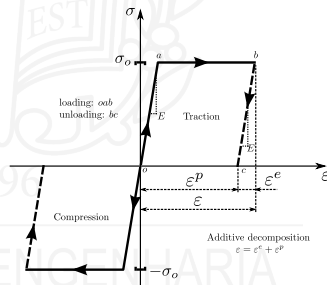
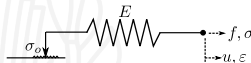
- Loading and unloading conditions
(Karush-Kuhn-Tucker conditions)

$$\dot{\gamma} \geq 0 \quad f \leq 0 \quad \dot{\gamma} f = 0$$

- Flow rule

$$\dot{\varepsilon}^p = \dot{\gamma} \frac{\partial f}{\partial \sigma} = \dot{\gamma} \text{sign}(\sigma)$$

Rheological model of an Elastic Perfectly Plastic Material

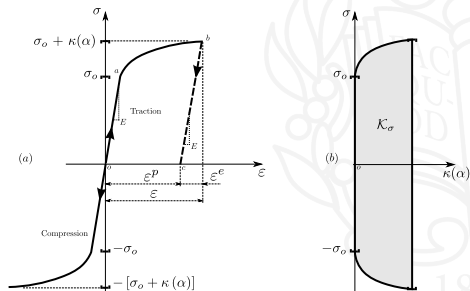


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1D elasto-plastic model with isotropic hardening

The isotropic hardening is responsible for the “size of yielding surface”. It is a variable that evolves with plastic deformation and updates the yielding value.



- Therefore, the *free energy*, Eq. (3.170), is now additively decomposed into a elastic ψ^e and inelastic ψ^i contributions as

$$\rho\psi(\varepsilon - \varepsilon^p, \alpha) = \rho\psi^e(\varepsilon - \varepsilon^p) + \rho\psi^i(\alpha). \quad (5.6)$$

with

$$\rho\psi^e(\varepsilon, \varepsilon^p) = \frac{1}{2}E(\varepsilon - \varepsilon^p)^2 \quad (5.7)$$

Again, we write

$$\sigma = \rho \frac{\partial \psi}{\partial \varepsilon} = \rho \frac{\partial \psi^e}{\partial \varepsilon} = E(\varepsilon - \varepsilon^p) \quad (5.8)$$

- Under the TIV framework, we introduce the isotropic hardening effect by setting a new internal variable, named α , that stands for the isotropic hardening effect in strains. Its counterpart, a “thermodynamic force” is the isotropic hardening effect κ in stress space.

but now

$$\kappa = \rho \frac{\partial \psi^i}{\partial \alpha}, \quad (5.9)$$

where κ is the “isotropic stress hardening” variable. Equations (5.8) and (5.9) are called *State Equations*.

Dissipation and flow rule

Dissipation and flow rule

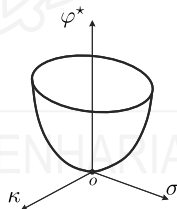
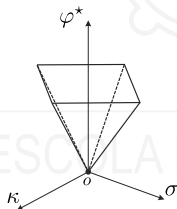
- By now, the explicit form of $\psi^i(\alpha)$ it is not required to be known. What we know is that whatever processes will be, it must yields

$$\Phi^{\text{mech}} = \sigma \dot{\varepsilon}^p - \kappa \dot{\alpha} \geq 0. \quad (5.10)$$

or using Eq. (5.8) and (5.9)

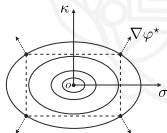
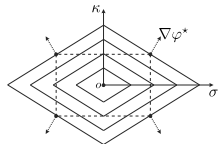
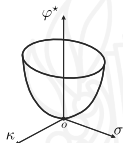
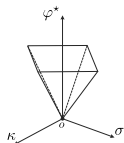
$$\Phi^{\text{mech}} = \underbrace{-\rho \frac{\partial \psi^e}{\partial \varepsilon^p}}_{\sigma} \dot{\varepsilon}^p - \underbrace{\rho \frac{\partial \psi^i}{\partial \alpha}}_{\kappa} \dot{\alpha} \geq 0. \quad (5.11)$$

- Therefore, the following question arises: Could one enforce the evolutions of $\dot{\varepsilon}^p$ and $\dot{\alpha}$ such that the dissipation inequality be always satisfied? To answer this question we introduce the *dissipation potential*, sometimes called *pseudo-dissipation potential*, φ^* being by construction a convex function with relation to variables $(\varphi^* = \varphi^*(\sigma, \kappa))$, non-negative, and zero valued function at origin ($\varphi^*(0, 0) = 0$).



Dissipation and flow rule

Dissipation and flow rule



- This figure shows examples of φ^* and the interpretation of the $\nabla \varphi^*$. It turns out that if a dissipation rule follows the hypothesis of *normal dissipation*.

- Using such type of construction it is possible satisfy the second law inequality (5.10) employing

$$\dot{\varepsilon}^p = \frac{\partial \varphi^*}{\partial \sigma} \quad (5.12)$$

$$\dot{\alpha} = -\frac{\partial \varphi^*}{\partial \kappa}. \quad (5.13)$$

- It quite easy to verify that if a potential function as φ^* is selected then the second law is verified. See the table bellow

σ	$\dot{\varepsilon}^p$	κ	$\dot{\alpha}$	$\sigma \dot{\varepsilon}^p$	$-\kappa \dot{\alpha}$	$\Phi^{\text{mech}} \geq 0$
+	+	+	-	+	+	✓
+	+	+	-	+	+	✓
-	-	-	+	+	+	✓
-	-	-	+	+	+	✓

Associative and non-associative flow rules

Associative and non-associative flow rules

- Many times the pseudo-potential of dissipation φ^* is taken as the yield function f . The formulations where $\varphi^* = f$ are called associative, associative flow rule, whereas the others ($\varphi^* \neq f$) are non-associative. See the discussion on section

$$\dot{\epsilon}^p = \dot{\gamma} \frac{\partial f}{\partial \sigma} \quad (5.14)$$

$$\dot{\alpha} = -\dot{\gamma} \frac{\partial f}{\partial \kappa}. \quad (5.15)$$

where $\dot{\gamma}$ is the plastic (inelastic) multiplier (recall that $\dot{\gamma} \geq 0$).

- By assuming the *maximum dissipation postulate* the evolution in Eqs. 5.14 and 5.15 are determined, see section ???. By now, let us assume such type of evolution.

Hardening material and yielding function

Hardening material and yielding function

- Let us rewrite the yield function to account for isotropic hardening. Here it assumes the following form

$$f(\sigma, \kappa) = |\sigma| - [\sigma_o + \kappa(\alpha)]. \quad (5.16)$$

Notice that:

$$\begin{cases} \kappa(\alpha) = 0 & \longrightarrow \text{Elastic-perfectly plastic material} \\ \kappa(\alpha) = \text{linear} & \longrightarrow \text{Elastic-linear hardening material} \\ \kappa(\alpha) = \text{nonlinear} & \longrightarrow \text{Elastic-nonlinear hardening material} \end{cases}.$$

- Moreover, $\kappa(\alpha) > 0$ represents hardening while $\kappa(\alpha) < 0$ will indicate softening. Assuming an associative plastic flow one finds

$$\dot{\epsilon}^p = \dot{\gamma} \frac{\partial f}{\partial \sigma} = \dot{\gamma} \text{sign}(\sigma) \quad (5.17)$$

$$\dot{\alpha} = -\dot{\gamma} \frac{\partial f}{\partial \kappa} = -\dot{\gamma}(-1) \rightarrow \dot{\alpha} = \dot{\gamma}. \quad (5.18)$$

and note that

$$\dot{\alpha} \geq 0. \quad (5.19)$$

Note that $\dot{\alpha} = \dot{\gamma} = \dot{\epsilon}^p$.

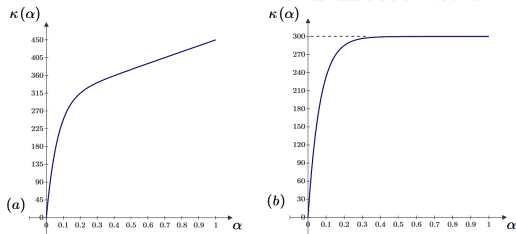
An example of isotropic hardening function

An example

- As an example of nonlinear isotropic hardening function we present a Voce type of *hardening law*

$$\kappa(\alpha) = H\alpha + (\sigma_{\infty} - \sigma_o) (1 - e^{-\delta\alpha}). \quad (5.20)$$

in which, H , σ_{∞} , σ_o , and δ are material constants. This kind of hardening law has been used in several works to account for the isotropic hardening effects.



- Isotropic hardening features given by Eq.(5.20) with $H = 150$ MPa, $\sigma_{\infty} = 750$ MPa, $\sigma_o = 450$ MPa, $\delta = 15$. a) Unbounded. b) Bounded by $\sigma_{\infty} - \sigma_o$ when $H = 0$.

- Notice that, for a material that exhibits such kind of isotropic hardening the Helmholtz free energy is

$$\rho\psi(\varepsilon - \varepsilon^p, \alpha) = \frac{1}{2}E(\varepsilon - \varepsilon^p)^2 + \frac{1}{2}H\alpha^2 + (\sigma_{\infty} - \sigma_o) \left(\alpha + \frac{1}{\delta}e^{-\delta\alpha} \right). \quad (5.21)$$

and

$$\rho\psi^i(\alpha) = \frac{1}{2}H\alpha^2 + (\sigma_{\infty} - \sigma_o) \left(\alpha + \frac{1}{\delta}e^{-\delta\alpha} \right). \quad (5.22)$$

Other isotropic hardening rules

Other isotropic hardening rules found in the literature

Some hardening laws ($C, n, \delta, \sigma_{sat}, \sigma_{\infty}$ are constants):

- Power-law hardening (Hollomon Law)):

$$\sigma = C (\bar{\varepsilon}^p)^n$$

in which n is known as the strain hardening exponent (also called the strain hardening index).

- Ludwik hardening:

$$\sigma = \sigma_o + C (\bar{\varepsilon}^p)^n$$

- Swift hardening:

$$\sigma = C (\varepsilon_o + \bar{\varepsilon}^p)^n$$

- Voce hardening:

$$\sigma = \sigma_o + \sigma_{sat} (1 - \exp(-\delta \bar{\varepsilon}^p))^n$$

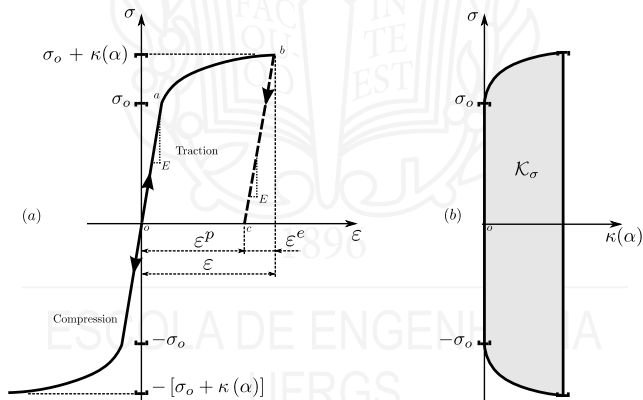
in which $\sigma_{sat} = (\sigma_{\infty} - \sigma_o)$.

Admissible stress set for isotropic hardening formulation

Admissible stress set

- The admissible stress set can be now defined as

$$\mathcal{K}_\sigma = \left\{ (\sigma, \kappa) \in \mathcal{R} \times \mathcal{R}^+ \mid f(\sigma, \kappa) = |\sigma| - [\sigma_o + \kappa(\alpha)] \leq 0 \right\}. \quad (5.23)$$



The elastoplastic tangent modulus (instantaneous/continuous)

Instantaneous/continuous elastoplastic tangent modulus

- The so-called elastoplastic tangent modulus is the instantaneous value of C^{ep} that satisfies the stress-strain rate relation

$$\dot{\sigma} = C^{ep} \dot{\varepsilon}. \quad (5.24)$$

- Let us derive the the instantaneous elastoplastic tangent modulus for linear isotropic plasticity. In such case Eq.(5.21) reduces to

$$\rho \psi(\varepsilon - \varepsilon^p, \alpha) = \frac{1}{2} E (\varepsilon - \varepsilon^p)^2 + \frac{1}{2} H \alpha^2 \quad (5.25)$$

and the state equations are Eq.(5.8) and Eq.(5.9) are

$$\sigma = \rho \frac{\partial \psi}{\partial \varepsilon} = \rho \frac{\partial \psi^e}{\partial \varepsilon} = E (\varepsilon - \varepsilon^p) \quad (5.26)$$

$$\kappa = \rho \frac{\partial \psi^i}{\partial \alpha} = H \alpha. \quad (5.27)$$

- Now assuming an associative plasticity model we have exactly the same evolution equations as in Eq.(5.17) and Eq.(5.18), that is:

$$\dot{\varepsilon}^p = \dot{\gamma} \frac{\partial f}{\partial \sigma} = \dot{\gamma} \text{sign}(\sigma)$$

$$\dot{\alpha} = -\dot{\gamma} \frac{\partial f}{\partial \kappa} = -\dot{\gamma} (-1) \rightarrow \dot{\alpha} = \dot{\gamma}.$$

Identification of the tangent modulus

Instantaneous elastoplastic tangent modulus identification

- Based on the *consistency condition* we can derive expressions for the plastic multiplier $\dot{\gamma}$ and for C^{ep} . Recall that

$$\dot{\gamma} \dot{f} = 0$$

and if $\dot{\gamma} > 0$ then $\dot{f} = 0$ (also from the KKT we have $f = 0$). The time derivative of Eq.(5.16) yields

$$\dot{f} = \frac{\partial f}{\partial \sigma} \dot{\sigma} + \frac{\partial f}{\partial \kappa} \frac{\partial \kappa}{\partial \alpha} \dot{\alpha} = 0$$

but $\frac{\partial \kappa}{\partial \alpha} = H$.

- Now introducing the rate of Eq.(5.8) and Eq.(5.18) into the last equation and keeping in mind Eq.(5.17) one can find that the plastic multiplier is

$$\dot{\gamma} = \frac{E}{E + H} \text{sign}(\sigma) \dot{\epsilon}. \quad (5.28)$$

- Now, by multiplying both sides of Eq.(5.28) by $\text{sign}(\sigma)$ yields

$$\dot{\epsilon}^p = \frac{E}{E + H} \dot{\epsilon}. \quad (5.29)$$

Substituting last equation into the rate of Eq.(5.8) yields

$$C^{ep} = \frac{EH}{E + H} \quad (5.30)$$

and

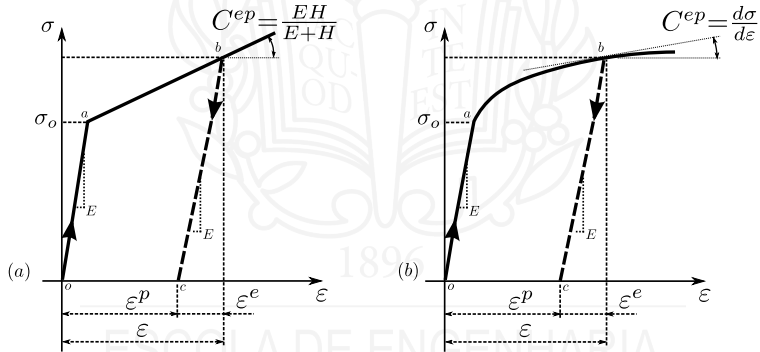
$$\dot{\sigma} = \begin{cases} E \dot{\epsilon} & \text{if } \dot{\gamma} = 0 \\ \frac{EH}{E+H} \dot{\epsilon} & \text{if } \dot{\gamma} > 0 \end{cases} \quad (5.31)$$

and the example is completed.

Tangent modulus: Graphical interpretation

Graphical interpretation of the tangent modulus

- Graphical interpretation of the tangent modulus for linear and nonlinear isotropic hardening.



Summary of the model

Summary of the one-dimensional isotropic model

- Based on what has been presented so far it is possible to realize that an elastoplastic formulation has the following ingredients:

- ▶ an elastic-plastic strain decomposition;
- ▶ an elastic law;
- ▶ a yield criterion, stated by the use of a yielding function;
- ▶ loading and unloading conditions;
- ▶ a hardening law characterizing the evolution of the internal variables;
- ▶ a flow rule, that is how the evolution of the plastic and hardening variables.

- Elastic-plastic strain decomposition

$$\varepsilon = \varepsilon^e + \varepsilon^p$$

- Elastic law

$$\sigma = \rho \frac{\partial \psi}{\partial \varepsilon} = \rho \frac{\partial \psi^e}{\partial \varepsilon} = E \left(\varepsilon - \varepsilon^p \right) = E \varepsilon^e$$

- Hardening law

$$\kappa = \rho \frac{\partial \psi^i}{\partial \alpha} = H \alpha + (\sigma_\infty - \sigma_o) \left(1 - e^{-\delta \alpha} \right)$$

- Yield criterion $f(\sigma, \kappa) = |\sigma| - [\sigma_o + \kappa(\alpha)] = 0$

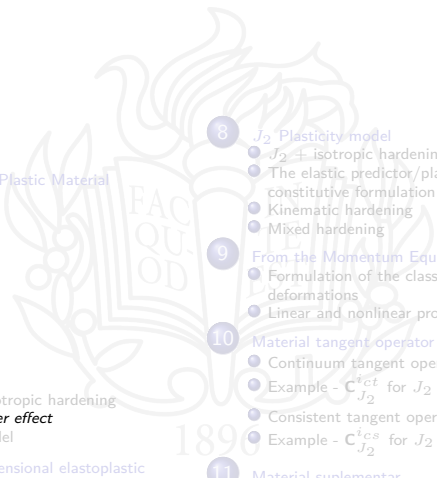
- Loading and unloading conditions (*KKT conditions*)

$$\dot{\gamma} \geq 0 \quad f \leq 0 \quad \dot{\gamma} f = 0$$

- Flow rule

$$\dot{\varepsilon}^p = \dot{\gamma} \frac{\partial f}{\partial \sigma} = \dot{\gamma} \operatorname{sign}(\sigma) \quad \text{and} \quad \dot{\alpha} = -\dot{\gamma} \frac{\partial f}{\partial \kappa} = \dot{\gamma}$$

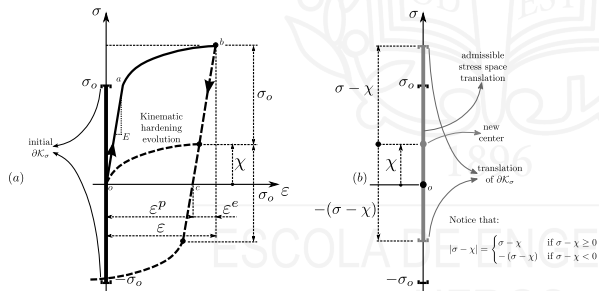
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Kinematic hardening model - The *Bauschinger effect*

The *Bauschinger effect*

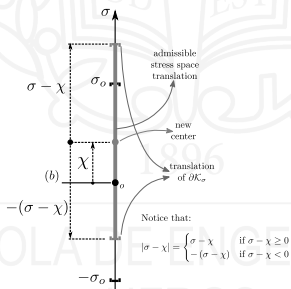
- It is noticed that after being loaded and hardened in one sense many materials, specially metals, presents a reduction in the yielding strength in the opposite sense. This phenomenon is known as the *Bauschinger effect* and can be modeled by the introduction of a kinematic hardening variable.
- The kinematic hardening, χ , is responsible for a translation of the yield surface, but without changing the yield surface. In other words, it is responsible for a rigid translation of the original f in the stress space.
- Quite often the kinematic variable, in the stress space, is called as *Back Stress*.



- Based on the idea, the yield criteria is

$$f(\sigma, \chi) = |\sigma - \chi| - \sigma_o. \quad (5.32)$$

Notice that the variable χ plays the role of shifting of the initial center (origin) of the coordinate axis σ .



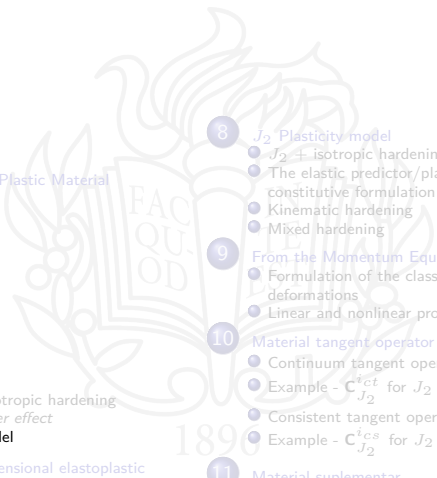
Notice that:

$$|\sigma - \chi| = \begin{cases} \sigma - \chi & \text{if } \sigma - \chi \geq 0 \\ -(\sigma - \chi) & \text{if } \sigma - \chi < 0 \end{cases}$$

Figure shows the geometric representation of the kinematic hardening effect in: (a) Stress-strain

diagram; (b) Admissible stress space evolution.

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Mixed isotropic and kinematic hardening model

Mixed formulation

- Rather than purely isotropic or purely kinematic hardening a material usually experiments both behaviors. In other words, the yield surface expands and translates simultaneously.
- The example of one-dimensional elastoplastic model with mixed isotropic/kinematic hardening presented here considers the nonlinear isotropic hardening law in Eq.(5.20)

$$\kappa(\alpha) = H\alpha + (\sigma_{\infty} - \sigma_o) (1 - e^{-\delta\alpha})$$

and the kinematic hardening by the *Armstrong-Frederic kinematic hardening law*

$$\chi(\beta) = a\beta \quad (5.33)$$

in which β is the *Back Strain* variable and a is a material parameter. Also, in the Armstrong-Frederic kinematic hardening law the flow rule is derived from the following pseudo-potential of dissipation

$$F = f + \frac{b}{2a}\chi^2 \quad (5.34)$$

in which b is a material parameter. The yielding function is

$$f(\sigma, \chi, \kappa) = |\sigma - \chi(\beta)| - [\sigma_o + \kappa(\alpha)] . \quad (5.35)$$

Helmholtz free energy and the stress space

Free energy

- The free energy potential for the mixed isotropic/kinematic hardening presented in this section is

$$\rho\psi(\varepsilon - \varepsilon^p, \alpha, \beta) = \frac{1}{2}E(\varepsilon - \varepsilon^p)^2 + \frac{1}{2}H\alpha^2 + (\sigma_\infty - \sigma_o) \left(\alpha + \frac{1}{\delta}e^{-\delta\alpha} \right) + \frac{a}{2}\beta^2. \quad (5.36)$$

or noticing that $\varepsilon^e = \varepsilon - \varepsilon^p$

$$\rho\psi(\varepsilon^e, \alpha, \beta) = \frac{1}{2}E(\varepsilon^e)^2 + \frac{1}{2}H\alpha^2 + (\sigma_\infty - \sigma_o) \left(\alpha + \frac{1}{\delta}e^{-\delta\alpha} \right) + \frac{a}{2}\beta^2. \quad (5.37)$$

This simple change is very straightforward meaning. The energy stored (and after removing the load then recovered) is always related with the elastic deformation.

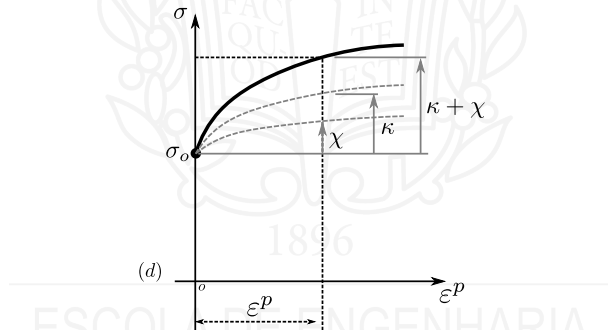
- The admissible stress space is given by the set

$$\mathcal{K}_\sigma = \left\{ (\sigma, \chi, \kappa) \in \mathcal{R} \times \mathcal{R} \times \mathcal{R}^+ \mid f(\sigma, \chi, \kappa) = |\sigma - \chi(\beta)| - [\sigma_o + \kappa(\alpha)] \leq 0 \right\}. \quad (5.38)$$

Mixed formulation: Hardening interpretation

Hardening interpretation in mixed formulation

- In a real material we have the combination of both effect (at the same time) what leads to the graphical interpretation given in figure below



Thermodynamics considerations

Free energy and state functions

- Assuming that the potential in Eq.(5.37) exists we can derive the following state equations

$$\sigma = \rho \frac{\partial \psi}{\partial \varepsilon^e} = \rho \frac{\partial \psi^e}{\partial \varepsilon^e} = E \varepsilon^e \quad (5.39)$$

$$\begin{aligned} \kappa &= \rho \frac{\partial \psi^i}{\partial \alpha} = H \alpha + (\sigma_\infty - \sigma_o) (1 - e^{-\delta \alpha}) \\ \chi &= \rho \frac{\partial \psi^i}{\partial \beta} = a \beta \end{aligned} \quad (5.40)$$

in which we assume that the free energy potential in Eq.(5.37) can be split additively into

$$\psi(\varepsilon^e, \kappa, \chi) = \psi^e(\varepsilon^e) + \psi^i(\kappa, \chi) \quad (5.41)$$

- Note that:

- ▶ $\psi^e(\varepsilon^e)$ - Energy stored in the material by elastic deformations and that will be released (recovered) during the unloading;
- ▶ $\psi^i(\kappa, \chi)$ - Energy stored (blocked) inside the material by isotropic & kinematic hardening.

Evolution, complementary equations

Evolution, complementary equations

- Once the dissipation potential was defined the evolution equations are derived based on the normality hypothesis as follows:

$$F = f + \frac{b}{2a}\chi^2 \quad \text{with} \quad f(\sigma, \chi, \kappa) = |\sigma - \chi(\beta)| - [\sigma_o + \kappa(\alpha)]$$

- Evolution:

$$\dot{\epsilon}^p = \dot{\gamma} \frac{\partial F}{\partial \sigma} = \dot{\gamma} \frac{\partial f}{\partial \sigma} = \dot{\gamma} \frac{\partial}{\partial \sigma} |\sigma - \chi| = \dot{\gamma} \text{sign}(\sigma - \chi) \quad (5.42)$$

$$\dot{\alpha} = -\dot{\gamma} \frac{\partial F}{\partial \kappa} = -\dot{\gamma} \frac{\partial f}{\partial \kappa} = \dot{\gamma} \quad (5.43)$$

$$\begin{aligned} \dot{\beta} &= -\dot{\gamma} \frac{\partial F}{\partial \chi} = -\dot{\gamma} \frac{\partial f}{\partial \chi} - \dot{\gamma} \frac{b}{a} \chi \\ &= \dot{\gamma} \left[\text{sign}(\sigma - \chi) - \frac{b}{a} \chi \right]. \end{aligned} \quad (5.44)$$

- Notice that as $\dot{\alpha} = \dot{\gamma}$ and $\dot{\gamma} = \dot{\bar{\epsilon}}^p$ then $\dot{\alpha}$ give us the accumulated plastic strain during the deformation process.

Numerical integration algorithm of the one dimensional elastoplastic constitutive formulation

Numerical integration algorithm

- See the file at moodle!



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Classroom Slides on Plasticity

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Maximum dissipation postulate

Let us introduce the linking between the plastic dissipation potential $\varphi^*(\sigma, \beta_k)$ and the yield criteria $f(\sigma, \beta_k)$. This is done by means of the *Maximum Intrinsic Dissipation* postulate.

- It is important to note that there is some discussion in the literature about whether *maximum dissipation* is a *principle of nature* (law of nature) or a *postulate* stated in order to satisfy dissipation inequality, see the discussion in [Ottosen & Ristinmaa, 2005].

Definition 10

MAXIMUM DISSIPATION POSTULATE: The *maximum dissipation postulate* states that among all pairs of $(\hat{\sigma}, \hat{\beta}_k)$ that one which solves the problem (σ, β_k) maximizes the intrinsic mechanical dissipation in inequality (4.14) for a given (fixed) $(\dot{\epsilon}^p, \dot{\alpha}_k)$. Mathematically, given a pair $(\dot{\epsilon}^p, \dot{\alpha}_k)$ we seek for $(\sigma, \beta_k) \in \mathcal{K}_\sigma$ such that

$$\Phi^{\text{mech}}(\sigma, \beta_k; \dot{\epsilon}^p, \dot{\alpha}_k) \geq \Phi^{\text{mech}}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k) \quad \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma \quad (7.1)$$

or

$$(\sigma, \beta_k) = \arg \max_{(\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma} \Phi^{\text{mech}}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k). \quad (7.2)$$

Constrained problem

Constrained problem

- Note that, this optimization problem is not unconstrained, in other words it must respect the admissibility of $(\hat{\sigma}, \hat{\beta}_k)$, that is, it is mandatory to enforce:

$$(\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_{\sigma} \quad (7.3)$$

where the *admissible stress set* is now redefined as

$$\mathcal{K}_{\sigma} = \{(\sigma, \beta_k) \mid f(\sigma, \beta_k) \leq 0\}. \quad (7.4)$$

in which

$$f(\sigma, \beta_k) \quad (7.5)$$

assumes a quite general format for the yield function.

Constrained optimization problems

Writing the problem:

- The mathematical problem can be announced as:
- This last problem is equivalent to the following one:

Problem 11

Assuming Φ^{mech} smooth find (σ, β_k) such that

$$(\sigma, \beta_k) = \arg \max_{(\hat{\sigma}, \hat{\beta}_k)} \Phi^{mech}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k) \quad (7.6)$$

subject to

$$(\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma \quad (7.7)$$

Problem 12

Assuming Φ^{mech} smooth find (σ, β_k) such that

$$\arg \max_{(\hat{\sigma}, \hat{\beta}_k)} \Phi^{mech}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k) \quad (7.8)$$

subject to

$$f(\hat{\sigma}, \hat{\beta}_k) \leq 0 \quad (7.9)$$

Rewriting the problem as a minimization

Maximization to minimization problem

- This maximization problem can now be written as a minimization just changing the sign of the objective function, that is

$$\Phi^{\text{mech}} = -\overline{\Phi^{\text{mech}}}$$

or using the mechanical dissipation defined in Eq. 4.14 one has

$$\overline{\Phi^{\text{mech}}}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k) = -\hat{\sigma} \cdot \dot{\epsilon}^p + \hat{\beta}_k \star \dot{\alpha}_k \quad (7.10)$$

and then the problem reduces to:

Problem 13

Assuming $\overline{\Phi^{\text{mech}}}$ smooth find (σ, β_k) such that

$$\arg \min_{(\hat{\sigma}, \hat{\beta}_k)} \overline{\Phi^{\text{mech}}}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k) \quad (7.11)$$

subject to

$$f(\hat{\sigma}, \hat{\beta}_k) \leq 0. \quad (7.12)$$

Lagrangian function

From the theory of optimization:

- By introducing the *Lagrangian* function (theory of optimization) we move from a constrained optimization problem, Problem 13, to an unconstrained one. That is,

$$\mathcal{L}^{\text{mech}}(\hat{\sigma}, \hat{\beta}_k, \dot{\gamma}; \dot{\epsilon}^p, \dot{\alpha}_k) = \overline{\Phi^{\text{mech}}}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k) + \dot{\gamma} f(\hat{\sigma}, \hat{\beta}_k) \quad (7.13)$$

submitted to the following conditions:

1 Stationarity of $\mathcal{L}(x, \cdot)$ - the gradient condition

$$\nabla \mathcal{L}^{\text{mech}}(\hat{\sigma}, \hat{\beta}_k, \dot{\gamma}; \dot{\epsilon}^p, \dot{\alpha}_k) = 0 \quad (7.14)$$

3 Complementary - Slackness:

$$\dot{\gamma} f(\hat{\sigma}, \hat{\beta}_k) = 0. \quad (7.16)$$

what leads to

$$\nabla \overline{\Phi^{\text{mech}}}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k) + \nabla [\dot{\gamma} f(\hat{\sigma}, \hat{\beta}_k)] = 0.$$

4 Non-Negativity of the Lagrange multipliers for inequalities:

$$\dot{\gamma} \geq 0. \quad (7.17)$$

2 Feasibility of the solution - respect the constraints:

$$f(\hat{\sigma}, \hat{\beta}_k) \leq 0. \quad (7.15)$$

In this context the multiplier $\dot{\gamma}$ receives the name of *Lagrange Multiplier*.

TIV problem

TIV problem

- The pair (σ, β_k) that respects conditions 7.14 to 7.17 simultaneously is the solution of the problem 13.
- As a result:

Problem 14

For a given $(\dot{\epsilon}^p, \dot{\alpha}_k)$, the solution of the problem 13 is the pair (σ, β_k) such that

$$\nabla \mathcal{L}^{mech}(\sigma, \beta_k, \dot{\gamma}; \dot{\epsilon}^p, \dot{\alpha}_k) = \mathbf{0} \quad (7.18)$$

and respects the Karush-Kuhn-Tucker conditions

$$\dot{\gamma} \geq 0, \quad f \leq 0, \quad \text{and} \quad \dot{\gamma} f = 0$$

Stationarity of $\mathcal{L}^{\text{mech}}$

Let us now compute the Stationarity of $\mathcal{L}^{\text{mech}}$ with relation to each variable $(\sigma, \beta_k, \dot{\gamma})$.

① The *first component* of $\nabla \mathcal{L}^{\text{mech}}$ is

$$\begin{aligned} [\nabla \mathcal{L}^{\text{mech}}]_1 &= [\nabla \overline{\Phi^{\text{mech}}}]_1 + [\nabla (\dot{\gamma} f(\sigma, \beta_k))]_1 = 0 \\ \frac{\partial \mathcal{L}^{\text{mech}}}{\partial \sigma} &= -\dot{\epsilon}^p + \dot{\gamma} \frac{\partial f(\sigma, \beta_k)}{\partial \sigma} = 0 \rightarrow \dot{\epsilon}^p = \dot{\gamma} \frac{\partial f(\sigma, \beta_k)}{\partial \sigma}. \end{aligned} \quad (7.19)$$

② The *second component* of $\nabla \mathcal{L}^i$ is

$$\begin{aligned} [\nabla \mathcal{L}^{\text{mech}}]_2 &= [\nabla \overline{\Phi^{\text{mech}}}]_2 + [\nabla (\dot{\gamma} f(\sigma, \beta_k))]_2 = 0 \\ \frac{\partial \mathcal{L}^{\text{mech}}}{\partial \beta_k} &= \dot{\alpha}_k + \dot{\gamma} \frac{\partial f(\sigma, \beta_k)}{\partial \beta_k} = 0 \rightarrow \dot{\alpha}_k = -\dot{\gamma} \frac{\partial f(\sigma, \beta_k)}{\partial \beta_k}. \end{aligned} \quad (7.20)$$

③ The *third component* of $\nabla \mathcal{L}^i$ is

$$\begin{aligned} [\nabla \mathcal{L}^{\text{mech}}]_3 &= [\cancel{\nabla \overline{\Phi^{\text{mech}}}}]_3 \overset{0}{+} [\nabla (\dot{\gamma} f(\sigma, \beta_k))]_3 = 0 \\ \frac{\partial \mathcal{L}^{\text{mech}}}{\partial \dot{\gamma}} &= f(\sigma, \beta_k) = 0. \end{aligned} \quad (7.21)$$

TIV problem: Evolution of the internal variables and KKT conditions

Evolution of the internal variables and KKT conditions

Fact 15

The maximum dissipation implies the imposition of

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{\gamma} \frac{\partial f(\boldsymbol{\sigma}, \boldsymbol{\beta}_k)}{\partial \boldsymbol{\sigma}}; \quad \dot{\boldsymbol{\alpha}}_k = -\dot{\gamma} \frac{\partial f(\boldsymbol{\sigma}, \boldsymbol{\beta}_k)}{\partial \boldsymbol{\beta}_k} \quad (7.22)$$

together with the Karush-Kuhn-Tucker conditions

$$\dot{\gamma} \geq 0, \quad f \leq 0, \quad \text{and} \quad \dot{\gamma} f = 0.$$

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Very important consequences

Consequences of the maximum dissipation postulate. From inequality 7.1:

$$\Phi^{\text{mech}}(\sigma, \beta_k; \dot{\epsilon}^p, \dot{\alpha}_k) \geq \Phi^{\text{mech}}(\hat{\sigma}, \hat{\beta}_k; \dot{\epsilon}^p, \dot{\alpha}_k), \quad (\sigma, \beta_k) \in \partial \mathcal{K}_\sigma; \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma$$

and using Eq. 4.14

$$\sigma \cdot \dot{\epsilon}^p - \beta_k \star \dot{\alpha}_k \geq \hat{\sigma} \cdot \dot{\epsilon}^p - \hat{\beta}_k \star \dot{\alpha}_k, \quad (\sigma, \beta_k) \in \partial \mathcal{K}_\sigma; \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma$$

what results in

$$(\sigma - \hat{\sigma}) \cdot \dot{\epsilon}^p - (\beta_k - \hat{\beta}_k) \star \dot{\alpha}_k \geq 0, \quad (\sigma, \beta_k) \in \partial \mathcal{K}_\sigma; \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma.$$

Now, using Eq.(7.19) and Eq.(7.20)

$$(\sigma - \hat{\sigma}) \cdot \dot{\gamma} \left. \frac{\partial f}{\partial \sigma} \right|_{(\sigma, \beta_k)} + (\beta_k - \hat{\beta}_k) \star \dot{\gamma} \left. \frac{\partial f}{\partial \beta_k} \right|_{(\sigma, \beta_k)} \geq 0, \quad (\sigma, \beta_k) \in \partial \mathcal{K}_\sigma; \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma$$

or by multiplying -1

$$(\hat{\sigma} - \sigma) \cdot \dot{\gamma} \left. \frac{\partial f}{\partial \sigma} \right|_{(\sigma, \beta_k)} + (\hat{\beta}_k - \beta_k) \star \dot{\gamma} \left. \frac{\partial f}{\partial \beta_k} \right|_{(\sigma, \beta_k)} \leq 0, \quad (\sigma, \beta_k) \in \partial \mathcal{K}_\sigma; \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma.$$

Realizing that 0 can be substituted by the KKT condition $\dot{\gamma} f(\hat{\sigma}, \hat{\beta}_k)$

$$(\hat{\sigma} - \sigma) \cdot \dot{\gamma} \left. \frac{\partial f}{\partial \sigma} \right|_{(\sigma, \beta_k)} + (\hat{\beta}_k - \beta_k) \star \dot{\gamma} \left. \frac{\partial f}{\partial \beta_k} \right|_{(\sigma, \beta_k)} \leq \dot{\gamma} f(\hat{\sigma}, \hat{\beta}_k), \quad (\sigma, \beta_k) \in \partial \mathcal{K}_\sigma; \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma$$

The case of $\dot{\gamma} = 0$ satisfies the above condition (trivially). However, when $\dot{\gamma} > 0$ one can write ($f(\sigma, \beta_k) = 0$)

$$(\hat{\sigma} - \sigma) \cdot \left. \frac{\partial f}{\partial \sigma} \right|_{(\sigma, \beta_k)} + (\hat{\beta}_k - \beta_k) \star \left. \frac{\partial f}{\partial \beta_k} \right|_{(\sigma, \beta_k)} \leq f(\hat{\sigma}, \hat{\beta}_k) - f(\sigma, \beta_k), \quad (\sigma, \beta_k) \in \partial \mathcal{K}_\sigma; \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma \quad (7.23)$$

Very important consequences

Consequences of the maximum dissipation postulate

- Interpreting the inequation 7.23

$$f(\hat{\sigma}, \hat{\beta}_k) - f(\sigma, \beta_k) \geq (\hat{\sigma} - \sigma) \cdot \left. \frac{\partial f}{\partial \sigma} \right|_{(\sigma, \beta_k)} + (\hat{\beta}_k - \beta_k) \star \left. \frac{\partial f}{\partial \beta_k} \right|_{(\sigma, \beta_k)} \quad (\sigma, \beta_k) \in \partial \mathcal{K}_\sigma; \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma$$

leads to the convexity of the yielding function f (according to theorem ??) f , see also in [Simo & Hughes, 1998, p. 102].

- Moreover, the maximum plastic dissipation also implies that the *admissible stress set* $\mathcal{K}_\sigma$ be convex. To show such an important mathematical feature we make use of the key result from the *level set theorem*, which is stated in a more complete form in ??. This result is a direct consequence of inequality 7.23. Again, as $(\sigma, \beta_k) \in \partial \mathcal{K}_\sigma$ then $f(\sigma, \beta_k) = 0$ resulting in

$$f(\hat{\sigma}, \hat{\beta}_k) \leq 0, \quad \forall (\hat{\sigma}, \hat{\beta}_k) \in \mathcal{K}_\sigma$$

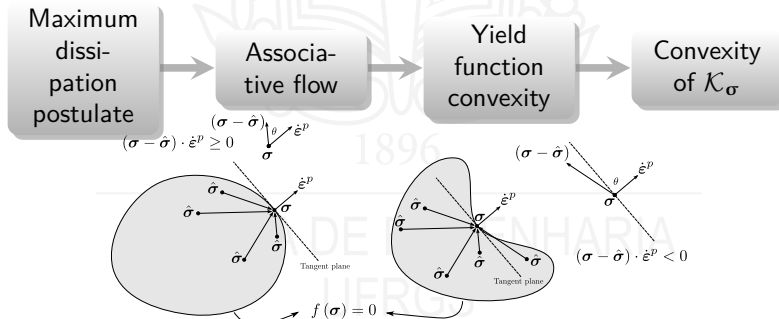
implying that $\mathcal{K}_\sigma$ is the *convex* level set given by the constant $c = 0$ in ??.

Very important consequences

Convexity

Fact 16

Notice that the postulate of maximum dissipation (constrained by the yield function) leads to associative plasticity which results into the convexity of the yield function (by means of the theorem ??) that leads (by means of the theorem ??) to the convexity of the admissible stress set $\mathcal{K}_\sigma$.



Classroom Slides on Plasticity

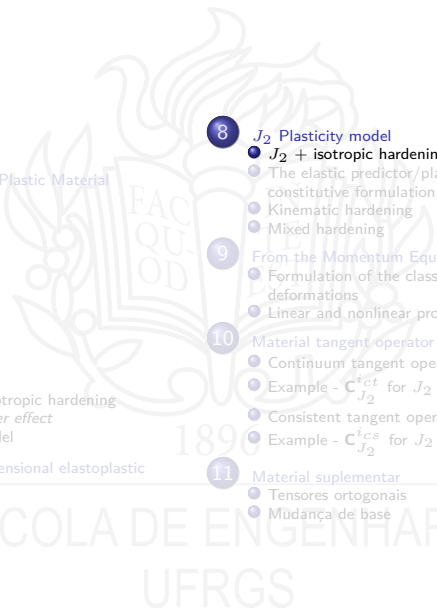
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Main aspects of the formulation

Model description:

- Additive decomposition of the infinitesimal strain tensor into an elastic and plastic (permanent) parts

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p \quad (8.1)$$

where now $\boldsymbol{\varepsilon}$ is the total strain tensor, $\boldsymbol{\varepsilon}^e$ is the elastic strain tensor, and $\boldsymbol{\varepsilon}^p$ plastic strain tensor.

- Linear elastic isotropic phase, that is, the Hooke's law is employed. It can be written in a compact form as

$$\boldsymbol{\sigma} = \mathbf{C} \boldsymbol{\varepsilon}^e \quad \left(\sigma_{ij} = C_{ijkl} \varepsilon_{kl}^e \right) \quad (8.2)$$

where

$$\mathbf{C} = 2\mu \mathbf{I} + \left(K - \frac{2}{3}\mu \right) \mathbf{I} \otimes \mathbf{I}.$$

- Yielding criterion - *von Mises* yield criterion (to be defined later in this section)

$$f = f(J_2(\boldsymbol{\sigma}), \kappa) \quad (8.3)$$

- An associative flow rule

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{\gamma} \frac{\partial f}{\partial \boldsymbol{\sigma}} \quad (8.4)$$

$$\dot{\alpha} = -\dot{\gamma} \frac{\partial f}{\partial \kappa} \quad (8.5)$$

- Loading and unloading conditions - *KKT* conditions (Eq. (2.18) and Eq. (2.19))

$$\dot{\gamma} \geq 0 \quad f \leq 0 \quad \dot{\gamma} f = 0.$$

Deviatoric and volumetric decomposition

- The name J_2 model comes from the second invariant of the deviatoric part of σ . By decomposing the stress tensor into its deviatoric and volumetric part as follows

$$\sigma = \sigma_d + \frac{1}{3} I_1 (\sigma) \mathbf{I}. \quad (8.6)$$

Analogously one can decompose the strain tensor as

$$\varepsilon = \varepsilon_d + \frac{1}{3} I_1 (\varepsilon) \mathbf{I}. \quad (8.7)$$

- The terms σ_d and ε_d are the deviatoric parts of the stress and total strain, respectively, whereas $\frac{1}{3} I_1 (\sigma)$ and $\frac{1}{3} I_1 (\varepsilon)$ are their respectively volumetric parts. The deviatoric part is related to the distortion (shear) and no variation of volume associated with it.

- Conversely, the volumetric part retains the whole volumetric information of σ or ε . The volumetric part of the stress and strain tensor fields have special names (notations) in solid mechanics. Usually the decomposition shown in Eq. 8.6 and Eq. 8.7 are written as

$$\sigma = \sigma_d + p \mathbf{I}. \quad (8.8)$$

$$\varepsilon = \varepsilon_d + \frac{1}{3} \varepsilon_v \mathbf{I}. \quad (8.9)$$

where

$$p = \frac{1}{3} I_1 (\sigma) \quad (8.10)$$

is known as *Hydrostatic Stress* and

$$\varepsilon_v = I_1 (\varepsilon) \quad (8.11)$$

is the *volumetric deformation*.

Invariants of a deviatoric tensor

Invariants

- The invariants of a deviatoric tensor σ_d are usually denominated by the letter J instead of I . Their definition is presented here using the stress tensor σ but it can be extended to any tensor.

1 First invariant - J_1

$$J_1(\sigma_d) \equiv I_1(\sigma_d) \quad (8.12)$$

but notice that

$$J_1(\sigma_d) = \text{tr}(\sigma_d) = \text{tr}\left(\sigma - \frac{1}{3}I_1(\sigma)\mathbf{I}\right) = 0. \quad (8.13)$$

2 Second Invariant - J_2

$$\begin{aligned} I_2(\sigma_d) &= \frac{1}{2} \left\{ [\text{tr}(\sigma_d)]^2 - [\text{tr}(\sigma_d^2)] \right\} \\ &= -\frac{1}{2} [\text{tr}(\sigma_d^2)] \end{aligned}$$

therefore the J_2 is defined as

$$J_2(\sigma_d) := -I_2(\sigma_d) = \frac{1}{2} \text{tr}(\sigma_d^2) = \frac{1}{2} \sigma_{dij} \sigma_{dij} = \frac{1}{2} \sigma_d \cdot \sigma_d \quad (8.14)$$

3 Third invariant - J_3

$$J_3(\sigma_d) \equiv I_3(\sigma_d). \quad (8.15)$$

J_2 in principal stresses

J_2

- Let us write the stress tensor σ in its principal directions and then take its deviatoric part.

$$[\sigma] = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \\ 0 & 0 & \sigma_3 \end{bmatrix} \Rightarrow [\sigma_d] = \begin{bmatrix} \sigma_1 - \frac{1}{3}I_1 & 0 & 0 \\ 0 & \sigma_2 - \frac{1}{3}I_1 & 0 \\ 0 & 0 & \sigma_3 - \frac{1}{3}I_1 \end{bmatrix}. \quad (8.16)$$

- Now taking the second invariant J_2 leads to

$$\begin{aligned} J_2(\sigma_d) &= \frac{1}{2} \sigma_{d_{ij}} \sigma_{d_{ij}} \\ &= \frac{1}{2} \left[\left(\frac{2}{3}\sigma_1 - \frac{1}{3}\sigma_2 - \frac{1}{3}\sigma_3 \right)^2 + \left(-\frac{1}{3}\sigma_1 + \frac{2}{3}\sigma_2 - \frac{1}{3}\sigma_3 \right)^2 + \left(-\frac{1}{3}\sigma_1 - \frac{1}{3}\sigma_2 + \frac{2}{3}\sigma_3 \right)^2 \right] \end{aligned} \quad (8.17)$$

and after a straightforward manipulation reduces to

$$\begin{aligned} J_2(\sigma_d) &= \frac{1}{3} (\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - \sigma_1\sigma_2 - \sigma_1\sigma_3 - \sigma_2\sigma_3) \\ &= \frac{1}{6} [(\sigma_1 - \sigma_2)^2 + (\sigma_1 - \sigma_3)^2 + (\sigma_2 - \sigma_3)^2] \end{aligned} \quad (8.18)$$

- Keep this last result, Eq. 8.18, in mind.

von Mises yielding criterion

von Mises

- The *von Mises* yielding criterion, also known as the maximum distortion strain energy criterion, is now introduced. The strain energy density associated with an isotropic linear elastic solid under deformation is

$$\rho\psi^e = w^e = \frac{1}{2E} \left[\sigma_1^2 + \sigma_2^2 + \sigma_3^2 - 2\nu (\sigma_1\sigma_2 + \sigma_1\sigma_3 + \sigma_2\sigma_3) \right]. \quad (8.19)$$

This energy can be additively decomposed into a volumetric w_v^e and deviatoric w_d^e parts as follows

$$w^e = w_v^e + w_d^e \quad (8.20)$$

where

$$w_v^e = \frac{1-2\nu}{6E} (\sigma_1 + \sigma_2 + \sigma_3)^2 = \frac{1-2\nu}{6E} I_1(\boldsymbol{\sigma})^2 \quad (8.21)$$

$$w_d^e = \frac{1}{12G} \left[(\sigma_1 - \sigma_2)^2 + (\sigma_1 - \sigma_3)^2 + (\sigma_2 - \sigma_3)^2 \right]. \quad (8.22)$$

Notice that based on Eq.(8.18) we can write the deviatoric part of the strain energy density, Eq.(8.22), as

$$w_d^e = \frac{1}{2G} J_2(\boldsymbol{\sigma}_d) \quad (8.23)$$

von Mises criterion

von Mises criterion

von Mises criterion: The yielding of a material begins when the deviatoric (distortional) strain energy density reaches a critical value.

- The critical value is understood as that deviatoric strain energy density associated with yielding point under uni-axial conditions, or in other words, the energy $w_d^e|_{1D}$ from the mechanical *traction test*.
- In such situation it is possible to write $\sigma_1 = \sigma_o$ and $\sigma_2 = \sigma_3 = 0$ and the critical value in the *von Mises* criterion becomes

$$w_d^e|_{1D} = \frac{1}{6G}\sigma_o^2. \quad (8.24)$$

- Now, the *von Mises* criterion can be mathematically written as

$$w_d^e = w_d^e|_{1D} \quad (8.25)$$

what reduces to

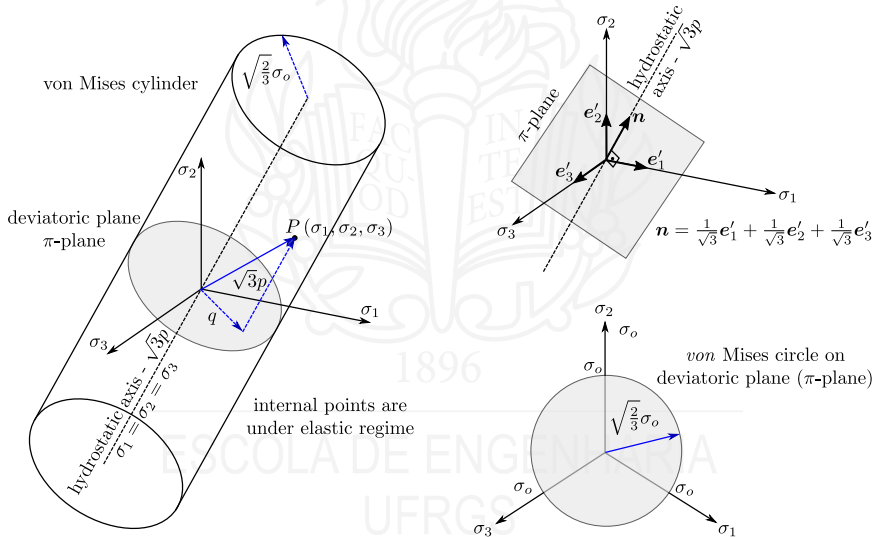
$$\sqrt{3J_2(\sigma_d)} = \sigma_o. \quad (8.26)$$

- The term $\sqrt{3J_2}$ is many times called as the *von Mises equivalent stress* given by

$$q(\sigma_d) = \sqrt{3J_2(\sigma_d)}. \quad (8.27)$$

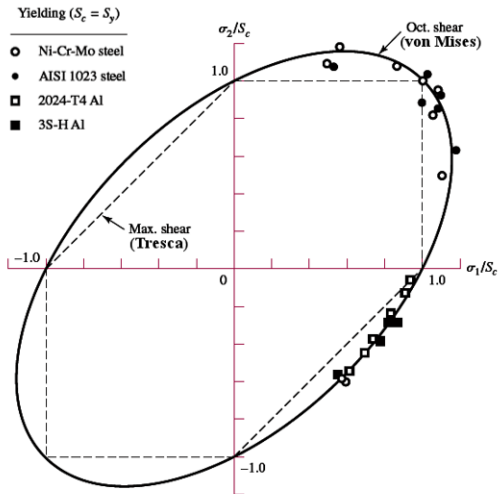
Graphical interpretation

von Mises cylinder



Assessment of the von Mises criterion in practice

von Mises criterion



von Mises yield function - J_2 Plasticity model

J_2 Plasticity model

We can now write the *von Mises* yield function as:

- Plasticity with isotropic hardening

$$f(\boldsymbol{\sigma}_d, \kappa) = q(\boldsymbol{\sigma}_d) - [\sigma_o + \kappa(\alpha)] \quad (8.28)$$

and notice that when $\kappa(\alpha) = 0$ one recovers the *perfect plasticity* model.

Associative plastic flow - *Prandtl-Reuss* plasticity

Associative plastic flow

- Employing the *von Mises* yield function and associative plasticity one writes

$$\dot{\epsilon}^p = \dot{\gamma} \frac{\partial f}{\partial \sigma}. \quad (8.29)$$

- In this case, we recover so-called *Prandtl-Reuss* plasticity. Consider the plasticity with the isotropic hardening case. In this case the plastic flow rule reduces to

$$\dot{\epsilon}^p = \dot{\gamma} \frac{\partial f}{\partial \sigma} = \dot{\gamma} \sqrt{\frac{3}{2}} \frac{\sigma_d}{\|\sigma_d\|} \quad (8.30)$$

or defining

$$\mathbf{N}_\sigma = \sqrt{\frac{3}{2}} \frac{\sigma_d}{\|\sigma_d\|} \quad N_{\sigma_{ij}} = \sqrt{\frac{3}{2}} \frac{\sigma_{d_{ij}}}{\|\sigma_d\|}. \quad (8.31)$$

the plastic flow is

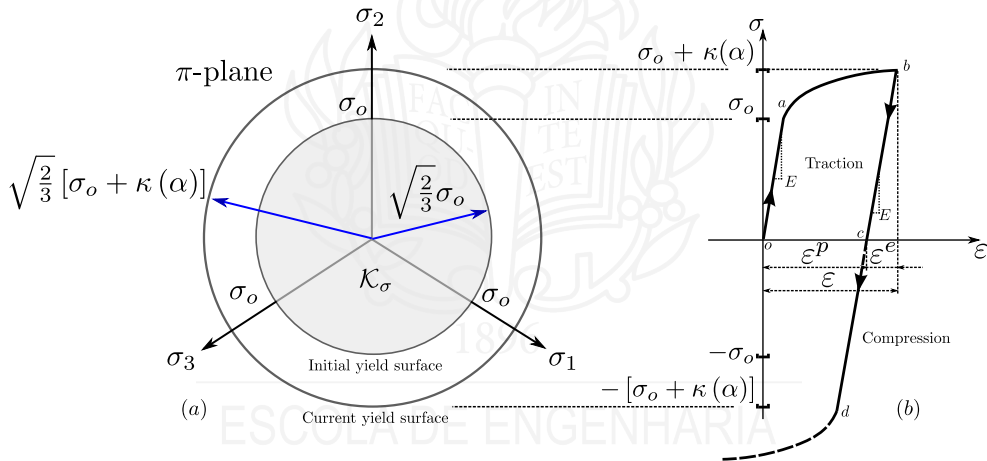
$$\dot{\epsilon}^p = \dot{\gamma} \mathbf{N}_\sigma \quad (8.32)$$

- Also, the evolution of the isotropic hardening variable is

$$\dot{\alpha} = -\dot{\gamma} \frac{\partial f}{\partial \kappa} = \dot{\gamma} = \dot{\epsilon}^p. \quad (8.33)$$

Moving yield surface

Moving yield surface - isotropic hardening



Thermodynamic aspects

Thermodynamic aspects

- There are no difference from the conceptual point of view from the one-dimensional model to the multiaxial except those that come from the generalization from the scalar quantities to the tensorial ones. Focusing in the energetic aspects the *Helmholtz free energy* potential is again additively decomposed into a elastic ψ^e and inelastic ψ^i contributions as, see Eq.(5.6),

$$\rho\psi(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p, \alpha) = \rho\psi^e(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) + \rho\psi^i(\alpha)$$

or

$$\rho\bar{\psi}(\boldsymbol{\varepsilon}^e, \alpha) = \rho\bar{\psi}^e(\boldsymbol{\varepsilon}^e) + \rho\psi^i(\alpha).$$

- Note that, for a material that exhibits isotropic hardening given by Eq.(5.20) the Helmholtz free energy is

$$\rho\psi(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p, \alpha) = \underbrace{\frac{1}{2}\mathbf{C}(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p) \cdot (\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p)}_{\rho\psi^e(\boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^p)} + \underbrace{\frac{1}{2}H\alpha^2 + (\sigma_\infty - \sigma_o)\left(\alpha + \frac{1}{\delta}e^{-\delta\alpha}\right)}_{\rho\psi^i(\alpha)} \quad (8.34)$$

or

$$\rho\bar{\psi}(\boldsymbol{\varepsilon}^e, \alpha) = \underbrace{\frac{1}{2}\mathbf{C}\boldsymbol{\varepsilon}^e \cdot \boldsymbol{\varepsilon}^e}_{\rho\bar{\psi}^e(\boldsymbol{\varepsilon}^e)} + \underbrace{\frac{1}{2}H\alpha^2 + (\sigma_\infty - \sigma_o)\left(\alpha + \frac{1}{\delta}e^{-\delta\alpha}\right)}_{\rho\psi^i(\alpha)}. \quad (8.35)$$

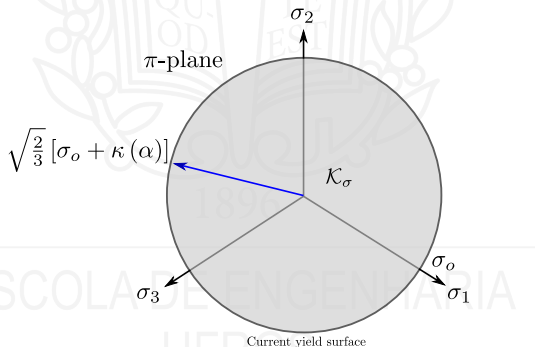
The admissible stress set in multiaxial formulation

$\mathcal{K}_\sigma$ in multiaxial formulation for isotropic hardening

- The admissible stress set can be now defined as

$$\mathcal{K}_\sigma = \{(\boldsymbol{\sigma}, \kappa) \in (\mathbb{R} \times \mathbb{R}) \times \mathbb{R} \mid f(\boldsymbol{\sigma}_d, \kappa) = q(\boldsymbol{\sigma}_d) - [\sigma_o + \kappa(\alpha)] \leq 0\}. \quad (8.36)$$

and its geometrical interpretation can be established only for special stress states as for in terms of the invariants of the stress tensor. Figure bellow shows the admissible stress set considering the principal values stress space.



Model summary

Model summary

- Elastic-plastic strain decomposition- Eq.(8.1)

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p$$

- Elastic law - Eq.(8.2)

$$\boldsymbol{\sigma} = \rho \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}} = \rho \frac{\partial \bar{\psi}^e}{\partial \boldsymbol{\varepsilon}^e} = \mathbf{C} \left(\boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}^p \right) = \mathbf{C} \boldsymbol{\varepsilon}^e \quad \left(\sigma_{ij} = C_{ijkl} \varepsilon_{kl}^e \right)$$

- Hardening law- Eq.(5.20)

$$\kappa = \rho \frac{\partial \psi^i}{\partial \alpha} = H\alpha + (\sigma_\infty - \sigma_o) \left(1 - e^{-\delta \alpha} \right)$$

- Yield criterion - Eq.(8.28)

$$f(\boldsymbol{\sigma}_d, \kappa) = q(\boldsymbol{\sigma}_d) - [\sigma_o + \kappa(\alpha)] = 0$$

- Loading and unloading conditions (*Karush-Kuhn-Tucker conditions*) - Eq. (2.18) and Eq. (2.19)

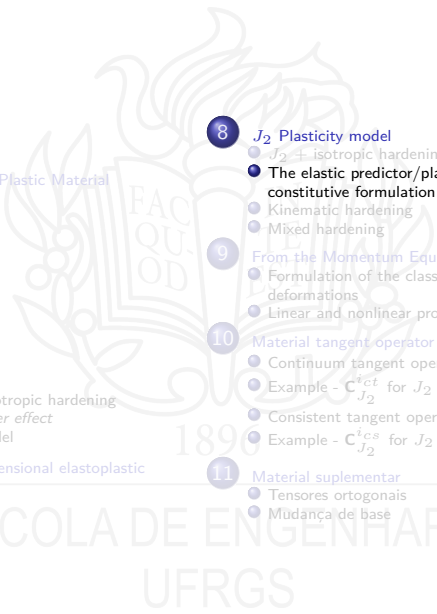
$$\dot{\gamma} \geq 0 \quad f \leq 0 \quad \dot{\gamma} f = 0$$

- Flow rule - Eq. (2.18) and Eq. (8.32)

$$\dot{\boldsymbol{\varepsilon}}^p = \dot{\gamma} \mathbf{N}_\sigma$$

$$\dot{\alpha} = -\dot{\gamma} \frac{\partial f}{\partial \kappa} = \dot{\gamma}$$

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The elastic predictor/plastic corrector for J_2 plasticity

J_2 plasticity

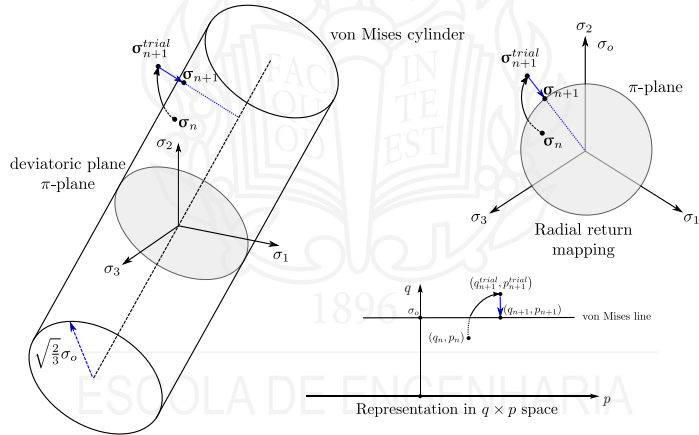
- See the attached file for a detailed description of:
- The statement of the J_2 problem in continuum and discretized versions;
- The algorithm considering multiple equations;
- The reduction to one single equation return mapping algorithm;
- Details on the geometric interpretation:

1896

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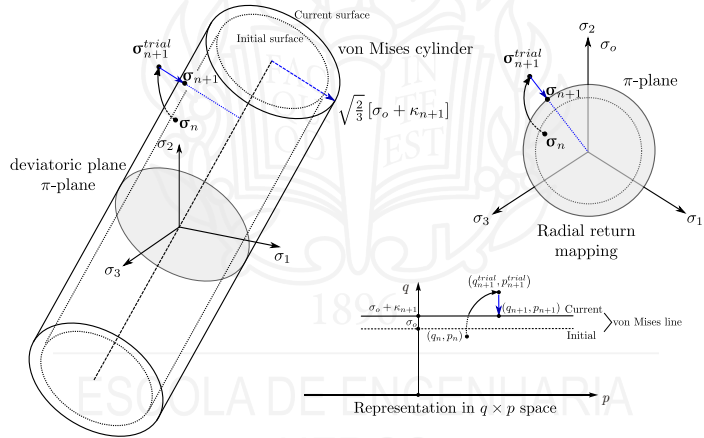
Geometric interpretation

Geometric interpretation of the return mapping algorithm for J_2+ isotropic hardening model. Elasto-perfectly plastic case:

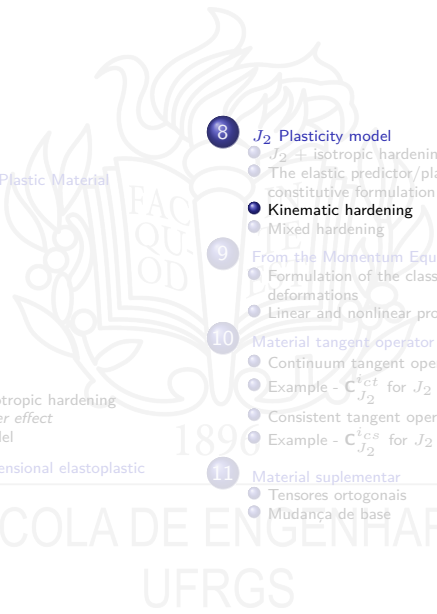


Geometric interpretation

Geometric interpretation of the return mapping algorithm for $J_2 +$ isotropic hardening model. Isotropic hardening case:

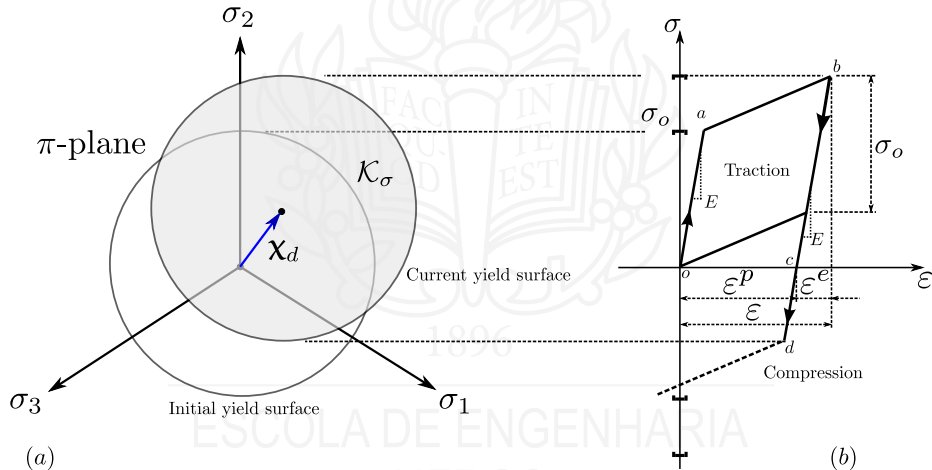


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Kinematic hardening effect in J_2 formulation

Geometric interpretation for linear kinematic hardening



Kinematic hardening

Model description

- Infinitesimal strains;
- Additive decomposition of the infinitesimal strain tensor into an elastic and plastic parts

$$\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^e + \boldsymbol{\varepsilon}^p \quad (8.37)$$

where $\boldsymbol{\varepsilon}$ is the total strain tensor, $\boldsymbol{\varepsilon}^e$ is the elastic strain tensor, and $\boldsymbol{\varepsilon}^p$ plastic strain tensor.

- Material that behaves in a linear and isotropic manner, that is, the Hooke's law can be employed. It can be written as

$$\boldsymbol{\sigma} = \mathbf{C}\boldsymbol{\varepsilon}^e \quad (\sigma_{ij} = C_{ijkl}\varepsilon_{kl}^e) \quad (8.38)$$

where

$$\mathbf{C} = 2\mu\mathbf{I} + \left(K - \frac{2}{3}\mu\right)\mathbf{I} \otimes \mathbf{I} \quad (8.39)$$

in which $\mu \equiv G$ is the shear modulus, K is bulk modulus.

- $\mathbb{I}$ is the fourth order identity tensor, and $\mathbf{I}$ is the second order identity tensor. In components we have

$$\mathbb{I}_{ijkl} = \frac{1}{2} (\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \quad (8.40)$$

$$(\mathbf{I} \otimes \mathbf{I})_{ijkl} = \delta_{ij}\delta_{kl}. \quad (8.41)$$

- Internal variables

- ▶ $\boldsymbol{\alpha}_d$ and $\boldsymbol{\chi}_d$ are the kinematic hardening variables. The first stands for the *kinematic back strain hardening* and the second is the thermodynamic force *kinematic back stress hardening*;

Kinematic hardening in J_2 formulation

Model description

- Yielding function - J_2 (*von Mises*) yielding function

$$\begin{aligned} f &= f(\boldsymbol{\sigma}_d, \boldsymbol{\chi}_d) \\ &= q(\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d) - \sigma_o \end{aligned} \quad (8.42)$$

in which

$$\begin{aligned} q(\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d) &= \sqrt{3J_2(\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d)} \\ &= \sqrt{\frac{3}{2}(\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d) \cdot (\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d)} \\ &= \sqrt{\frac{3}{2}} \|\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d\| \end{aligned} \quad (8.43)$$

with

$$\|\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d\| = \sqrt{(\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d) \cdot (\boldsymbol{\sigma}_d - \boldsymbol{\chi}_d)} \quad (8.44)$$

Kinematic hardening in J_2 formulation

Model description

- A non-associative flow rule ($F \neq f$), meaning that

$$\dot{\epsilon}^p = \dot{\lambda} \frac{\partial F}{\partial \sigma} \quad (8.45)$$

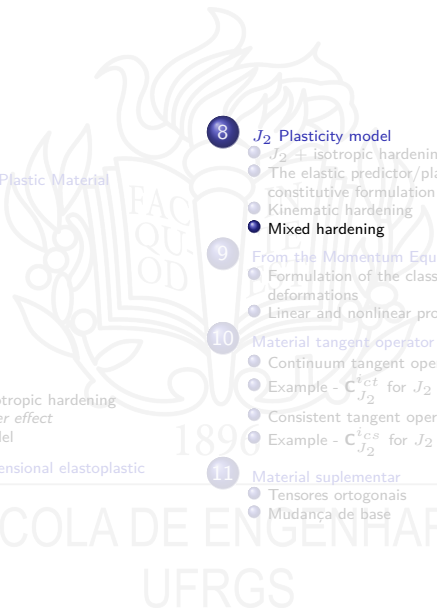
$$\dot{\alpha}_d = -\dot{\lambda} \frac{\partial F}{\partial \chi_d} \quad (8.46)$$

- Loading and unloading conditions - *KKT* conditions (Eq. (2.18) and Eq. (2.19))

$$\dot{\gamma} \geq 0 \quad f \leq 0 \quad \dot{\gamma} f = 0. \quad (8.47)$$

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Under construction

Under construction

- Under construction
- Under construction
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Classroom Slides on Plasticity

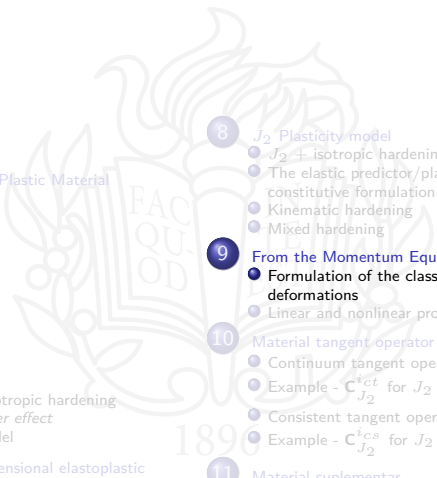
Prof. Rodrigo Rossi¹

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Draft version - - - Under development

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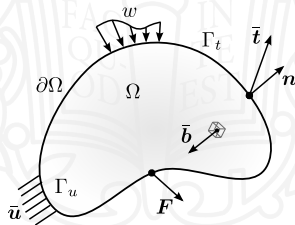
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Classical BVP under small deformations

Model description:

- Let $\Omega \subset \mathbb{R}^3$ be a bounded domain with a *Lipschitz* boundary $\partial\Omega$, subjected to a prescribed body force $\bar{b}$ defined on Ω , a prescribed surface traction $\bar{t}$ defined on Γ_t and a prescribed displacement $\bar{u}$ defined on Γ_u . n is a unit outward normal defined on $\partial\Omega$ and $\partial\Omega = \Gamma_u \cup \Gamma_t$ with $\Gamma_u \cap \Gamma_t = \emptyset$.



- At Γ_t the Cauchy relation is enforced

$$\sigma n = \bar{t}, \text{ at } \Gamma_t$$

while in the boundary Γ_u the prescribed displacement must be satisfied

$$u = \bar{u}, \text{ at } \Gamma_u.$$

BVP of a Body in Equilibrium in terms of Stress

BVP:

- The classical Boundary Value Problem (BVP) associated with equilibrium in terms of stresses may be stated as:

Problem 17

BOUNDARY VALUE PROBLEM OF A BODY IN EQUILIBRIUM IN TERMS OF STRESS: *Find $\mathbf{u}$ so that*

$$\begin{aligned}\operatorname{div} \boldsymbol{\sigma} + \rho \bar{\mathbf{b}} &= \mathbf{0}, \quad \forall \mathbf{x} \in \Omega \\ \boldsymbol{\sigma} \mathbf{n} &= \bar{\mathbf{t}}, \quad \forall \mathbf{x} \in \Gamma_t. \\ \mathbf{u} &= \bar{\mathbf{u}}, \quad \forall \mathbf{x} \in \Gamma_u\end{aligned}\tag{9.1}$$

Here, $\mathbf{n}$ is the outer normal to the Γ surface, $\boldsymbol{\sigma}$ is the Cauchy stress tensor and

$$\boldsymbol{\varepsilon}(\mathbf{u}) = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]\tag{9.2}$$

and notice that $\boldsymbol{\sigma}(\boldsymbol{\varepsilon}(\mathbf{u}))$.

- In the presentation of Problem 17 we have omitted the constitutive problem, but remember that we need to find a stress state where
- $$\boldsymbol{\sigma}(\boldsymbol{\varepsilon}(\mathbf{u}(\mathbf{x}, t))) = \rho \left. \frac{\partial \psi}{\partial \boldsymbol{\varepsilon}^e} \right|_t\tag{9.3}$$

together with a set of evolution equations for $\boldsymbol{\varepsilon}^p$ and for $\boldsymbol{\alpha}_k$.

From BVP to VBVP

The Method of Weighted Residuals

- Employing the weighted residuals method one writes

$$\mathbf{r} = \operatorname{div} \boldsymbol{\sigma}(\mathbf{u}) + \rho \bar{\mathbf{b}}$$

where $\mathbf{r}$ has the interpretation of the residual of the differential equation. So, let's define an inner product between $\mathbf{r}$ and $\mathbf{v}$, $\mathbf{v} \in \mathcal{H}_0$, and orthogonalize the residue with relation to $\mathbf{v}$, that is:

$$(\mathbf{r}, \mathbf{v})_{L^2} = \int_{\Omega} \mathbf{r} \cdot \mathbf{v} \, d\Omega = \int_{\Omega} (\operatorname{div} \boldsymbol{\sigma}(\mathbf{u}) + \rho \bar{\mathbf{b}}) \cdot \mathbf{v} \, d\Omega = 0, \quad \forall \mathbf{v} \in \mathcal{H}_0 \quad (9.4)$$

with

$$\mathcal{H}_0 = \left\{ \mathbf{u} \mid u_i \in \mathcal{H}^1(\Omega), \mathbf{u} = \mathbf{0} \text{ at } \Gamma_u \right\}. \quad (9.5)$$

- Let's use the following results: First

$$\operatorname{div} (\mathbf{T}^T \mathbf{a}) = \mathbf{a} \cdot (\operatorname{div} \mathbf{T}) + \mathbf{T} \cdot \nabla \mathbf{a} \quad (9.6)$$

and then the divergence theorem for vector fields

$$\int_{\Omega} \operatorname{div} (\mathbf{T}^T \mathbf{a}) \, d\Omega = \int_{\Gamma} (\mathbf{T}^T \mathbf{a}) \cdot \mathbf{n} \, d\Gamma. \quad (9.7)$$

From BVP to VBVP

The Method of Weighted Residuals

- Thus, we have as intermediate result:

$$\begin{aligned}\int_{\Omega} \operatorname{div} \boldsymbol{\sigma}(\mathbf{u}) \cdot \mathbf{v} \, d\Omega &= \int_{\Omega} \left(\operatorname{div} (\boldsymbol{\sigma}^T \mathbf{v}) - \boldsymbol{\sigma}(\mathbf{u}) \cdot \nabla \mathbf{v} \right) d\Omega \\ &= \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \mathbf{v} \, d\Gamma - \int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) \cdot \nabla \mathbf{v} \, d\Omega, \quad \forall \mathbf{v} \in \mathcal{H}_0.\end{aligned}\tag{9.8}$$

- Replacing this result in Eq. 9.4 yields

$$\int_{\Omega} \left(\operatorname{div} \boldsymbol{\sigma}(\mathbf{u}) + \rho \bar{\mathbf{b}} \right) \cdot \mathbf{v} \, d\Omega = - \int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) \cdot \nabla \mathbf{v} \, d\Omega + \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \mathbf{v} \, d\Gamma + \int_{\Omega} \rho \bar{\mathbf{b}} \cdot \mathbf{v} \, d\Omega, \quad \forall \mathbf{v} \in \mathcal{H}_0.$$

- Recalling that

$$\boldsymbol{\sigma} \cdot \nabla \mathbf{v} = \boldsymbol{\sigma} \cdot \left\{ \frac{1}{2} [\nabla \mathbf{v} + (\nabla \mathbf{v})^T] + \frac{1}{2} [\nabla \mathbf{v} - (\nabla \mathbf{v})^T] \right\} = \boldsymbol{\sigma} \cdot \left\{ \frac{1}{2} [\nabla \mathbf{v} + (\nabla \mathbf{v})^T] \right\} = \boldsymbol{\sigma} \cdot \boldsymbol{\varepsilon}(\mathbf{v})$$

we write

$$\int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) \cdot \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega = \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \mathbf{v} \, d\Gamma + \int_{\Omega} \rho \bar{\mathbf{b}} \cdot \mathbf{v} \, d\Omega, \quad \forall \mathbf{v} \in \mathcal{H}_0.\tag{9.9}$$

Which is known as the variation formulation of the problem.

Weak formulation, variational formulation

The Variational Boundary Value Problem (VBVP)

- The weak formulation, variational formulation, related to Eq.(9.1) can be stated as:

Problem 18

WEAK FORM OF THE BOUNDARY VALUE PROBLEM OF A BODY IN EQUILIBRIUM IN TERMS OF STRESS: *Find $\mathbf{u} \in \mathcal{H}$, $t \in (t_o, t_f)$, so that*

$$\mathcal{R}(\mathbf{u}, \mathbf{v}) = \int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}) \cdot \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega - \int_{\Omega} \rho \bar{\mathbf{b}} \cdot \mathbf{v} \, d\Omega - \int_{\Gamma_t} \bar{\mathbf{t}} \cdot \mathbf{v} \, d\Gamma = 0, \quad \forall \mathbf{v} \in \mathcal{H}_0 \quad (9.10)$$

where $\mathcal{R}(\mathbf{u}, \mathbf{v})$ stands for the residue of the weak form.

- Let

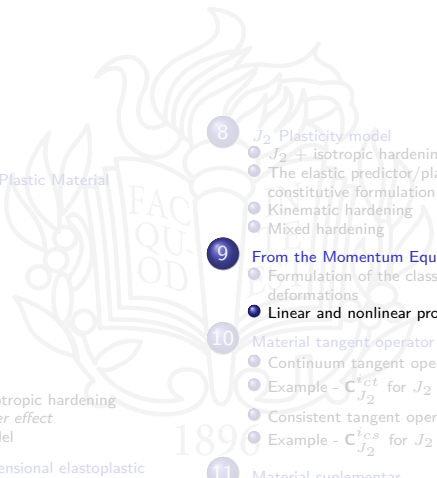
$$\mathcal{H} = \left\{ \mathbf{u} \mid u_i \in \mathcal{H}^1(\Omega), \mathbf{u} = \bar{\mathbf{u}} \text{ at } \Gamma_u \right\} \quad (9.11)$$

denote the set of the kinematically admissible displacements and

$$\mathcal{H}_0 = \left\{ \mathbf{u} \mid u_i \in \mathcal{H}^1(\Omega), \mathbf{u} = \mathbf{0} \text{ at } \Gamma_u \right\} \quad (9.12)$$

the space of admissible variations.

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Linear constitutive relationship

Linear problem

- For sake of initial argumentation let's assume that the constitutive relationship between $\sigma \times \varepsilon$ is linear and isotropic, therefore the *Hooke's law* is employed. Its tensorial representation is given by

$$\sigma = \mathbf{C} \varepsilon \quad (9.13)$$

in which

$$\mathbf{C} = 2\mu \mathbf{I} + \left[\kappa - \frac{2}{3}\mu \right] \mathbf{I} \otimes \mathbf{I}. \quad (9.14)$$

where $\mathbf{I}$, $\mathbf{I}$, μ and κ are the fourth-order symmetric identity tensor, the second order identity tensor, the shear and bulk modulus, respectively.

- Clearly Eq.(9.10) reduces to

$$\mathcal{R}(u, v) = \int_{\Omega} \mathbf{C} \varepsilon(u) \cdot \varepsilon(v) \, d\Omega - \int_{\Omega} \rho \bar{b} \cdot v \, d\Omega - \int_{\Gamma_t} \bar{t} \cdot v \, d\Gamma = 0, \quad \forall v \in \mathcal{H}_0. \quad (9.15)$$

Nonlinear Problem

Nonlinear Problem

- When the constitutive relationship is nonlinear Eq.(9.10) will be nonlinear as well. Notice that the non-linearity in Eq.(9.10) resides in the dependance $\sigma(\epsilon(u))$. In order to solve such problem we invoke the Directional (Gateaux) derivative.

Remark

DIRECTIONAL (GATEAUX) DERIVATIVE: Assuming $\mathcal{R}$ being sufficiently regular in Eq.(9.10) and expanding it in a *Taylor* series around a known value u_o , where

$$u = u_o + \tilde{u}, \quad (9.16)$$

and keeping only the first term we obtain a first order approximation for Eq.(9.10), that is

$$\begin{aligned} \mathcal{R}(u, v) &= \mathcal{R}(u_o + \tilde{u}, v) = 0, \quad \forall v \in \mathcal{H}_0 \\ \mathcal{R}(u_o + \tilde{u}, v) &\approx \mathcal{R}(u_o, v) + D\mathcal{R}(u_o, v)[\tilde{u}] = 0, \quad \forall v \in \mathcal{H}_0. \end{aligned} \quad (9.17)$$

It yields

$$D\mathcal{R}(u_o, v)[\tilde{u}] = -\mathcal{R}(u_o, v), \quad \forall v \in \mathcal{H}_0. \quad (9.18)$$

in which

$$D\mathcal{R}(u_o, v)[\tilde{u}] = \lim_{\epsilon \rightarrow 0} \frac{\mathcal{R}(u_o + \epsilon \tilde{u}, v) - \mathcal{R}(u_o, v)}{\epsilon} = \left. \frac{d\mathcal{R}(u_o + \epsilon \tilde{u}, v)}{d\epsilon} \right|_{\epsilon=0}, \quad \forall v \in \mathcal{H}_0. \quad (9.19)$$

Directional (Gateaux) derivative

Linearized term

- Now we are able to determine the linearized contribution given by $D\mathcal{R}(u_o, v)[\tilde{u}]$. Notice that

$$\begin{aligned}\left. \frac{d\mathcal{R}(u_o + \epsilon \tilde{u}, v)}{d\epsilon} \right|_{\epsilon=0} &= \left. \frac{d}{d\epsilon} \int_{\Omega} \sigma(u_o + \epsilon \tilde{u}) \cdot \varepsilon(v) \, d\Omega \right|_{\epsilon=0}, \quad \forall v \in \mathcal{H}_0 \\ &= \int_{\Omega} \left. \frac{d}{d\epsilon} \sigma(u_o + \epsilon \tilde{u}) \right|_{\epsilon=0} \cdot \varepsilon(v) \, d\Omega, \quad \forall v \in \mathcal{H}_0.\end{aligned}\tag{9.20}$$

but

$$\sigma = \sigma(\varepsilon(u))\tag{9.21}$$

then

$$\left. \frac{d}{d\epsilon} \sigma(u_o + \epsilon \tilde{u}) \right|_{\epsilon=0} = \left. \frac{d\sigma}{d\varepsilon} \right|_{u_o} \left. \frac{d}{d\epsilon} \varepsilon(u_o + \epsilon \tilde{u}) \right|_{\epsilon=0},\tag{9.22}$$

and keeping in mind the infinitesimal strain tensor definition, Eq.(9.2), we write

$$\left. \frac{d}{d\epsilon} \varepsilon(u_o + \epsilon \tilde{u}) \right|_{\epsilon=0} = \varepsilon(\tilde{u})\tag{9.23}$$

due to the linearity of the operator ε .

Directional (Gateaux) derivative

Linearized term

- Going backwards we write

$$\begin{aligned} \left. \frac{d\mathcal{R}(\mathbf{u}_o + \epsilon \tilde{\mathbf{u}}, \mathbf{v})}{d\epsilon} \right|_{\epsilon=0} &= \left. \frac{d}{d\epsilon} \int_{\Omega} \boldsymbol{\sigma}(\mathbf{u}_o + \epsilon \tilde{\mathbf{u}}) \cdot \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega \right|_{\epsilon=0}, \quad \forall \mathbf{v} \in \mathcal{H}_0 \\ &= \int_{\Omega} \left. \frac{d\boldsymbol{\sigma}}{d\boldsymbol{\varepsilon}} \right|_{\mathbf{u}_o} \boldsymbol{\varepsilon}(\tilde{\mathbf{u}}) \cdot \boldsymbol{\varepsilon}(\mathbf{v}) \, d\Omega, \quad \forall \mathbf{v} \in \mathcal{H}_0. \end{aligned} \quad (9.24)$$

- The term $\left. \frac{d\boldsymbol{\sigma}}{d\boldsymbol{\varepsilon}} \right|_{\mathbf{u}_o}$ in Eq.(9.24) is known as the infinitesimal material tangent modulus and must be identified for each material model. It is defined as

$$\hat{\mathbf{C}} = \left. \frac{d\boldsymbol{\sigma}}{d\boldsymbol{\varepsilon}} \right|_{\mathbf{u}_o}. \quad (9.25)$$

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Tangent modulus and tangent operator

Tangent modulus and tangent operator

- In the context of infinitesimal strains the material tangent modulus coincides to the so-called tangent operator, which can be consistent or continuum;
- In other words, such differences will be pronounced when finite strains are required;
- Despite the terms are slightly different in name, its correct derivation plays a very important role inside the solution framework once the convergence rate depends on it.
- Notice that in the case of an elastoplastic formulation we got two different values for Eq.(9.25), i.e., when the state is elastic ($f(\boldsymbol{\sigma}, \boldsymbol{\beta}_k) \leq 0$) then $\hat{\mathbf{C}} = \mathbf{C}$, the linear elastic isotropic fourth order tensor, see Eq.(9.14), but in case of violation of the admissible stress set, ($f(\boldsymbol{\sigma}, \boldsymbol{\beta}_k) > 0$), we have $\hat{\mathbf{C}} = \mathbf{C}^i$.
- In summary:

$$\hat{\mathbf{C}} = \begin{cases} \mathbf{C} & \text{if } f(\boldsymbol{\sigma}, \boldsymbol{\beta}_k) \leq 0 \\ \mathbf{C}^i & \text{if } f(\boldsymbol{\sigma}, \boldsymbol{\beta}_k) > 0 \end{cases} \quad (9.26)$$

Classroom Slides on Plasticity

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Draft version - - - Under development

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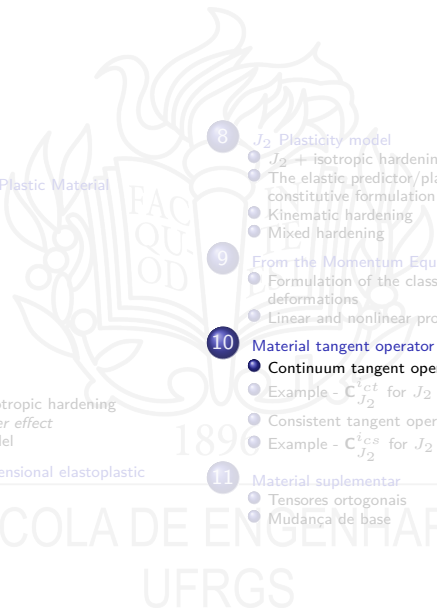
Material tangent operator derivation

Material tangent operator derivation

- As key points in modeling the mechanical response of materials, one may cite the correct choice of the constitutive model, the assessment of experimental data and the choice of a proper numerical solution method. In terms of computational efficiency, *CPU* time consuming and convergence rates, it is imperative to correctly derive the so-called *material tangent modulus*.
- Based on we ha seen so far, two are the possibilities of derivation:
 - ① The so-called *continuum* (instantaneous) $\mathbf{C}^{i_{ct}}$ or
 - ② The so-called *consistent* $\mathbf{C}^{i_{cs}}$ tangent operator.

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Continuum tangent operator - $\mathbf{C}^{i_{ct}}$

Continuum tangent operator - $\mathbf{C}^{i_{ct}}$

- The derivation of the so called continuum tangent operator follows the idea presented on page 136. As demonstrated there, we begin by writing the following rate equation

$$\dot{\sigma} = \mathbf{C} \dot{\varepsilon}^e \quad (10.1)$$

or

$$\dot{\sigma} = \mathbf{C} (\dot{\varepsilon} - \dot{\varepsilon}^p) = \mathbf{C} \left(\dot{\varepsilon} - \dot{\gamma} \frac{\partial f(\sigma, \beta_k)}{\partial \sigma} \right). \quad (10.2)$$

- To the derivation of the continuum tangent operator we have to determine the plastic multiplier. No problem, it is also derived based on the same idea presented on page 137. That is, by enforcing the *consistency condition*

$$\dot{\gamma} \dot{f} = 0$$

that is, $\dot{\gamma} > 0$ then $\dot{f} = 0$.

Continuum tangent operator - $\mathbf{C}^{i_{ct}}$

Continuum tangent operator - $\mathbf{C}^{i_{ct}}$

- The time derivative of Eq.(7.5) yields

$$\frac{d}{dt} f(\boldsymbol{\sigma}, \beta_k) = \dot{f} = \frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \dot{\boldsymbol{\sigma}} + \frac{\partial f}{\partial \beta_k} \star \dot{\beta}_k = 0.$$

Furthermore based on Eq.(4.13) we write

$$\dot{\beta}_k = \frac{d}{dt} \left(\rho \frac{\partial \psi}{\partial \alpha_k} \right) \quad (10.3)$$

but

$$\psi = \psi^e(\boldsymbol{\varepsilon}^e) + \psi^i(\boldsymbol{\alpha}_k) \quad (10.4)$$

therefore assuming $\rho = cte$

$$\dot{\beta}_k = \rho \frac{d}{dt} \left(\frac{\partial \psi}{\partial \alpha_k} \right) = \rho \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \dot{\alpha}_k. \quad (10.5)$$

- Using this last result together with Eq.(10.2) and Eq.(7.20) we can write

$$\frac{\partial f}{\partial \boldsymbol{\sigma}} \cdot \mathbf{C} \left(\dot{\boldsymbol{\varepsilon}} - \dot{\gamma} \frac{\partial f}{\partial \boldsymbol{\sigma}} \right) - \dot{\gamma} \rho \frac{\partial f}{\partial \beta_k} \star \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \frac{\partial f}{\partial \beta_k} = 0$$

and recalling that $\mathbf{N}_{\boldsymbol{\sigma}} = \frac{\partial f}{\partial \boldsymbol{\sigma}}$ and $\mathbf{N}_{\beta_k} = \frac{\partial f}{\partial \beta_k}$ then or

$$\dot{\gamma} = \frac{\mathbf{N}_{\boldsymbol{\sigma}} \cdot \mathbf{C} \dot{\boldsymbol{\varepsilon}}}{\mathbf{N}_{\boldsymbol{\sigma}} \cdot \mathbf{C} \mathbf{N}_{\boldsymbol{\sigma}} + \rho \frac{\partial f}{\partial \beta_k} \star \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \frac{\partial f}{\partial \beta_k}}. \quad (10.6)$$

Continuum tangent operator - $\mathbf{C}^{ict}$

Continuum tangent operator - $\mathbf{C}^{ict}$

- Returning to Eq.(10.2) and substitute Eq.(10.6) and write

$$\begin{aligned}\dot{\sigma} &= \mathbf{C} \left(\dot{\varepsilon} - \frac{\mathbf{N}_{\sigma} \cdot \mathbf{C} \dot{\varepsilon}}{\mathbf{N}_{\sigma} \cdot \mathbf{C} \mathbf{N}_{\sigma} + \rho \mathbf{N}_{\beta_k} \star \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \mathbf{N}_{\beta_k}} \mathbf{N}_{\sigma} \right) \\ &= \mathbf{C} \dot{\varepsilon} - \frac{\mathbf{N}_{\sigma} \cdot \mathbf{C} \dot{\varepsilon}}{\mathbf{N}_{\sigma} \cdot \mathbf{C} \mathbf{N}_{\sigma} + \rho \mathbf{N}_{\beta_k} \star \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \mathbf{N}_{\beta_k}} \mathbf{C} \mathbf{N}_{\sigma}\end{aligned}\quad (10.7)$$

and let us keep a close look at the following product $(\mathbf{N}_{\sigma} \cdot \mathbf{C} \dot{\varepsilon}) \mathbf{C} \mathbf{N}_{\sigma}$

$$\begin{aligned}(\mathbf{N}_{\sigma} \cdot \mathbf{C} \dot{\varepsilon}) \mathbf{C} \mathbf{N}_{\sigma} &= (\mathbf{N}_{\sigma} \cdot \mathbf{C} \dot{\varepsilon}) \mathbf{C} \mathbf{N}_{\sigma} \\ &= (\dot{\varepsilon} \cdot \mathbf{C} \mathbf{N}_{\sigma}) \mathbf{C} \mathbf{N}_{\sigma} \\ &= (\mathbf{C} \mathbf{N}_{\sigma} \otimes \mathbf{C} \mathbf{N}_{\sigma}) \dot{\varepsilon}\end{aligned}\quad (10.8)$$

where we made use of the generalization to high order tensors of following results $\mathbf{T} \mathbf{u} \cdot \mathbf{v} = \mathbf{u} \cdot \mathbf{T}^T \mathbf{v}$ and $(\mathbf{a} \otimes \mathbf{b}) \mathbf{c} = (\mathbf{b} \cdot \mathbf{c}) \mathbf{a}$.

Continuum tangent operator - $\mathbf{C}^{ict}$

Continuum tangent operator - $\mathbf{C}^{ict}$

- Now, by introducing Eq.(10.8) into Eq.(10.7) yields

$$\begin{aligned}\dot{\sigma} &= \mathbf{C}\dot{\varepsilon} - \frac{(\mathbf{C}\mathbf{N}_{\sigma} \otimes \mathbf{C}\mathbf{N}_{\sigma}) \dot{\varepsilon}}{\mathbf{N}_{\sigma} \cdot \mathbf{C}\mathbf{N}_{\sigma} + \rho \mathbf{N}_{\beta_k} \star \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \mathbf{N}_{\beta_k}} \\ &= \left(\mathbf{C} - \frac{\mathbf{C}\mathbf{N}_{\sigma} \otimes \mathbf{C}\mathbf{N}_{\sigma}}{\mathbf{N}_{\sigma} \cdot \mathbf{C}\mathbf{N}_{\sigma} + \rho \mathbf{N}_{\beta_k} \star \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \mathbf{N}_{\beta_k}} \right) \dot{\varepsilon}\end{aligned}\quad (10.9)$$

- Finally, and by identification, based on Eq.(9.25) and Eq.(10.9) we write

$$\mathbf{C}^{ict} = \mathbf{C} - \frac{\mathbf{C}\mathbf{N}_{\sigma} \otimes \mathbf{C}\mathbf{N}_{\sigma}}{\mathbf{N}_{\sigma} \cdot \mathbf{C}\mathbf{N}_{\sigma} + \rho \mathbf{N}_{\beta_k} \star \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \mathbf{N}_{\beta_k}}. \quad (10.10)$$

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Example - J_2 plasticity model - $\mathbf{C}_{J_2}^{ict}$

$\mathbf{C}_{J_2}^{ict}$ for J_2 plasticity model

- The normal $\mathbf{N}_\sigma$ for J_2 plasticity model with isotropic hardening is given by Eq.(8.31) that is shown again bellow

$$\mathbf{N}_\sigma = \sqrt{\frac{3}{2}} \frac{\boldsymbol{\sigma}_d}{\|\boldsymbol{\sigma}_d\|}$$

and considering the isotropic hardening shown in Eq.(5.22) the term $\rho \mathbf{N}_{\beta_k} \star \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \star \mathbf{N}_{\beta_k}$ reduces to

$$\rho \left(N_{\beta_1} \right)^2 \frac{\partial^2 \psi^i}{\partial \alpha_k^2} = \left(\frac{\partial f(\boldsymbol{\sigma}, \kappa(\alpha))}{\partial \kappa} \right)^2 \frac{\partial^2 \psi^i}{\partial \alpha_k^2} \quad (10.11)$$

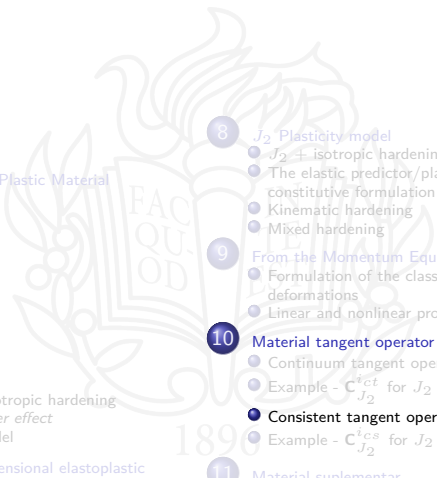
and as

$$\frac{\partial f(\boldsymbol{\sigma}, \kappa(\alpha))}{\partial \kappa} = -1 \quad \text{and} \quad \rho \frac{\partial^2 \psi^i}{\partial \alpha_k^2} = H + \delta(\sigma_\infty - \sigma_o) e^{-\delta \alpha} \quad (10.12)$$

therefore the continuum tangent operator Eq.(10.10) for J_2 plasticity model with isotropic hardening is

$$\mathbf{C}_{J_2}^{ict} = \mathbf{C} - \frac{\frac{3}{2\boldsymbol{\sigma}_d \cdot \boldsymbol{\sigma}_d} \mathbf{C} \boldsymbol{\sigma}_d \otimes \mathbf{C} \boldsymbol{\sigma}_d}{\frac{3}{2\boldsymbol{\sigma}_d \cdot \boldsymbol{\sigma}_d} \boldsymbol{\sigma}_d \cdot \mathbf{C} \boldsymbol{\sigma}_d + H + \delta(\sigma_\infty - \sigma_o) e^{-\delta \alpha}} \quad (10.13)$$

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Consistent tangent operator - $\mathbf{C}^{ics}$

Consistent tangent operator - $\mathbf{C}^{ics}$

- The consistent derivation is somewhat different. The consistent term stands for *consistent with the integration algorithm*. Therefore, the identification of $\mathbf{C}^{ics}$ is performed based on the non-linear system of equations of the return mapping algorithm.
- Considering a some kind of general material under associative plasticity this non-linear system of equations is such that $(f(\boldsymbol{\sigma}_{n+1}^{trial}, \beta_{k_n}^{trial}) > 0)$.

$$\begin{cases} \boldsymbol{\varepsilon}_{n+1}^e - \boldsymbol{\varepsilon}_{n+1}^{e^{trial}} + \Delta\gamma \mathbf{N}_{\boldsymbol{\sigma}_{n+1}} \\ \boldsymbol{\alpha}_{k_{n+1}} - \boldsymbol{\alpha}_{k_n} - \Delta\gamma \mathbf{N}_{\beta_{k_{n+1}}} \\ f(\boldsymbol{\sigma}_{n+1}, \beta_{k_{n+1}}) \end{cases} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \quad (10.14)$$

- Now we perform a linearization of system of equations in Eq. (10.14). Such linearization leads to the following set of equations:

$$\begin{cases} d\boldsymbol{\varepsilon}_{n+1}^e - d\boldsymbol{\varepsilon}_{n+1}^{e^{trial}} + d(\Delta\gamma) \mathbf{N}_{\boldsymbol{\sigma}_{n+1}} + \Delta\gamma d\mathbf{N}_{\boldsymbol{\sigma}_{n+1}} \\ d\boldsymbol{\alpha}_{k_{n+1}} - d(\Delta\gamma) \mathbf{N}_{\beta_{k_{n+1}}} - \Delta\gamma d\mathbf{N}_{\beta_{k_{n+1}}} \\ df(\boldsymbol{\sigma}_{n+1}, \beta_{k_{n+1}}) \end{cases} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (10.15)$$

Consistent tangent operator - $\mathbf{C}^{ics}$

Consistent tangent operator - $\mathbf{C}^{ics}$

- Notice that in set of Eq.(10.15) we have:

$$d\mathbf{N}_{\sigma_{n+1}} = \frac{\partial \mathbf{N}_{\sigma_{n+1}}}{\partial \sigma_{n+1}} d\sigma_{n+1} + \frac{\partial \mathbf{N}_{\sigma_{n+1}}}{\partial \beta_{k_{n+1}}} d\beta_{k_{n+1}} \quad (10.16a)$$

$$d\mathbf{N}_{\beta_{k_{n+1}}} = \frac{\partial \mathbf{N}_{\beta_{k_{n+1}}}}{\partial \sigma_{n+1}} d\sigma_{n+1} + \frac{\partial \mathbf{N}_{\beta_{k_{n+1}}}}{\partial \beta_{k_{n+1}}} d\beta_{k_{n+1}} \quad (10.16b)$$

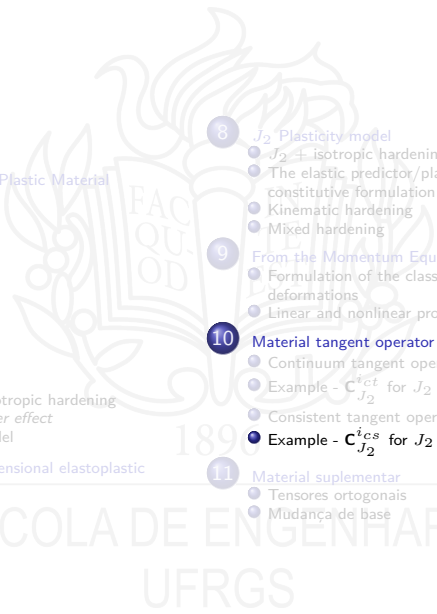
$$df(\sigma_{n+1}, \beta_{k_{n+1}}) = \mathbf{N}_{\sigma_{n+1}} \cdot d\sigma_{n+1} + \frac{\partial f}{\partial \beta_{k_{n+1}}} \star d\beta_{k_{n+1}}. \quad (10.16c)$$

- Our identification begins realizing that the linearization of the elastic deformation can be written in terms of

$$d\boldsymbol{\varepsilon}_{n+1}^e = \mathbf{C}^{-1} d\boldsymbol{\sigma}_{n+1}. \quad (10.17)$$

- Substituting Eq.(10.17) and Eqs.(10.16) into Eqs.(10.15) allows the identification of $\mathbf{C}^{ics}$.

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Example - J_2 plasticity model - $\mathbf{C}_{J_2}^{ics}$

$\mathbf{C}_{J_2}^{ics}$ for J_2 plasticity model

- The identification of $\mathbf{C}_{J_2}^{ics}$ for the J_2 plasticity model with isotropic hardening requires the linearization of the following set of equations:

$$\begin{cases} d\boldsymbol{\varepsilon}_{n+1}^e + d(\Delta\gamma) \mathbf{N}_{\boldsymbol{\sigma}_{n+1}} + \Delta\gamma \frac{\partial \mathbf{N}_{\boldsymbol{\sigma}_{n+1}}}{\partial \boldsymbol{\sigma}_{n+1}} d\boldsymbol{\sigma}_{n+1} \\ \quad d\alpha_{n+1} - d(\Delta\gamma) \\ \mathbf{N}_{\boldsymbol{\sigma}_{n+1}} \cdot d\boldsymbol{\sigma}_{n+1} + \frac{\partial f(\boldsymbol{\sigma}_{n+1}, k(\alpha_{n+1}))}{\partial \alpha_{n+1}} d\alpha_{n+1} \end{cases} = \begin{bmatrix} d\boldsymbol{\varepsilon}_{n+1}^{trial} \\ 0 \\ 0 \end{bmatrix} \quad (10.18)$$

- After a straightforward algebraic manipulation, the consistent tangent operator - $\mathbf{C}_{J_2}^{ics}$ is identified as

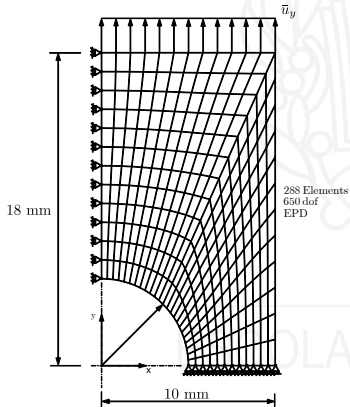
$$\mathbf{C}_{J_2}^{ics} = \frac{d\boldsymbol{\sigma}_{n+1}}{d\boldsymbol{\varepsilon}_{n+1}^{trial}} = \left(\mathbf{C}^{-1} + \Delta\gamma \frac{\partial \mathbf{N}_{\boldsymbol{\sigma}_{n+1}}}{\partial \boldsymbol{\sigma}_{n+1}} - \frac{1}{\frac{\partial f}{\partial \alpha_{n+1}}} \mathbf{N}_{\boldsymbol{\sigma}_{n+1}} \otimes \mathbf{N}_{\boldsymbol{\sigma}_{n+1}} \right)^{-1} \quad (10.19)$$

what can be rearranged to other formats using some tensor algebra.

A numeric example

Effects of the use of $\mathbf{C}_{J_2}^{ict}$ or $\mathbf{C}_{J_2}^{ics}$ in practice

- Example: Rectangular plate with a hole



- The prescribed displacement at the top of the plate is $\bar{u}_y = 0.5\text{mm}$;
- The material parameters are:
 $E = 200\text{GPa}$, $\nu = 0.3$, $H = 129.24\text{MPa}$,
 $\delta = 16.93$, $\sigma_\infty = 715\text{MPa}$ and
 $\sigma_o = 450\text{MPa}$.
- Also, the convergence criterion is the number of iterations to reach the admissible error given by
 $\|\mathbf{r}_{n+1}\|_\infty^{adm} \leq 10^{-6}$.

	Step	1	2	3	4	5
EPD	$\mathbf{C}_{J_2}^{ict}$	3	19	38	54	52
	$\mathbf{C}_{J_2}^{ics}$	4	6	7	7	6
AxyS	$\mathbf{C}_{J_2}^{ict}$					
	$\mathbf{C}_{J_2}^{ics}$					

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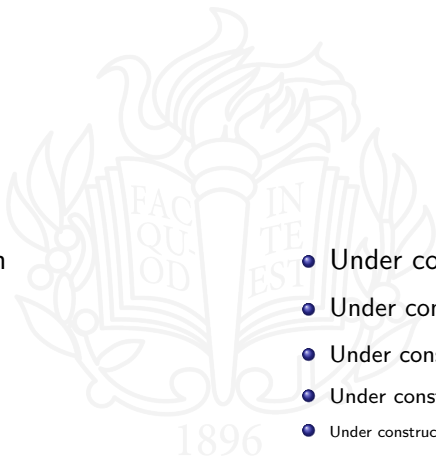
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Classroom Slides on Plasticity

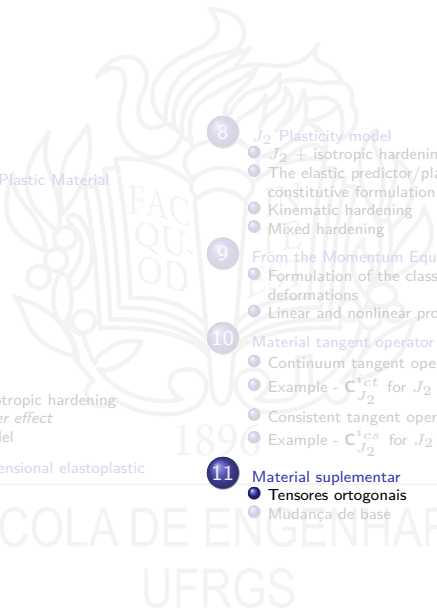
Prof. Rodrigo Rossi¹

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Draft version - - - Under development

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Tensores ortogonais

Uma classe muito especial de tensores de segunda ordem

- Let us define now a very important class of second-order tensors, the so-called *orthogonal tensors*, which are denoted by $\mathbf{Q}$ in this text. An orthogonal tensor holds the following properties

$$\mathbf{Q}^T \mathbf{Q} = \mathbf{Q} \mathbf{Q}^T = \mathbf{I} \quad (11.1)$$

what is seen as the orthogonality property, what implies that

$$\mathbf{Q}^T = \mathbf{Q}^{-1}. \quad (11.2)$$

Another important property is that the determinant of any orthogonal tensor equals either $+1$ or -1 , that is

$$\det \mathbf{Q} = \pm 1.$$

- The set of all orthogonal tensors will be denoted $\mathcal{O}$. An orthogonal tensor $\mathbf{Q}$ with $\det \mathbf{Q} = +1$ is called a *proper orthogonal tensor*. The set of all *proper orthogonal tensor* will be denoted $\mathcal{O}^+$.

Tensores ortogonais

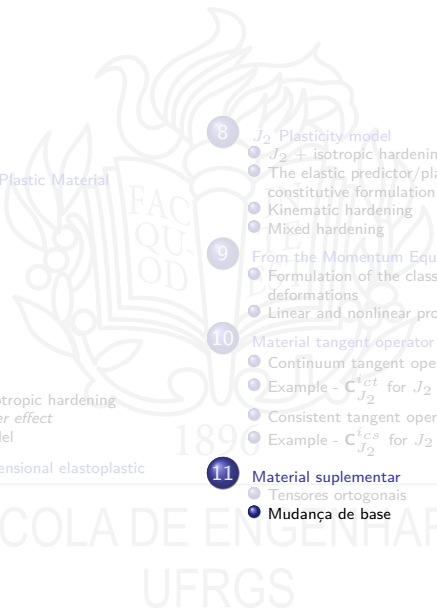
Uma classe muito especial de tensores de segunda ordem

- Proper orthogonal tensors represent a *rotation*. Orthogonal tensors that represent a rotation will be denoted in this text by $\mathbf{R}$ to *simplify* their identification. Orthogonal tensors with $\det \mathbf{Q} = -1$ represents a *reflection* in one of the axis, see Fig. (*missing*). Also, see [Malvern 1969] p. 32 for more details.

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Mudança de base

Uma aplicação muito importante de tensores ortogonais é a mudança de base... Primeiro, vamos investigar (revisitar) a mudança de base de um vetor

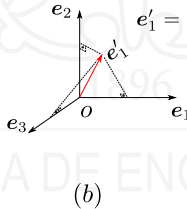
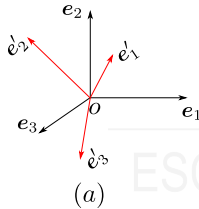
- The vector $\mathbf{u}$ is written in terms of the basis $\mathbf{e}_1, \mathbf{e}_2$, and $\mathbf{e}_3$, that is a basis $\mathbf{b} = \{\mathbf{e}_1, \mathbf{e}_2, \mathbf{e}_3\}$, as

$$\mathbf{u} = u_i \mathbf{e}_i.$$

As the choice of the basis is somewhat arbitrary, THE SAME VECTOR can be written in a different basis such as $\mathbf{e}'_1, \mathbf{e}'_2$, and $\mathbf{e}'_3$, named $\mathbf{b}' = \{\mathbf{e}'_1, \mathbf{e}'_2, \mathbf{e}'_3\}$

$$\mathbf{u} = u'_i \mathbf{e}'_i$$

$$\mathbf{e}'_1 = (\mathbf{e}'_1 \cdot \mathbf{e}_1) \mathbf{e}_1 + (\mathbf{e}'_1 \cdot \mathbf{e}_2) \mathbf{e}_2 + (\mathbf{e}'_1 \cdot \mathbf{e}_3) \mathbf{e}_3$$



Mudança de base

Vamos introduzir a notação matricial onde escrevemos explicitamente os componentes na base a que nos referimos:

- So

$$[\mathbf{u}]_b = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix} \quad (11.3)$$

is the vector $\mathbf{u}$ written in terms of the original basis $\mathbf{e}_i$, whereas

$$[\mathbf{u}]_{b'} = \begin{bmatrix} u'_1 \\ u'_2 \\ u'_3 \end{bmatrix} \quad (11.4)$$

is written in terms of the original basis $\mathbf{e}'_i$.

- The relation between both basis:

$$\mathbf{e}'_1 = (\mathbf{e}'_1 \cdot \mathbf{e}_1) \mathbf{e}_1 + (\mathbf{e}'_1 \cdot \mathbf{e}_2) \mathbf{e}_2 + (\mathbf{e}'_1 \cdot \mathbf{e}_3) \mathbf{e}_3.$$

Clearly

$$\begin{aligned} \mathbf{e}'_i &= (\mathbf{e}'_i \cdot \mathbf{e}_1) \mathbf{e}_1 + (\mathbf{e}'_i \cdot \mathbf{e}_2) \mathbf{e}_2 + (\mathbf{e}'_i \cdot \mathbf{e}_3) \mathbf{e}_3 \\ &= (\mathbf{e}'_i \cdot \mathbf{e}_j) \mathbf{e}_j. \end{aligned} \quad (11.5)$$

- Now, define the components Q_{ij} as

$$Q_{ij} := \mathbf{e}_i \cdot \mathbf{e}'_j. \quad (11.6)$$

Mudança de base

Vamos introduzir a notação matricial onde escrevemos explicitamente os componentes na base a que nos referimos:

- That is, the cartesian components are written as

$$[Q] := \begin{bmatrix} e_1 \cdot e'_1 & e_1 \cdot e'_2 & e_1 \cdot e'_3 \\ e_2 \cdot e'_1 & e_2 \cdot e'_2 & e_2 \cdot e'_3 \\ e_3 \cdot e'_1 & e_3 \cdot e'_2 & e_3 \cdot e'_3 \end{bmatrix}. \quad (11.7)$$

- Then Eq. (11.5) is re-written as

$$e'_i = (e'_i \cdot e_j) e_j = Q_{ji} e_j \quad (11.8)$$

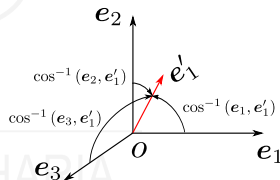
$$\stackrel{\text{Eq. (??)}}{=} Q e_i \quad (11.9)$$

or

$$e_i = Q^T e'_i. \quad (11.10)$$

- Based on Eq. (11.6) one shows that the components of Q are, in fact, the direction cosines between both basis. That is

$$Q_{ij} := \cos(e_i, e'_j). \quad (11.11)$$



Mudança de base

- Components of $\mathbf{u}$ with relation to the original basis in terms of the new basis $\mathbf{e}'_i$, that is

$$\begin{aligned} u_i &= \mathbf{u} \cdot \mathbf{e}_i = \mathbf{u} \cdot Q_{ij} \mathbf{e}'_j = \\ &= Q_{ij} (\mathbf{u} \cdot \mathbf{e}'_j) = Q_{ij} u'_j, \end{aligned} \quad (11.12)$$

or in analog form

$$\begin{aligned} u'_i &= \mathbf{u} \cdot \mathbf{e}'_i = \mathbf{u} \cdot Q_{ji} \mathbf{e}_j \\ &= Q_{ji} (\mathbf{u} \cdot \mathbf{e}_j) = Q_{ji} u_j \end{aligned} \quad (11.13)$$

- Equations (11.12)-(11.13) in terms of a matrix notation

$$[\mathbf{u}]_b = [\mathbf{Q}] [\mathbf{u}]_{b'}, \quad (11.14a)$$

$$[\mathbf{u}]_{b'} = [\mathbf{Q}]^T [\mathbf{u}]_b \quad (11.14b)$$

where

$$\begin{aligned} [\mathbf{Q}] &= \begin{bmatrix} \mathbf{e}'_1 \cdot \mathbf{e}_1 & \mathbf{e}'_2 \cdot \mathbf{e}_1 & \mathbf{e}'_3 \cdot \mathbf{e}_1 \\ \mathbf{e}'_1 \cdot \mathbf{e}_2 & \mathbf{e}'_2 \cdot \mathbf{e}_2 & \mathbf{e}'_3 \cdot \mathbf{e}_2 \\ \mathbf{e}'_1 \cdot \mathbf{e}_3 & \mathbf{e}'_2 \cdot \mathbf{e}_3 & \mathbf{e}'_3 \cdot \mathbf{e}_3 \end{bmatrix} \\ &= \begin{bmatrix} | & | & | \\ \mathbf{e}'_1 & \mathbf{e}'_2 & \mathbf{e}'_3 \\ | & | & | \end{bmatrix} \end{aligned} \quad (11.15)$$

meaning that the vectors of the b' are the columns of $[\mathbf{Q}]$.

Mudança de base

Tensores de segunda ordem: Regra de transformação entre duas bases diferentes

- Consider the mapping in the basis e_i :

$$\mathbf{v} = \mathbf{T}\mathbf{u}. \quad (11.16)$$

The same mapping with relation to a new basis e'_i is

$$\mathbf{v}' = \mathbf{T}'\mathbf{u}'. \quad (11.17)$$

- How does $\mathbf{T}'$ relate to $\mathbf{T}$? Write $\mathbf{u}$ and $\mathbf{v}$ in the new basis, i.e.

$$\mathbf{Q}\mathbf{v}' = \mathbf{T}\mathbf{Q}\mathbf{u}' \rightarrow \mathbf{v}' = \mathbf{Q}^T\mathbf{T}\mathbf{Q}\mathbf{u}'. \quad (11.18)$$

Substituting Eq. (11.17) into Eq. (11.18) yields

$$\mathbf{T}'\mathbf{u}' = \mathbf{Q}^T\mathbf{T}\mathbf{Q}\mathbf{u}' \rightarrow (\mathbf{T}' - \mathbf{Q}^T\mathbf{T}\mathbf{Q})\mathbf{u}' = \mathbf{0} \quad \forall \mathbf{u}' \quad (11.19)$$

resulting in

$$\mathbf{T}' = \mathbf{Q}^T\mathbf{T}\mathbf{Q} \quad \text{or} \quad \mathbf{T} = \mathbf{Q}\mathbf{T}'\mathbf{Q}^T \quad (11.20)$$

or in components

$$T'_{ij} = Q_{ki}T_{kl}Q_{lj} = Q_{ki}Q_{lj}T_{kl}. \quad (11.21)$$

Mudança de base

Tensores de segunda ordem: Regra de transformação entre duas bases diferentes

- Equation 11.14b and Eq. 11.20 are given, in Cartesian components, by

$$\begin{bmatrix} u'_1 \\ u'_2 \\ u'_3 \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{21} & Q_{31} \\ Q_{12} & Q_{22} & Q_{32} \\ Q_{13} & Q_{23} & Q_{33} \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$\begin{bmatrix} T'_{11} & T'_{12} & T'_{13} \\ T'_{21} & T'_{22} & T'_{23} \\ T'_{31} & T'_{32} & T'_{33} \end{bmatrix} = \begin{bmatrix} Q_{11} & Q_{21} & Q_{31} \\ Q_{12} & Q_{22} & Q_{32} \\ Q_{13} & Q_{23} & Q_{33} \end{bmatrix} \begin{bmatrix} T_{11} & T_{12} & T_{13} \\ T_{21} & T_{22} & T_{23} \\ T_{31} & T_{32} & T_{33} \end{bmatrix} \begin{bmatrix} Q_{11} & Q_{12} & Q_{13} \\ Q_{21} & Q_{22} & Q_{23} \\ Q_{31} & Q_{32} & Q_{33} \end{bmatrix}.$$

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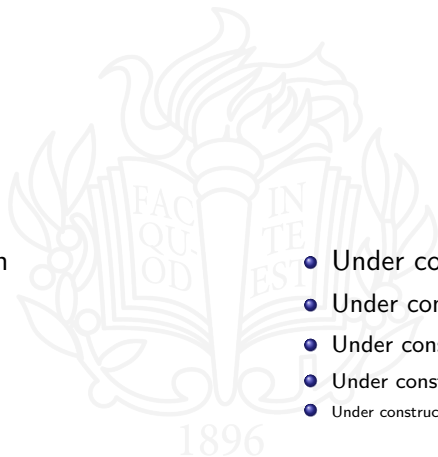
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